

SEMESTER 3

INSTRUMENTATION AND CONTROL ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof),	9

	Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER: S3

ELECTRONIC CIRCUITS AND NETWORKS

Course Code	PCICT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104: Introduction To Electrical And Electronics Engineering	Course Type	Theory

Course Objectives:

1. To develop an in-depth knowledge of operation of transistors and perform small signal analysis of amplifiers
2. To apply network theorems, Laplace and phasor analysis to solve network problems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	BJT: Review of biasing- AC and DC load lines, Q point, fixing of operating point, biasing stability, stability factors (<i>1 Hour</i>) simple biasing circuit design -various biasing methods of BJT: fixed, voltage divider, biasing independent of power supply. Thermal stability. (<i>1 Hour</i>) MOSFET: n channel- structure and operation, regions of operation, pinch-off. Biasing methods: by fixing gate source voltage, fixing gate voltage and connecting source resistance, using two power supplies, drain-gate feedback resistor. MOSFET as amplifier. channel length modulation (<i>3 Hours</i>) SMALL SIGNAL ANALYSIS OF MOSFET: voltage transfer characteristics, small signal operation-equivalent circuit, hybrid pi model. CS, CG, CD configurations, expressions for characteristic parameters: input and output impedance, voltage gain and transconductance. (<i>6 Hours</i>)	11

2	MULTISTAGE AND DIFFERENTIAL AMPLIFIERS: Multi stage amplifier - Differential amplifier- basic MOSFET differential pair operation. <i>(3 Hours)</i> FEEDBACK AMPLIFIERS: Effect of negative feedback on bandwidth, gain and noise reduction. Voltage, Current, Transconductance, and Transresistance amplifiers. <i>(5 Hours)</i> POWER AMPLIFIERS: classifications (basic operation using input output signal and circuit diagram) and power efficiency - class A (series fed/ transformer coupled) - class B (transformer coupled push pull amplifier, complementary symmetry) - class C - power amplifier <i>(3 Hours)</i>	11
3	BASIC CIRCUITS ANALYSIS: Review of fundamental laws: Ohm's law, resistors in series and parallel circuits voltage and current division, Source transformation technique, star - delta transformation <i>(1 Hour)</i> Kirchoff's laws - mesh current and node voltage - method of mesh and nodal analysis <i>(1 Hour)</i> NETWORK THEOREMS FOR RESISTIVE CIRCUITS WITH DEPENDENT AND INDEPENDENT SOURCES: Thevenin's and Norton theorems, Superposition theorem, Maximum power transfer theorem, Reciprocity theorem, Millman's theorem. related problems <i>(8 Hours)</i> LAPLACE TRANSFORM: Review of Laplace transform and inverse Laplace transform of common functions and properties. solution of differential equations using Laplace transform <i>(1 Hour)</i>	11
4	Steady state And transient response of RL, RC and RLC circuits , application of Laplace transform in RLC circuit analysis. Analysis of electrical and mechanical systems-Mass Spring Damper system,Electric motor,Liquid level System <i>(7 Hours)</i> TWO PORT NETWORKS: Types Of Two Port Networks- Z , Y, H, ABCD. Concept of symmetry and reciprocity, Conversions. <i>(4 Hours)</i>	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand biasing circuits for BJTS and MOSFETS, their operation, and perform small signal analysis to optimize amplifier performance.	K2
CO2	Understand multistage and differential amplifiers, the effects of negative feedback on amplifier performance, and the operation and efficiency of various classes of power amplifiers.	K2
CO3	Understand fundamental circuit laws and analysis techniques, apply network theorems to resistive circuits, and use Laplace transforms to solve differential equations in circuit analysis.	K2
CO4	Understand rotational and translational mechanical systems and mechanical systems ,applying principles of dynamics and differential equations to model and solve related problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											2
CO3	3											2
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microelectronics	Adel S. Sedra, Kenneth C. Smith, Tony Chan Carusone, Vincent Gaudet	Oxford University Press	Edition, 2020
2	Electronic Circuits And Theory	Robert Boyelstad, Louis Nashelsky	Prentice Hall	7 th Edition,
3	Network Analysis And Synthesis	Ravish R.	Mc Graw Hill	2 nd Edition, 2015
4	Network Analysis	Valkenburg V.	Prentice Hall	3 rd Edition 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic circuits, analysis and design	Neamen D.	TMH	3 rd edition, 2007
2	Electronic Devices and Circuits	David A Bell	Oxford university press	5 th Edition, 2008
3	Circuits And Networks- Analysis And Synthesis	Sudhakar A., S. P. Shyammohan	Mc Graw Hill	5 th Edition, 2015
4	Networks- Analysis And Synthesis	Franklin F. Kuo	Wiley India	2 nd Edition, 2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108102112 , https://nptel.ac.in/courses/108105158
2	https://nptel.ac.in/courses/108105158
3	https://nptel.ac.in/courses/108105112
4	https://nptel.ac.in/courses/108105112

SEMESTER: S3

TRANSDUCERS AND MEASUREMENTS

Course Code	PCICT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Learn various transducer types and their principles. Differentiate between primary and secondary transducers.
2. Understand principles and functions of electrical measuring instruments ,current and potential transformers and resistance measurement techniques with bridges and ohmmeters.
3. Master techniques for measuring force, acceleration, torque, power, and fluid properties.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Transducers Definition of transducers - primary and secondary transducers. Principles of variable resistance transducers – Potentiometers - Strain gauge Piezo electric transducers, materials and properties, modes of deformation, Hall effect transducers. Principle, type and construction of variable inductive transducers, Different types of self and mutual inductance transducers, LVDT and RVDT. Uses, advantages and disadvantages of inductive transducers. Principle, types and construction of different types of variable capacitance Transducers. Uses, advantages and disadvantages of capacitive transducers, LDR.	9
2	Electrical Measuring Instruments Indicating Instruments: Principle, types of control and damping; Moving coil Instruments: Types (permanent magnet, dynamometer type meters); Moving Iron Instruments: Attraction and repulsion type, Principles and torque equation; Wattmeters: Dynamometer type wattmeter, Principles and torque	11

	equation; Measurement of single phase and three phase power; true RMS meter; Errors and Compensation.	
3	Transformers Current transformers and Potential transformers; use of instrument transformers with wattmeter. Bridges and Potentiometers Measurement of resistance: Ohmmeter, Megger, Wheatstone bridge; Kelvin's double bridge; Phasor diagram; AC bridges: Measurements of inductance using Maxwell and Anderson bridges; Measurements of capacitance using Schering Bridge; Potentiometer: General principle; Modern form of dc potentiometers; Vernier dial principle; Standardization.	11
4	Force Measurement - Platform balance - Force to displacement conversion - Proving ring - Conversion of force to hydraulic pressure - Piezoelectric force transducer. Measurement of acceleration Preliminary ideas - Characteristics of a spring - mass - damper system - Piezoelectric accelerometer - Laser Doppler Vibrometer - Fiber Optic Accelerometer Measurement of torque and power - Mechanical brake arrangement - Prony brake - Electric generator as a dynamometer - Measure shear stress on the shaft - Tachometer - Mechanical Device - Non-contact optical RPM meter. Fluid properties: Density, surface tension, capillarity and viscosity; Newton's law of viscosity; Fluid Statics; Pascal's law, Centre of pressure, Buoyancy, Metacentre; Basic equations of fluid flow; continuity, momentum and energy equations; Bernoulli's equations	13

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles, types, construction, and applications of various transducers, enabling them to effectively select, design, and implement appropriate sensing technologies across engineering disciplines.	K2
CO2	Possess comprehensive understanding of the principles, types, construction, and applications of electrical measuring instruments.	K2
CO3	Understand the theory, operation, and practical application of transformers, instrument transformers with wattmeters and various types of bridges.	K2
CO4	Understand the principles and operation of measuring systems used for force, and torque.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical Measurements (for module 4)	S.P.Venkateshan	Ane Books	2 nd edition, 2015
2	A course in Electrical and Electronics Measurements and Instrumentation	A K SAWHNEY	Dhanpat Rai & Co. (P) Limited	19th edition, 2015
3	Mechanical Measurements	Thomas G. Beckwith, Roy D. Marangoini, John H. Lienhard V	Pearson	5th edition, 1993

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Metrology	R.K. Jain	Khanna Publishers	2022
2	Electrical and Electronics Measurements and Instrumentation	Prithwiraj Purkait, Budhaditya Biswas, Santanu Das	McGraw Hill Education	2017

Video Links (NPTEL, SWAYAM...)	
Sl. No	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107242/

SEMESTER: S3

DESIGN OF LOGIC CIRCUITS

Course Code	PBICT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Students will gain expertise in using number systems, Boolean algebra, logic gates and minimization techniques
2. This course aims to equip students with the knowledge to analyse and design both combinational and sequential logic circuits using fundamental concepts and practical techniques
3. This course will equip students to independently or collaboratively design and build practical digital circuits

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Number Systems: Binary, Octal, and Hexadecimal - Representation of negative numbers in binary -binary arithmetic Binary Codes: BCD, BCD addition, Excess-3, Gray code. Concept of parity Logic Gates: Introduction to logic families. Basic gates and universal gates, Realisation of logic gates using transistors and diodes, internal diagram and basic working of TTL NAND and CMOS NOT gates.	9
2	Minimization Techniques Boolean Algebra - Operations, Laws and theorems, De Morgan's theorems, SOP and POS Boolean expressions, truth tables, Karnaugh map (up to 4 variables), Minterm and Maxterm expansions, incompletely specified functions, formation of switching functions from word statement, realization using basic gates and universal gates Functions of combinational logic Adders and subtractors, ripple carry adders, BCD adder, binary to gray and	13

	gray to binary code converters, comparators, multiplexers, demultiplexers, decoders, encoders.	
3	Sequential Logic circuits Latches – SR Latch. Flip- Flops – SR, JK, D & T Flip Flops – Level & Edge triggered flip flops – Synchronous & Asynchronous inputs - Conversion between flip flops. Master slave flip flops, waveform diagrams Shift Registers SISO, SIPO, PISO, PIPO shift registers, Right & Left shifts, Universal shift registers	11
4	Counters Asynchronous counters- Up, Down and Up/ Down counter, Mod n counters – Ring counter- Johnson counter. Introduction to design of synchronous sequential circuits using Finite State Machines - Mealy & Moore types with single input-single out problems-design of sequence detectors, synchronous counters and sequence generators	11

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	• Each question carries 6 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Perform arithmetic operations and conversions in various number systems such as binary, octal, hexadecimal and binary codes	K3
CO2	Apply Boolean algebraic laws and theorems, DeMorgans Theorems and Karnaugh map to minimize Boolean expression, design combinational logic circuits and implement using basic gates and universal gates, and electronics design software tools	K3
CO3	Explain the working of latches, flip-flops and shift registers	K2
CO4	Design synchronous circuits using Mealy and Moore methods and Design asynchronous counters, and implement using flip-flop chips and electronic design software tools	K3
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3				3							3
CO3	3											3
CO4	3				3							3
CO5	3	3	3		3				3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Logic Design	Charles H Roth Jr	CENGAGE Learning Custom Publishing	7 th /2013
2	Fundamentals of Digital Logic With Verilog Design	Stephen Brown, Zvonko Vranesic	McGraw Hill	3 rd /2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Design Principles and Practices with Verilog	John F Wakerly	Pearson Education	5 th /2018
2	Digital principles and applications	Donald P Leach, Albert Paul Malvino	McGraw Hill	8 th /2014
3	Digital Design	M. Morris Mano	Visionias	6 th /2022

Video Links (NPTEL, SWAYAM...)	
Sl. No	Link ID
1	https://youtube.com/playlist?list=PLbRMhDVUMnge4gDT0vBWjCb3Lz0HnYKkX&si=VUIVRuaZ7Of_WXZZ

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
 - Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum	11

	Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
	b)	Point out any three merits of NPV method.	CO4	(3)												

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy	6

	and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-	6

	world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER: S3**LOGIC CIRCUIT LAB**

Course Code	PCICL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. The lab exposes students to the practical aspects of digital electronics design.
2. Students will learn how to translate theoretical knowledge into real-world circuits, debug malfunctions, and optimize circuit performance for specific applications.

Expt .No.	Experiments using hardware components-Combinational circuits (Minimum 5 Experiments)
1	Realization of logic gates using diodes and transistors
2	Study of basic logic gates and realization of logic gates using universal gates.
3	Half and full adders and subtractors using basic gates and universal gates
4	Study of adder IC and implementation of binary adders, adder cum subtractors & BCD adder using adder IC.
5	Design and implementation of code converters.
6	Design and implementation of comparator circuits
7	Realization of simple Mux, Demux, Decoder and Encoder using basic gates and study of their ICs
8	Combinational logic design using Multiplexers and Decoders
Expt. No.	Experiments using hardware components-Sequential circuits (Minimum 5 Experiments)
1	Realization of Flip-Flop Circuits (SR, JK, T, D and Master Slave JK) using basic gates.
2	Study of flip flop ICs and counter ICs
3	Asynchronous Counters

4	Johnson and Ring Counters
5	Synchronous counters.
6	Shift registers using flip flops (Serial in Left shift, right shift, SIPO, PISO, bidirectional)
Expt. No.	Experiments using electronic design software tools (Minimum 2 Experiments)
1	A sequence generator circuit display the sequence in seven segment display
2	A sequence detector Circuit
3	Design and implement a BCD counter and display the count in seven segment display

**Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Realise logic gates using diodes and transistors, and universal gates	K3
CO2	Simplify, design and implement combinational circuits using basic and universal gates	K3
CO3	Design and implement synchronous and asynchronous sequential circuits	K3
CO4	Implement digital circuits using electronic design software tools	K3
CO5	Demonstrate skill in technical communication and team work	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3				3							3
CO5	3								3	3		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Logic Design	Charles H Roth Jr	CENGAGE Learning Custom Publishing	7 th /2013
2	Fundamentals of Digital Logic With Verilog Design	Stephen Brown, Zvonko Vranesic	McGraw Hill	3 rd /2013
3	Digital Fundamentals	Floyd and Jain	Pearson Education	11 th /2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Design Principles and Practices with Verilog	John F Wakerly	Pearson Education	5 th /2018
2	Digital principles and applications	Donald P Leach, Albert Paul Malvino	McGraw Hill	8 th /2014
3	Digital Design	M. Morris Mano	Visionias	6 th /2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLbRMhDVUMnge4gDT0vBWjCb3Lz0HnYKkX&si=VUIVRuaZ7Of_WXZZ

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER: S3**ELECTRONIC DEVICES AND CIRCUITS LAB**

Course Code	PCICL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Electronics and Electrical Engineering	Course Type	Lab

Course Objectives:

1. To study the characteristics of diodes and transistors
2. To design clipping and clamping circuits for desired specifications
3. To realise low pass and high pass filters, integrator and differentiator circuits
4. To study the operation of amplifiers and obtain frequency response

Expt.No.	Experiments
1	V-I characteristics of diodes and zener diodes.
2	Series voltage regulator.
3	Rectifier circuits: Half wave, full wave and bridge
4	Filters: RC low pass and high pass filter
5	Integrator and differentiator circuits
6	V I characteristics of BJT and MOSFET
7	Biasing of BJT
8	Single stage RC coupled CE amplifier
9	Biasing of MOSFET
10	Clippers
11	Clampers
12	Cascade amplifier- gain and frequency response
13	Differential amplifier using transistors
14	Two stage RC Coupled amplifier
15	Class AB power amplifier

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse characteristics of Discrete components	K4
CO2	Design various wave shaping circuits using diodes and study the working of voltage regulators and RC filters	K3
CO3	Design bjt/mosfet biasing circuits and study the characteristics of different amplifiers	K3
CO4	Design power supply circuits, differentiator, integrator using discrete components	K3
CO5	Demonstrate skills in team work, technical communication and documenting an experimental work	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3									3
CO2	3		3									3
CO3	3		3									3
CO4	3		3									3
CO5									3	3		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microelectronics	Adel S. Sedra, Kenneth C. Smith, Tony Chan Carusone, Vincent Gaudet	Oxford University Press	8 th Edition, 2020
2	Electronic Circuits And Theory	Robert Boyelstad, Louis Nashelsky	Prentice Hall	9 th Edition, 2006
3	The Art of Electronics	Paul Horowitz, Winfield Hill	Cambridge University Press	3 rd Edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Semiconductor Physics and Devices	Donald A Neaman	Tata Mc Graw Hill	4/e, 2012
2	Electronic devices and circuits	Millman and Halkias,	McGraw Hill Publication	4/e, 2015
3	Schaum's Outline of Electronic Devices and Circuits	Jimmy J Cathey	McGraw-Hill Education	2/e, 2002
4	Electronics lab manual	K A Navas	Rajath Publications	4/e, 2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117107095
2	https://nptel.ac.in/courses/108105158
3	https://nptel.ac.in/courses/108105158 , https://nptel.ac.in/courses/108101094
4	https://nptel.ac.in/courses/108101094

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

**INSTRUMENTATION AND CONTROL
ENGINEERING**

SEMESTER S4
MATHEMATICS FOR ELECTRICAL SCIENCE – 4

(Group B)

Course Code	GBMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To expose the students to the basics of random processes essential for their subsequent study of analog and digital communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Random process concept, classification of process, Methods of Description of Random process, Special classes, Average Values of Random Process, Stationarity- SSS, WSS, Autocorrelation functions and its properties, Ergodicity, Mean-Ergodic Process, Mean-Ergodic Theorem, Correlation Ergodic Process, Distribution Ergodic Process. [Text 2: Relevant topics from Chapter 6]	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Probability, Statistics and Random Processes	T Veerarajan	The McGraw-Hill	3 rd edition, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Probability and Random Processes	Palaniammal, S.	PHI Learning Private Limited	3 rd edition, 2015
4	Introduction to Probability	David F. Anderson, Timo, Benedek	Cambridge	1 st edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/

SEMESTER: S4**INDUSTRIAL INSTRUMENTATION 1**

Course Code	PCICT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT205 BASICS OF INSTRUMENTATION ENGINEERING	Course Type	Theory

Course Objectives:

1. To familiarize the students about various types of temperature, pressure, level, pH, density, viscosity, turbidity and conductivity measurement systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Measurement of Temperature Definitions and units- Standards of temperature – Thermometry and pyrometry- Thermocouples- Peltier effect- Thomson effect- Seebeck effect- Types of thermocouples (Chromel- copper. Chromel- alumel. copper- constantan, iron-constantan, platinum-rhodium- platinum) Cold junction compensation-Signal conditioning for thermocouple-Thermowells— Thermopiles- Resistance thermometers- Principle of operation- Platinum RTD- Construction of RTDs- Lead compensation- 3wire method – 4 wire method- Thermistors – Quartz crystal sensors-Expansion thermometers- Bimetallic thermometers – Liquid filled thermometers- Gas filled thermometers- Vapor pressure thermometers- Pyrometry- Stefan Boltzmann’s law- Black body radiation- Total radiation pyrometers-Optical radiation pyrometers-Disappearing filament photo electric pyrometer Conductivity Measurement Theory of operation, Two electrode cells, Four electrode measurement, electrodeless cell	11

2	Measurement of Pressure Introduction – units and definitions – standards of pressure – bellows type pressure sensors: motion balance and force balance pressure sensors, C-bourdon, spiral and helical bourdon pressure sensors-Diaphragm or capsule type sensors: motion balance and force balance pressure sensors-Differential pressure elements: dry, force balance and dry, motion balance- Electronic pressure sensors: strain gauge, Capacitance, Potentiometric, Resonant wire, Piezoelectric, Magnetic and Optical transducers, High-Pressure sensors: Dead-Weight piston gauge, Bulk modulus cells, Manometers: U tube manometer – inclined manometer- ring balanced type manometer, Micromanometers. Vacuum pressure measuring elements- McLeod gauge – ionization gauges – Knudsen gauge – momentum transfer gauges- thermal conductivity gauges – Pirani’s gauge.	11
3	Level measurement Bubblers-Capacitance level sensors-Level switches; field effect and diaphragm switch, Differential pressure level detector: electronic d/p transmitters, displacer level detectors: torque tube displacer, interface measurement, Float level devices: float level switch ,Laser level detectors: triangular measurement, time-of-reflection measurement. Level gauges: Flat glass gauge, magnetic level gauge, remote reading gauge, Optical level devices; Light reflection, transmission and refraction. Radar level transmitters: time-of flight determination, radiation level sensor; continuous level measurement, Ultrasonic level measurement,Resistance tape level measurement, Rotating paddle switch	11
4	Density Measurement Displacement and float type, hydrometer, hydrostatic densitometers. Radiation densitometer, vibrating densitometer, gas densitometers. pH measurement Theory-Measurement electrodes-Fibre-optic pH measurement-digital pH meters- amplifiers for pH electrodes. Viscosity measurement Laboratory types: Rotational, Saybolt, Redwood , Falling ball and Capillary tube viscometers. Industrial types –Capillary type of viscosity measurements- Falling piston-float viscometer –Oscillating blade. Turbidity measurement Forward scattering type, scattered light detectors, backscatter turbidity analyzer	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Course Outcomes (COs)

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain working principle and measurement techniques of various types of temperature measuring and conductivity measuring instruments	K2
CO2	Discuss working principle and measurement techniques of various types of pressure measuring instruments.	K2
CO3	Explain working principle and measurement techniques of various types of level measuring instruments	K2
CO4	Describe working principle and measurement techniques of various types of density, pH, viscosity and turbidity measuring instruments.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineers Handbook	Liptak.B.G	Chilton book.co.	Fourth,Volume 1, 2003
2	Instrument technology	Jones.E.B	Scientific Pub	Volume1,Edition3, 1974
3	Instrumentation Reference Book	Walt Boyes	Elsevier	4/e.2010
4	Applied Instrumentation in Process Industries	Andrew and Williams	Gulf publications	Volume 1,2.Edition2,1979

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical and Industrial Measurements	Jain.R.K	Khanna Publishers	Edition11,1995
2	Principles of Industrial Instrumentation	Patranabis.D	Tata McGraw Hill	Edition2,2008
3	Measurement System - Application and Design	Doebelin.E.O	McGraw hill	Edition4,1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec28.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec29.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec27.mp4
2	https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec21.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec22.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec23.mp4
3	https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec46.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec47.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec48.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec49.mp4
4	https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod11lec54.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod11lec55.mp4

SEMESTER: S4

SIGNALS AND SYSTEMS

Course Code	PCICT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of the fundamental concepts of signals and systems, including various types of signals and system properties.
2. Equip students with the ability to mathematically analyze signals and systems using tools such as convolution, Fourier series, Fourier transform, Laplace transform, and Z-transform.
3. Understand the fundamental concepts of sampling and its importance in signal processing.
4. Learn about sampling theorem, aliasing, and reconstruction of signals

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of Signals and Systems: Definitions and Classification of Signals (Continuous-time and Discrete-time, Periodic and Aperiodic, Even and Odd, Energy and Power signals) Basic Operations on Signals (Scaling, Shifting, Folding) Systems: Definitions and Classification of Systems (Linear and Nonlinear, Time-Invariant and Time-Variant, Causal and Non-Causal, Stable and Unstable) Examples of Systems System Properties: Linear Time-Invariant (LTI) Systems ,Stability and Causality in LTI Systems Representation of systems: Differential equation representation of continuous time systems. Difference equation representation of discrete systems.	12

	Discrete time LTI systems and linear convolution. Stability and causality of LTI systems.	
2	Overview of Fourier Series: Representation of Periodic Signals ,Convergence of Fourier series. Continuous-Time Fourier Transform (CTFT)-Properties Review of Laplace Transform, ROC of Transfer function, Properties of ROC, Stability and causality conditions. Relation between Fourier and Laplace transforms.	10
3	Discrete time fourier series for discrete periodic signals. Properties of DTFS. Discrete time fourier transform (DTFT) and its properties. Analysis of discrete time LTI systems using DTFT. Z transform , Mapping of s plane to Z plane ,ROC , Inverse transform, properties, Relation between DTFT and Z-Transform, Analysis of discrete time LTI systems using Z transforms, Stability and causality using Z transform.	13
4	Sampling Definition and importance of sampling in signal processing Nyquist-Shannon Sampling Theorem, Conditions for perfect reconstruction, Definition and causes of aliasing, Effects of undersampling, Methods to prevent aliasing, Ideal vs. Practical Sampling, Ideal sampling (impulse sampling), Practical sampling methods (zero-order hold, first-order hold) Process of quantization, Quantization error and its effects, Uniform and non- uniform quantization, Reconstruction of continuous-time signals from discrete-time samples, Interpolation methods (zero-order hold, first-order hold, sinc interpolation), Oversampling: benefits and applications, Under sampling: conditions and applications Sampling in the frequency domain, Sampling of band pass signals Introduction to multi rate systems, Decimation and interpolation. Applications in digital signal processing	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and classify various types of signals and systems.	K2
CO2	Apply Fourier series, Fourier transform, Laplace transform, and Z-transform for signal analysis.	K3
CO3	Model and solve signal processing problems using differential and difference equations	K3
CO4	Understand the fundamental principles and practical techniques of sampling, including the application of the sampling theorem, methods to prevent aliasing, and techniques for signal reconstruction.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Signals and Systems	Alan V. Oppenheim, Alan S. Willsky, and S. Hamid Nawab	Prentice Hall	2 nd edn 1997
2	Signal and Systems	Simon Haykin and Barry Van Veen	Wiely	2 nd edition 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Linear Systems and Signals,	B. P. Lathi	Oxford University Press	2005
2	Signals and Systems, Scham's outlines	H. P Hsu, R. Ranjan	TMH	2006
3	Principles of Signals and Systems	Fred J. Taylor	McGraw-Hill Education	1994

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ee15/preview,
2	https://onlinecourses.nptel.ac.in/noc20_ee15/preview
3	https://onlinecourses.nptel.ac.in/noc20_ee15/preview
4	https://onlinecourses.nptel.ac.in/noc20_ee15/preview

SEMESTER: S4

INTEGRATED CIRCUITS AND SYSTEMS

Course Code	PBICT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PC40T302	Course Type	Theory

Course Objectives:

1. Provide students with a comprehensive understanding of integrated circuit fabrication techniques and operational amplifier theory.
2. Enable students to apply their theoretical understanding of operational amplifiers to design and analyze fundamental circuit configurations.
3. To develop a comprehensive understanding of advanced operational amplifier circuits and their applications in signal processing, waveform generation, and oscillation.
4. To comprehensively understand the theory, design principles, and practical implementation of various types of analog filters using operational amplifiers.
5. To develop a comprehensive understanding of analog-to-digital converters (ADCs), digital-to-analog converters (DACs), voltage regulators, 555 timers, voltage- controlled oscillators (VCOs), and phase-locked loops (PLLs)

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Integrated Circuit Fabrication Process Brief overview of integrated circuit fabrication technology – Silicon wafer preparation, Epitaxial Growth, Oxidation, Photolithography, Diffusion, Ion Implantation, isolation techniques, metallization and packaging. Operational amplifiers Introduction of op amps- block level diagram of op amp –Basic information of op amp (741 op amp) Characteristics of Operational amplifiers Ideal op amp characteristics-DC characteristics-input bias current, input offset current, input offset voltage, thermal drift, CMRR, PSRR-AC characteristics-	12

	frequency response, stability of op amp, slew rate. Applications of Operational amplifiers Basic applications-inverting amplifier, non-inverting amplifier, differential amplifier, scale changer, summing amplifier, subtractor, adder, voltage follower, V-I converter (grounded load type and floating load type), Howland constant current circuit and I-V converter.	
2	Instrumentation amplifier (3 op amp design)-op amp integrator- op amp differentiator- op amp circuit using diodes-rectifier (half and full wave), peak detector, sample and hold circuit, log and antilog amplifier, multiplier and divider. Comparator (inverting and non-inverting type)-applications of comparator-zero crossing detector, window detector, time marker generator. Waveform generators and Oscillators using Operational amplifier Schmitt trigger- Astable multivibrator- Monostable multivibrator- Triangular waveform generator. Oscillator-introduction, Block level diagram-RC phase shift oscillator- Wien Bridge Oscillator.	12
3	Filters Introduction-LPF-first order filter, second order filter (sallen-key Butterworth), higher order filter-HPF- first order filter, second order filter (sallen-key Butterworth), higher order filter- BPF-wide BPF, Narrow band pass filter-BRF-wide BRF, narrow BRF (Twin T Notch design)-All pass filter.	9
4	Analog to Digital Converter and Digital to Analog Converter DAC-weighted resistor type DAC, R-2R DAC, ADC-Direct type -Flash type, counter type, Servo type, successive approximation type- Integrating type ADC - dual slope type- specifications of DAC and ADC. Voltage regulator-Introduction, series op amp regulator-IC regulators-78XX and 79XX characteristics. 555 timer-introduction-functional diagram-Mono stable operation and its applications- Astable operation and its applications. Voltage controlled oscillator and PLL (basic principle only)	11

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Achieve proficiency in Integrated Circuit Fabrication and Operational Amplifier Theory	K2
CO2	Design linear, nonlinear circuits, oscillators and waveform generators.	K3
CO3	Design active filters using operational amplifiers.	K3
CO4	Understand the working of Analog-to-Digital, Digital-to-Analog Conversion Techniques, Voltage Regulators, 555 Timer, VCO and PLL.	K2
CO5	Design and implement an op-amp based circuit for specific application and prepare a brief report.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3		3		3							3
CO3	3		3		3							3
CO4	3											3
CO5	3	3	3		3				3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Linear Integrated Circuits	D Roy Choudhury and Shail B Jain	New Age International Publishers	6 th Edition, 2021
2	Laboratory Manual for Operational Amplifiers and Linear ICs	David A. Bell	OUP Canada	2 nd edition 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op Amps: Design, Application, and Troubleshooting	David Terrell	Newnes	2nd edition 1996
2	Op-Amps and Linear Integrated Circuits	Ramakant A. Gayakwad	Pearson Education	4th edition 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://nptel.ac.in/courses/117103063/
2	https://nptel.ac.in/courses/117/101/117101106/
3	https://nptel.ac.in/courses/117/107/117107094/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER: S4

POWER ELECTRONICS FOR CONTROL ENGINEERING

Course Code	PEICT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYESL106 BASIC ELECTRICAL AND ELECTRONICS	Course Type	Theory

Course Objectives:

1. Identify and describe the characteristics and operations of various power semiconductor devices.
2. Comprehend the operation and design of basic power converter circuits.
3. Implement control strategies in power electronic converters.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Power semiconductor devices: Power diodes, power transistors, thyristor family, SCRs, Triac, GTOs, power MOSFETs, IGBTs, MCTs-static and dynamic characteristics, protection circuits, series and parallel connections, turn-on characteristics, turn off characteristics	9
2	AC-DC Converters (Rectifiers) Operation of half-wave and full-wave rectifiers. Bridge rectifiers and performance parameters (ripple, efficiency). DC-DC Converters Types of DC-DC converters (buck, boost, buck-boost, Cuk). Mode of operation, duty cycle calculations, and efficiency analysis. Design and application examples in power supplies and renewable energy systems. DC-AC Converters (Inverters) Basic principles of inverters and types (square wave, sine wave, PWM inverters).	12

	Performance parameters and harmonic analysis. Applications in UPS, motor drives, and grid-tied renewable energy systems.	
3	Control Techniques in Power Electronics -Pulse Width Modulation (PWM) Introduction to PWM techniques. Types of PWM (single-pulse, multiple-pulse, sinusoidal PWM). Applications in voltage regulation and motor control. Feedback Control Systems Basics of feedback control systems in power electronics. Implementation of feedback loops in power converters	6
4	Industrial Applications and Practical Considerations Power Electronics in Renewable Energy Role of power electronics in solar PV and wind energy systems. Maximum Power Point Tracking (MPPT) techniques. Grid integration and storage solutions. Motor Drives and Control Types of motors used in industrial applications (DC, induction, synchronous). Role of power electronics in motor control. Variable frequency drives (VFD) and their design considerations. Practical Issues and Emerging Trends Thermal management, EMI/EMC, and protection strategies in power electronics. Reliability and safety standards. Emerging technologies and future trends in power electronics (wide bandgap semiconductors, smart grids, IoT integration).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts and significance of power electronics in modern technology.	K2
CO2	Design and analyze AC-DC rectifiers, DC-DC converters, and DC-AC inverters.	K3
CO3	Identify and describe the characteristics and operations of various power semiconductor devices.	K2
CO4	Apply pulse-width modulation and feedback control techniques in power electronic systems.	K3
CO5	Understand the applications of power electronics in renewable energy systems and motor drives.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power Electronics: Converters, Applications, and Design	Ned Mohan, Tore M. Undeland, and William P. Robbins.	WILEY INDIA	3 RD EDITION 2002
2	Power Electronics Handbook	Muhammad H. Rashid.	BH ELSEVIER	5 TH EDITION 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	IEEE Power Electronics Transcations			
2	Power Electronics: Circuits, Devices and Applications	Muhammad H. Rashid	BH ELSEVIER	5 TH EDITION 2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/102/108102145/
2	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee97/
3	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee97/
4	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee97/

SEMESTER: S4

NETWORK THEORY

Course Code	PEICT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Electronic Circuits and Networks	Course Type	Theory

Course Objectives:

1. To develop a comprehensive understanding of three-phase circuit analysis
2. To impart knowledge of S-domain analysis, transient analysis, and phasor analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Three phase circuits: Phase sequence- Star and delta connection-Relation between line and phase voltages and currents in balanced systems Analysis of balanced and unbalanced 3 phase circuits-Measurement of active and reactive power.	9
2	S Domain: Transfer Function, driving point and transfer functions for one port and two port network. Impulse Function & Response. Concept And Significance Of Poles And Zeros. Time Domain Response From Pole Zero Plot. Concept Of Stability, Routh-Hurwitz Criterion- analysis of marginally stable, unstable and stable systems	9
3	AC Transient Analysis: Behaviour Of Energy Storage Elements In The Presence Of Sources. Analysis Of Source Free Networks, Transient Response of R-L, R-C, R-L-C Circuits (Series And Parallel Combination) For Sinusoidal Excitations-Initial Conditions-Solution Method Using Laplace Transforms	9
4	Phasor Analysis: Application Of Phasor Analysis In RLC Networks Filters: Low Pass, High Pass, Band Pass, Band Elimination, Filter Design and implementation	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze and solve problems related to three-phase circuits, including the relationships between line and phase voltages and currents in balanced systems, and measure active and reactive power .	K3
CO2	Apply S-domain analysis techniques to determine transfer functions, understand the significance of poles and zeros, and derive time-domain responses from pole-zero plots to assess system behavior.	K3
CO3	Evaluate the stability of electrical systems using the Routh-Hurwitz criterion and categorize systems as stable, marginally stable, or unstable based on this analysis.	K3
CO4	Perform AC transient analysis of R-L, R-C, and R-L-C circuits under sinusoidal excitations using Laplace transforms, and apply phasor analysis for the effective study of RLC networks, including the design and implementation of various types of filters.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3										3
CO2	2	3										3
CO3	2	3										3
CO4	2	3										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Network Analysis	Valkenburg V.	Prentice Hall	3 rd Edition 2011
2	Engineering Circuit Analysis	W. H. Hayt and J. E. Kemmerly	McGraw Hill Education	8 th Edition 2013
3	Fundamentals of Electric Circuits	C. K. Alexander and M. N. O. Sadiku	McGraw Hill Education	5 th edition 2004
4	Network Analysis	Joseph A. Edminister, Mahmood Maqvi,	Schaum's Outline Series, TMH	4 th edition 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Circuits And Networks- Analysis And Synthesis	Sudhakar A., S. P. Shyammoan	Mc Graw Hill	5 th Edition, 2015
2	Networks- Analysis And Synthesis	Franklin F. Kuo	Wiley India	2 nd Edition, 2012
3	Network Analysis And Synthesis	Ravish R.	Mc Graw Hill	2 nd Edition, 2015
4	Electronic circuits, analysis and design	Neamen D.	TMH	3 rd edition, 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108108076
2	https://nptel.ac.in/courses/107106081
3	https://archive.nptel.ac.in/courses/108/102/108102185/
4	https://nptel.ac.in/courses/122106025

SEMESTER: S4
ANALOG CIRCUITS

Course Code	PEICT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Introduction To Electrical and Electronics Engineering GYEST104	Course Type	Theory

Course Objectives:

1. To develop knowledge of small signal analysis of BJTs
2. To design wave shaping circuits according to requirements and to develop an in-depth understanding of communication systems and power devices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Small Signal Analysis of BJT: CE Configuration Using Small Signal Hybrid-Pi Model For Mid Frequency And Low Frequency. (Gain, Input And Output Impedance). High Frequency Equivalent Circuits Of BJT, Miller Effect, Analysis Of High Frequency Response Of CE Amplifier.	9
2	Wave Shaping Circuits: Clippers And Clampers Multivibrators: Types of multivibrators- Monostable, Bistable and Astable multivibrators- duty cycle, frequency Schmidt Trigger Circuit Oscillators: Barkhausen's Criterion For Oscillation - Types Of Oscillators - Phase Shift, Wien Bridge, Hartley, Colpitt's And Crystal Oscillators.	9
3	Power Devices: Thyristor Family: UJT, SCR, Triac, Diac- Operation And VI Chara. Transferred Electron Devices, Hot Electron Devices, Varactor Diode, Power Diodes, Power Transistors, IGBT and GTO- Operation and VI Characteristics	9

4	Linear Voltage Regulators: load & line regulation, types of voltage regulators, , Block diagram of a series voltage regulator, series voltage regulator (working principle and design for a specific output voltage and current), Short circuit protection and fold back protection, Output current boosting using Darlington pair transistor.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand CE configuration using small signal hybrid-pi model, high frequency equivalent circuits, Miller effect, and CE amplifier response.	K2
CO2	Understand clippers, clampers, multivibrators, Schmidt trigger, and various oscillators	K2
CO3	Comprehend operation and VI characteristics of UJT, SCR, Triac, Diac, transferred electron devices, hot electron devices, varactor diode, power diodes, power transistors, IGBT, and GTO.	K2
CO4	Understand load and line regulation, types of voltage regulators, and design series voltage regulators with protection and output current boosting.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											2
CO3	3											2
CO4	3											2
CO5	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microelectronic Circuits	A.S.Sedra and K.C.Smith	Oxford University Press	6/e, 2006
2	Electronic Devices and Circuits	R.L. Boylestad and Louis Nashelsky	Prentice Hall of India	9/e, 2006
3	Semiconductor Physics and Devices	Donald A Neaman	Tata Mc Graw Hill	4/e, 2012
4	Electronic devices and circuits	Millman and Halkias	McGraw Hill Publication	4/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Devices	Thomas L. Floyd	Pearson Education Inc.	8 th edition, 2007
2	The Art of Electronics	Paul Horowitz, Winfield Hill	Cambridge University Press	3 rd edition, 2011
3	Schaum's Outline Of Electronic Devices And Circuits	Jimmy J Cathey	McGraw-Hill Education	2 nd edition, 2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105158
2	https://nptel.ac.in/courses/108105158 , https://nptel.ac.in/courses/108101094
3	https://nptel.ac.in/courses/108101094
4	https://nptel.ac.in/courses/117105143 , https://nptel.ac.in/courses/117102059

SEMESTER: S4

ELECTRICAL MACHINES

Course Code	PEICT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand the basic concepts of different types of electrical machines and their performance.
2. To know the different methods of starting D.C motors and induction motors
3. Analyse the performance of Synchronous and special machines

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	DC Machines-principle of operation-e.m.f equation-types of Excitations. General idea of armature reaction, OCC and load characteristics - simple numerical problems.Principles of DC motors-Back e.m.f and its significance, motors-torque and speed equations-Characteristics of DC motor	9
2	Transformers – principles of operations – emf equation-losses and efficiency – OC and SC tests. Equivalent circuits-efficiency calculations-maximum efficiency – all day efficiency –simple numerical problems. Auto transformers	7
3	Three phase induction motors- slip ring and squirrel cage types-principles of operation – rotating magnetic field- torque slip characteristics- no load and blocked rotor tests. Starting of Induction motor. Relation between Torque, Mechanical Power and Rotor Output Single phase motors- principle of operation of single-phase induction motor – split phase motor – capacitor start motor- stepper motor-universal motor	11

4	Synchronous machines types –construction- emf equation of alternator – regulation of alternator by emf method- Principles of operation of synchronous motors- methods of starting- V curves-synchronous condenser Stepper motors: Principle of Operation, multitask variable reluctance motors, single stack variable reluctance motors, Hybrid stepper motors, Linear stepper motor, comparison, Torque speed characteristics	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles of operation, e.m.f equations, types of excitations, armature reaction, OCC and load characteristics, back e.m.f, torque and speed equations, and characteristics of DC motors.	K2
CO2	Understand the principles of operation, e.m.f equation, losses, efficiency, OC and SC tests, equivalent circuits, and solve efficiency-related numerical problems.	K2
CO3	Understand the principles of operation, types, torque-slip characteristics, no load and blocked rotor tests, starting methods, relationship between torque, mechanical power, and rotor output for three-phase induction motors, and understand the principles and types of single-phase motors, including split phase, capacitor start, stepper, and universal motors.	K2
CO4	Understand the construction, e.m.f equations of alternators, regulation by emf method, principles of synchronous motors, starting methods, V curves, synchronous condensers, and understand the principles of operation, types and torque-speed characteristics..	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2			2						2
CO2	3	2				2						2
CO3	3	2	2			2						2
CO4	3	2	2			2						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrical Machinery	Bimbira P. S.,	Khanna Publishers,	2 nd edition 2011
2	Theory of AC Machines,	Nagrath J. and D. P. Kothari,	Tata McGraw Hill	5 th edition 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Performance and Design of AC Machines,	Say M G	CBS Publishers, New Delhi	3rd edition, 2002.
2	Electric Machines	Ashfaq Husain, Haroon Ashfaq,	Dhanpat Rai and Co	3rd edition, 2002.
3	Theory and Performance of Electrical Machines	Gupta J B,	S K Kataria & Sons	14th edition, 2013
4	Electrical Technology	B. L. Theraja	, S. Chand Publications.	Vol II

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ee103/preview
2	https://onlinecourses.nptel.ac.in/noc24_ee103/preview
3	https://nptel.ac.in/courses/108108076
4	https://nptel.ac.in/courses/108108076

SEMESTER: S4**MECHANICAL INSTRUMENTATION**

Course Code	PEICT416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PC40T303	Course Type	Theory

Course Objectives:

1. To familiarize with the devices for measuring displacement, vibration, force and torque
2. To familiarize with the devices for angular and linear measurements
3. To understand the concept of fluid mechanics and its applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The generalized measurement system, direct comparison and indirect comparison, types of input quantities, measurement standards, uncertainty, introduction to uncertainty , propagating uncertainty Kline and McIntock approach. Simple problems in uncertainty. Vernier calliper- principle and measurement. Micrometer parts-measurements.	6
2	Displacement measurement- strain gauge type, Inductance type, Capacitance type, Eddy current type, Measurement of force- Analytical balance, Spring balance, Load cells- Magneto-elastics load cell, piezo-electric type, Elastic load cell, Velocity measurement- Capacitive tachometer, Drag-cup type tachometer. Acceleration- null balance accelerometer. Measurement of torque – Torque measurement by relative angular twist.	6
3	Linear and angular measurement;- Linear displacement transducer- Moire fringe techniques, slip gauges stack of slip gauge; method of selecting slip gauges; adjustable slip gauge; Measurement of angles; sine bar checking unknown angles; sine center; sources of error; angle gauges. Basic acoustical parameters. Sound measuring apparatus and techniques. Basic theory of	12

	seismic instrument. Detection of nuclear radiation-GM counter, Ionization chamber, Scintillation counter etc, Measurement of solar radiation.	
4	Fluid Mechanics - Fluid properties, density, surface tension, capillarity and viscosity. Newton's law of viscosity. Fluid Statics- Basic equations of fluid flow- continuity, momentum and energy equations, Applications of Bernoulli's equations- venturimeter, orifice meter, flow nozzles and Pitot-tube. Hot wire anemometer. measurement of surface roughness – surface texture – primary texture – secondary texture and the lay specification for surface textures – methods of measuring surface finish . The Talysurf instrument – the profilograph – Tomlinson surface meter – Tracer type profilograph-gear tooth measurement-tooth thickness-tooth spacing-pitch circle diameter-parkinsons gear tester	12

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles and applications of vernier calipers, micrometers, displacement measurement, load cells, and piezoelectric transducers.	K2
CO2	Understand the use of slip gauges, sine bars, and angle gauges for accurate linear and angular measurement, and understand sources of error.	K2
CO3	Understand fluid properties, fluid statics, and basic fluid flow equations,	K2
CO4	Apply Bernoulli's equation using devices like venturi meters, orifice meters, flow nozzles, and Pitot-tubes	K3
CO5	Understand the principles behind surface finish measurement	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											2
CO3	3											2
CO4	3											2
CO5	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Course in Mechanical Measurements and Instrumentation	A.K.Sawhney, Puneet Sawhney		

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Experimental Methods for Engineers	Holman J.P.	McGrawHill Publishing Company	
2	Mechanical Measurements	Thomas G. Beckwith and N. Lewis Buck	Pearson	6 th edition
3	Principles of Measurements Systems	John Bentley	Pearson Education	4 th edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112106179
2	https://archive.nptel.ac.in/courses/112/103/112103261/
3	https://archive.nptel.ac.in/courses/112/103/112103261/
4	https://archive.nptel.ac.in/courses/112/104/112104118/#

SEMESTER S4
DIGITAL SYSTEM DESIGN

Course Code	PEICT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PC40T303	Course Type	Theory

Course Objectives:

1. Fundamental Understanding of Verilog HDL
2. Design and Simulation Skills for Combinational and Sequential Circuits
3. Advanced Digital Design and FPGA Implementation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Verilog: Overview of HDL-Importance and applications of Verilog-Basic syntax and structure of a Verilog Program-Data types and operators Combinational Logic Design: Design and simulation of combinational circuits using Verilog-Examples: Half Adder, Full Adder, Multiplexer, Demultiplexer Sequential Logic Design: Design and simulation of sequential circuits using Verilog-D Flip-Flop, T Flip-Flop, JK Flip-Flop, 4-bit Counter	9
2	Verilog Programming Constructs: Continuous assignments - Procedural assignments (always blocks) - Blocking and non-blocking assignments - Case statements and if-else constructs Modules and Hierarchical Design: Modules and ports-Module instantiation-Hierarchical design concepts-Examples: Designing a simple ALU, 4-bit Adder/Subtractor Testbenches and Simulation: Writing testbenches for Verilog designs - Introduction to simulation tools -Timing and delays in Verilog - Simulation examples and debugging	9

3	Hazards in digital circuits :Static and dynamic hazards-Design of hazard free circuits-Elementary ideas of Clock Skew-Synchronizer failure and metastability Synchronous sequential circuit design using finite state machines (FSM): Mealy and Moore machines – ASM Charts - Design and implementation of FSMs using Verilog - Examples: Sequence Detector, Traffic Light Controller Asynchronous sequential circuits: Asynchronous Behavior-Analysis of asynchronous circuits-synthesis of asynchronous circuits- Race condition-State reduction – State assignment – Transition diagram	9
4	Designing with Programmable devices: ROM-Programmable Logic Arrays-Programmable Array Logic- combinational PLDs- (PAL14L4 & PAL12H6) - Sequential PLDs (PAL16R4)-Simple PLDs (22V10)-Complex Programmable Logic Devices (XC9500)-Field Programmable Gate Arrays (XC 4000 & FLEX 10k) FPGA design flow- Implementing Verilog designs on FPGAs- Examples: LED Blinker, Basic Counter on FPGA	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design and simulate combinational and sequential logic circuits using Verilog HDL	K3
CO2	Apply Verilog Constructs and Hierarchical Design Techniques in digital design and write testbenches for Verilog design	K3
CO3	Design synchronous circuits using Verilog, Design hazard free circuits, and design and analyse asynchronous circuits.	K4
CO4	Design and Implement Digital Systems Using Programmable Logic Devices and FPGAs	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				3							3
CO2	3				3							3
CO3	3				3							3
CO4	3		2		3			3	3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamental of digital logic with Verilog design	Stephen Brown and Zvonko Vranesic	McGraw Hill Education	2 nd /2017
2	Digital Design: Principles and Practices	John Wakerly	Pearson Education	5 th /2018
3	Verilog HDL A guide to digital design and synthesis	Samir Palnitkar	Pearson India	2 nd /2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog	M. Morris Mano , Michael D. Ciletti	Pearson Education	6 th /2018
2	Verilog by Example: A Concise Introduction for FPGA Design	Blaine Readler	Full ARC Press	2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLJ5C_6qdAvBELELTSPgzYkQg3HgclQh-5&si=dhhv9YMeWdYqBRPw
2	https://youtube.com/playlist?list=PLwdnzlV3ogoVGq4TIpX4NH6QEFYiAnyvA&si=P2T_bpZNoXecGiyH
3	https://youtube.com/playlist?list=PLwdnzlV3ogoVIY7iVqr-FhWUQEX7JDdiP&si=nXHoToMpd3HcaQFq
4	https://youtube.com/playlist?list=PL59HO_adcCOicvXhvCgcUwA-OmchjSkxX&si=Kg9BikELhsyBZRoM

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4
ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes,	6

	Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and	6

	human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method

(CIE: 50 marks, ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8

	the project, including methodologies, findings, and reflections)	2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4

CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3
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Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006

3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.

- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).

Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER: S4

TRANSDUCER AND MEASUREMENTS LAB

Course Code	PCICL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT303	Course Type	Lab

Course Objectives:

1. To make the students aware of basic concepts of measurement and operation of different types of transducers.
2. To make the students conscious about static and dynamic characteristics of different types of transducer.
3. To make the students to calibrate the electrical instruments

Expt. No.	Experiments
1	Dynamic response of (1) first order liquid level system (2) RC circuit.
2	Response Characteristics and V-I Characteristics of LDR.
3	Measurement of level using Capacitive transducers.
4	(1) Vibration measurement using Piezoelectric transducers. (2) Measurement of Speed (3) IR Range sensor
5	Calibration of DC ammeter, voltmeter and wattmeter.
6	Measurement of resistance using DC bridges.
7	Characteristics of Strain gauge and load cell.
8	Loading effect of analog voltmeter
9	Use of universal LCR Bridge and digital LCR meter for measurement of inductance, capacitance and resistance
10	Characteristics of Hall Effect transducer

11	Characteristics of displacement sensor (RVDT and LVDT) and measure an unknown linear and rotary displacement
12	Characteristics (loading and unloading) of Temperature Sensors (RTD, Thermocouple, Thermistor) and compare different sensors
13	Characteristics of Synchros
14	Familiarisation of Virtual Instrumentation software used (E.g.: - LabVIEW) a) Loops and charts. Case and sequence structures. (mandatory experiment)

**Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Calibrate an instrument.	K3
CO2	Determine the dynamic response of a measurement system.	K3
CO3	Measure temperature, displacement, force, strain, light, level, speed, vibration and flux.	K3
CO4	Extent the range & determine electrical parameters and loading effect and Familiarisation of Virtual Instrumentation software	K3
CO5	Demonstrate skills in technical communication and team work.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3
CO5	3	2							3	3		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Measurement Systems: Application and Design	Ernest.O Doebelin	McGraw- Hill.	Edition4,1990
2	A course in Mechanical Measurement and Instrumentation	A.K Sawhney	Dhanpat Rai & Co.	Edition20,2015
3	A course in Electrical and Electronic Measurements and Instrumentation	A K Sawhney	Dhanpat Rai & Co.	Edition25,2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrumentation Devices & Systems	C.S. Rangan, G.R. Sarma, V.S.V. Mani	Tata McGraw- Hill publishing company Ltd.	Edition2,1993
2	Transducers and Instrumentation	DVS Murthy	PHI	Edition2,2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107214/ https://archive.nptel.ac.in/courses/108/105/108105053/ https://archive.nptel.ac.in/courses/112/103/112103261/ https://archive.nptel.ac.in/courses/108/105/108105064/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER: S4

ANALOG CIRCUITS LAB

Course Code	PCICL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBICT404	Course Type	Lab

Course Objectives:

1. Gain practical experience in analyzing and designing fundamental op-amp circuits such as inverting, non-inverting, and voltage follower configurations using the 741 op-amp.
2. Explore more complex op-amp circuits and their applications.
3. Gain proficiency in designing and analyzing op-amp based filters.

Expt. No.	Experiments
1	Study of 741 op amp and implementation of basic circuits using 741 – Inverting, non-inverting, voltage follower.
2	Measurement of opamp parameters.
3	Summing and difference amplifiers using op-amp.
4	Design of Instrumentation amplifier.
5	Design of Active Integrator & differentiator circuits.
6	Comparator circuits using op-amp.
7	Design of Waveform generators (triangular & saw tooth).
8	Design of Wienbridge Oscillator.
9	Design of Astable multivibrator and Monostable multivibrator using opamp.
10	Design of Active first and second order high pass & low pass filters.
11	Design of Narrow Band pass filter.
12	Design of Notch filter.

13	Implement the following circuit using any simulation software and obtain the time domain and frequency domain signal at each specified point and write a report based on the result (Mandatory experiment) <pre> graph LR O1[Oscillator 10Hz] --> A1[Non Inverting Amplifier, G=3] O2[Oscillator 100Hz] --> A2[Non Inverting Amplifier, G=5] O3[Oscillator 1KHz] --> A3[Non Inverting Amplifier, G=7] A1 -- A --> AC[Adder circuit] A2 -- B --> AC A3 -- C --> AC AC -- D --> F1[First Order LPF 15Hz] AC -- D --> F2[Twin T Notch Filter 100Hz] AC -- D --> F3[First Order HPF 900Hz] F1 -- E --> Out1[] F2 -- F --> Out2[] F3 -- G --> Out3[] </pre>
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**Course Assessment Method (CIE: 50 marks,
ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop proficiency in designing fundamental amplifier circuits using op-amps and accurately estimating various offset parameters.	K3
CO2	Acquire the skills to design a wide range of both linear and non-linear circuits utilizing operational amplifiers.	K3
CO3	Acquire skill to design diverse active filter configurations, understand what makes each type of filter different and use them in different situations	K3
CO4	Gain the ability to design waveform generators and applying them effectively in practical applications.	K3
CO5	Demonstrate skills in technical communication and team work.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3		3									3
CO3	3		3		3							3
CO4	3		3		3							3
CO5									3	3		2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Laboratory Manual for Operational Amplifiers and Linear ICs	David A. Bell	OUP Canada	2 nd edition 2006
2	Linear Integrated Circuits	D Roy Choudhury	New Age International Private Limited	6 th edition 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op Amps: Design, Application, and Troubleshooting	David Terrell	Newnes	2 nd edition 1996
2	Op-Amps and Linear Integrated Circuits	Ramakant A. Gayakwad	Pearson Education	4 th edition 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
for all labs	https://nptel.ac.in/courses/117/107/117107094/
Virtual Lab	
	http://vlabs.iitkgp.ernet.in/bc/index.html#
	https://ae-iitr.vlabs.ac.in/List%20of%20experiments.html

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.

- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

SEMESTER 5

**INSTRUMENTATION AND CONTROL
ENGINEERING**

SEMESTER S5
CLASSICAL CONTROL ENGINEERING

Course Code	PCICT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0 :0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT302	Course Type	Theory

Course Objectives:

1. To introduce students to the foundational concepts of control systems, including the differences between open-loop and closed-loop systems and the historical development of automatic control.
2. To develop the ability to analyse control systems' time-domain and frequency-domain behaviours, focusing on the response characteristics and stability criteria.
3. To equip students with the tools and techniques for modelling, reducing, and analysing control systems using block diagrams, signal flow graphs, root locus, and frequency response methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Control System (3 Hours) Introduction to control systems; open-loop and closed-loop systems; Feedback and non-feedback systems; servomechanism; history and development of automatic control. Feedback Characteristics of Control Systems include reduction of parameter variation, control over system dynamics, and control of the effects of disturbance signals. Review of Mathematical Modelling and Transfer Function (1 Hour) Block Diagram Algebra & Signal Flow Graphs (7 Hours) Block diagram of an open-loop system; Block diagram of a closed loop system; block diagram and transfer function of a typical first-order system and second-order system. Block diagram reduction techniques. Concept of signal flow graph; construction of signal flow graph; Finding transfer function from Signal flow graph using Mason's gain formula.	11
2	Time-domain Analysis of First-order Systems (5 Hours) Time response, transient and steady-state response; standard test signals	12

	and their application in system analysis. Open-loop and closed-loop transfer functions, type and order of systems; the concept of poles and zeros; type and order of the system; Response of first-order system to unit impulse and unit step input (with derivation); Time response specifications of first-order systems. Time-domain Analysis of Second-order Systems (5 Hours) Response of under-damped second-order system to unit-step input (with derivation); Response of under-damped second-order system to unit impulse and unit ramp input (Derivation not required); Response of over-damped, critically damped and un-damped second-order system to unit step input (derivation not required); time response specifications of under-damped second order system. Steady-state Errors and Error Constants (2 Hours) Steady-state error, static error constants.	
3	Stability of Linear Systems (4 Hours) Stability of linear systems: Definitions, characteristic polynomial and characteristic equation, general stability criterion with reference to s-plane; Absolute and relative stability; necessary conditions for stability. Routh's criterion of stability. The Root Locus Technique (6 Hours) The root-locus concept; Evan's criteria; Construction of root locus; determination of roots from root locus; Effect of addition of zeros and poles on root locus; root contours.	10
4	Frequency Response Analysis (6 Hours) Frequency response representation; sinusoidal transfer function; Frequency domain specifications. Polar plot and bode plot: Construction. Stability in Frequency Domain (5 Hours) Stability in frequency domain; gain margin and phase margin; stability from bode plots. Nyquist stability criterion; assessment of relative stability using Nyquist criterion.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the fundamental principles of control systems, including the distinctions between open-loop and closed-loop systems.	K2
CO2	Apply block diagram reduction techniques or Mason's gain formula to simplify a given control system and determine its transfer function.	K3
CO3	To analyse the transient and steady-state response of first and second-order control systems.	K4
CO4	To apply Routh Stability Criterion or root locus technique for determining system stability.	K3
CO5	To analyse system stability using gain margin, phase margin, and the Nyquist stability criterion	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	3											2
CO3	3	2										3
CO4	3	3										3
CO5	3	2										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Control Systems Engineering	I. J. Nagrath, Madan Gopal	New Age International Private Limited	7 & 2021
2	Modern Control Engineering	Katsuhiko Ogata	Pearson Education India	5 & 2015
3	Automatic Control Systems	Farid Golnaraghi and Benjamin C. Kuo	McGraw Hill Education (India) Pvt. Ltd	10 & 2017
4	Control Systems: Principles and Design	M Gopal	McGraw Hill Education	4 & 2012
5	Linear Control Systems with MATLAB Applications	B.S. Manke	Khanna Publishers	12 & 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Control Systems	Richard Dorf, Robert Bishop	Pearson	14 & 2021
2	Control Systems	A. Anand Kumar	PHI Learning	2 & 2014
3	Modern Control Engineering	Choudhury, Dr. Roy	Prentice Hall India Learning Private Limited	1 & 2005
4	Schaum's Outline of Feedback and Control Systems	Joseph Distefano III, Allen R. Stubberud, Ivan J. Williams	McGraw Hill	2 & 2013
5	Nise's Control Systems Engineering	Norman S. Nise	Wiley	1 & 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Swayam - Control Systems by Prof. C. S. Shankar Ram, IIT Madras
2	Swayam - Control Systems by Prof. C. S. Shankar Ram, IIT Madras
3	Swayam - Control Systems by Prof. C. S. Shankar Ram, IIT Madras
4	Swayam - Control Systems by Prof. C. S. Shankar Ram, IIT Madras

SEMESTER S5

PROCESS CONTROL INSTRUMENTATION

Course Code	PCICT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Learn the basics of process control, including process variables, control actions, and process control block diagrams.
2. Study control system objectives, evaluation criteria, and final control operations involving signal conversion, actuators, and control elements.
3. Explore discontinuous and continuous controller modes, including proportional, integral, and derivative controls, as well as composite modes like PI, PD, and PID.
4. Gain proficiency in multiple loop control systems, electronic and pneumatic controllers, process loop tuning methods, PLC programming, DCS architecture, and SCADA supervisory functions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to process control Process, process variables, need for process control, basic control actions, process control block diagram. Control system evaluation Control system objective, control system evaluation criteria. Final control operation Signal conversion (analogue and digital electrical signals), actuators (electric, pneumatic and hydraulic), control elements (mechanical and electrical valves). Controller principles Process characteristics, control system parameters.	10

2	Discontinuous controller modes Two position mode, multi- position mode, floating control mode. Continuous controller modes Proportional controller mode, integral controller mode, derivative controller mode, composite controller modes-PI, PD and three mode. Control systems with multiple loops Feed forward control, practical aspects of the design of feed forward control, feed forward- feedback control, ratio control.	11
3	Electronic controllers General features. Single mode and composite controller modes (P and PI controller design using OP AMP). Pneumatic controllers Continuous controller modes (Principle and Design of P, PI, PD, PID control modes). Process loop tuning Tuning of controllers, Cohen-Coon method of tuning (process reaction curve method). Ultimate cycle method (Ziegler Nichol's method). Damped oscillation method. Frequency response method. Comparing tuning methods.	12
4	Programmable logic controller (PLC) PLC versus relay, evolution of PLC, functions of PLC, advantages. PLC Programming -Basics of Ladder Programming, on/off instructions, internal relay, jump instructions, data handling instruction, data manipulation instructions, Arithmetic and Comparison ,PID and other important instructions .Timers and Counters in PLC Distributed Control System (DCS) Introduction to DCS, evolution of DCS, DCS flow sheet symbols, architecture of DCS. SCADA Introduction to Supervisory computer tasks and DCS configuration: supervisory computer functions.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the elements of process control loops, final control, and discrete-state process control.	K2
CO2	Explain and compare the principle and concept of continuous and discontinuous control modes	K2
CO3	Design electronic circuits for controller with respect to the process needs and use pneumatic control modes to satisfy different process needs.	K3
CO4	Understand how to use PLCs, DCS, and SCADA systems to control and automate different industrial processes.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	3		2									3
CO4	3	3	3									3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Control Instrumentation Technology.	Curtis Johnson	Pearson, 8/e.	8 TH EDITION, 2005
2	Handbook on Process Control	B. G. Liptak	Chiller Book and Co.	3 RD EDITION 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Control An Introduction to Theory and Practice.	George Stephanopoulos	Pearson Education India, 1/e.	1 ST EDITION 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103105064
2	https://archive.nptel.ac.in/courses/103/101/103101142/
3	https://nptel.ac.in/courses/103105064
4	https://nptel.ac.in/courses/108105063

SEMESTER S5
INDUSTRIAL INSTRUMENTATION II

Course Code	PCICT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT402	Course Type	Theory

Course Objectives:

1. To provide a thorough understanding of the principles and techniques involved in flow measurement, including the various types of flow meters and their applications.
2. To equip students with the knowledge of fluid mechanics, flow meter linearization and compensation, differential pressure flow meters, and control valve systems, enhancing their ability to apply this knowledge in practical scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to the physics of flow rate and total flow measurement volumetric flow rate-mass flow rate- rate of flow measurement- classification of flow meters. Fluid mechanics concept viscosity- Reynolds number-flow regimes-laminar flow-turbulent flow- transitional flow- law of continuity - Bernoulli's equation. Linearization and compensation of flow meters Linear and Nonlinear Flow meters - Gas Flow Pressure and Temperature Compensation - Liquid Temperature Compensation - Pressure and Temperature Tap Location - Flow Computers - Multivariable Flowmeters.	7
2	Differential pressure flow meter- theory, construction and applications Venturi meter-types of venturi meter - limitations-pressure tapping in	10

	venturi meter - orifice meter - types of orifice meter - pressure taps - limitations. pitot tubes - limitations - method of use for circular duct and rectangular duct-precautions during installation of pitot tube- Dall tube - installation- flow nozzle- different types of flow nozzle-long radius nozzle-ISA 1932 nozzle- generalised features of differential pressure type flow meter- installation requirement with differential pressure flow meter-target flow meter. Positive displacement flow meters -theory, construction and applications reciprocating piston flow meter - 2 piston type and 4 piston type- Oscillating piston flow meter - nutating disc flow meter - helix meter - rotating vane flow meter - oval wheel flow meter. Inferential flow meters -theory, construction and applications single and multi-jet flow meter - propeller type flow meter - double rotor propeller type flow meter - turbine flow meter.	
3	Variable area flow meter - theory, construction and applications rotameter - rotameter basic equation - rotameter classification - rotameter components - installation. Miscellaneous flow meters- theory, construction and applications Electromagnetic flow meters -Ultrasonic flow meters- Vortex shedding flow meters - thermal flow meter- open channel flow meter-weir type and Parshall Flumes type Gas Analyser Thermal conductivity type analyser - Heat of reaction method - Paramagnetic method - Zirconia oxygen analyser - Electro chemical reaction method - spectroscopic analysis: IR radiation absorption type.	9
4	Control valves General diagram - final control operation - basic parts - valve body - Bonnet - stem - Actuator - Packing - Seat - Disk - electrical and pneumatic signal conversion - electric, pneumatic & hydraulic actuators - valve characteristics. Different types of control valves- disc, stem and seat design and applications gate valve - globe valve - plug valve - ball valve - butterfly valve - diaphragm valve - pinch valve - needle valve - safety and relief valves. Control valves accessories	10

	Positioners - Filters - Pressure regulator - Limit switch - proximity switch – solenoids. Control valve noise and methods of its reduction Mechanical Vibration-Aerodynamic Noise - Hydrodynamic Noise.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the physics of flow measurement, linearization, compensation and calibration in flow measurement.	K2
CO2	Understand different types of differential pressure type and mechanical type flow measurement devices and its selection.	K2
CO3	Understand the working of variable area flowmeter and special types of flow measurement devices like electromagnetic, ultrasonic, vortex flowmeter thermal flow meters and open channel flowmeter and discuss different types of gas analysers and its requirements	K2
CO4	Demonstrate comprehensive knowledge of control valves, and their accessories, including different valve types, operational characteristics and methods of noise reduction,	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Flow measurement techniques	Y.L Dr.Arora	Universal book corporation	2 nd edition, 1999
2	Industrial flow measurement basics and practices (for module-1)	Frenzel, H Grothey et. al	ABB Automation products	2011
3	Industrial Flow Measurement (for Module-1)	David W. Spitzer	Instrument Society of America	3 rd edition, 2005
4	Valve Selection Handbook (for module-4)	Peter Smith and R W Zappe	Gulf Publishing	5 th edition, 2004
5	Valve Handbook	Philip Skousen	McGraw-Hill Education	3 rd edition 2011
6	Principles of Industrial Instrumentation (for gas analysers)	D Patranabis	McGraw Hill Education	3 rd edition 2017
7	Flow measurement techniques	Y.L Dr.Arora	Universal book corporation	2 nd edition, 1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Lessons In Industrial Instrumentation	Tony R Kuphaldt	Samurai Media Limited	2017
2	Industrial Instrumentation Fundamentals	Austin E. Fribance	McGraw-Hill Inc.,US	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/103/103103147/
2	https://nptel.ac.in/courses/108105064

SEMESTER S5 MICROCONTROLLER

Course Code	PBICT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand Microcontroller Fundamentals and Architectures
2. Develop Proficiency in Programming Microcontrollers
3. Master Interfacing and Advanced Applications using microcontrollers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Intel 8051 Microcontroller – Overview-Architecture- Instruction set - Addressing modes Evolution of microprocessors and microcontrollers- Overview of Intel 8051 family of microcontroller - Architecture of Intel 8051 - ALU, internal RAM and ROM, Oscillator and Reset Circuits, Stack and stack pointer - Special Function Registers of 8051 - Ports - Instruction set of 8051 - Data transfer instructions, Arithmetic, Logical, Compare and Rotate instructions, Bit processing instructions - Addressing modes of 8051	10
2	Microcontroller languages - ports - timers Comparison of machine language, assembly language and high-level languages - Assembler directives and C data types for 8051 - An introduction to programming 8051 in Assembly and C language programming of 8051 - Examples illustrating the use of basic types of instructions - Programming the ports of 8051 - Timers of 8051 - Modes of operations - Programming the timers in assembly language and C language	10

3	Serial communication and Interrupts Introduction to serial communication - Synchronous and asynchronous communication – RS 232 – MAX232 - Modes of operations - Serial communication programming in assembly language and C language - Introduction to interrupts in 8051 - Priority of interrupts - Interrupt handling and execution for timer - External and serial interrupts - Programming to handle all types of interrupts in assembly language and C language. 8051 Microcontroller Interfacing Interfacing external memory to Intel 8051 - Examples of Interfacing data memory and code Memory - Interfacing LCD to 8051 - Interfacing matrix keyboard Interfacing - ADC and DAC interfacing - Interfacing sensors - Interfacing relays, stepper motors and DC motors.	13
4	Overview of AVR family - RISC and Harvard architecture of the AVR – salient features - The general-purpose registers in the AVR – AVR data memory – Using instruction with data memory – AVR status register – Data RAM / ROM memory space allocation in AVR – AVR data format and directives - I/O ports – Counters and Timers – Interrupts – Serial ports - (no programming in AVR) Atmega 32 microcontroller block diagram Arduino – Block diagram description of the system– Programming Arduino- Examples of practical solutions using Arduino	10

Suggestion on Project Topics

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	demonstrate a comprehensive understanding of the Intel 8051 microcontroller family, including its architecture, instruction set, and various addressing modes.	K2
CO2	Demonstrate the skills to effectively program the Intel 8051 microcontroller using both assembly and C languages in ports and timers of 8051.	K2
CO3	Program 8051 serial communication and interrupt using assembly and C languages and also gain knowledge in Interfacing of memory, LCD, matrix keyboard, stepper motor, DC motor, ADC, DAC, sensors, and relays to 8051 microcontrollers.	K3
CO4	Explain the architecture of AVR and Arduino ATmega 32 microcontrollers and its features.	K2
CO5	Design and implement a microcontroller-based project in instrumentation and control engineering area and prepare a brief report.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C01	3											3
C02	3											3
C03	3											3
C04	3											3
C05	3	3	3		3	2			3	3	2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontrollers & Embedded Systems	Muhammed Ali Mazidi and Janice GillispieMazidi	Pearson Education India	2 nd edition, 2007
2	The 8051 Microcontroller Based Embedded Systems	Manish K Patel	McGraw Hill Education	1 st edition, 2017
3	AVR Microcontroller and Embedded Systems: Using Assembly and C,	Muhammad Ali Mazidi, ZarmadNaimi	Pearson Education India	1 st edition, 2013
4	Arduino: A Technical Reference	J. M. Hughes	O'Reilly	1 st edition, 2016
5	The 8051 Microcontrollers & Embedded Systems	Muhammed Ali Mazidi and Janice GillispieMazidi	Pearson Education India	2 nd edition, 2007
6	The 8051 Microcontroller Based Embedded Systems	Manish K Patel	McGraw Hill Education	1 st edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller	Kenneth Ayala	Delmar Cengage Learning	3 rd edition, 2004
2	Microcontrollers: Architecture, Programming, Interfacing and System Design	Raj Kamal	Pearson	2 nd edition 2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/105/108105102/
2	https://nptel.ac.in/courses/117104072

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5
ANALYTICAL INSTRUMENTATION

Course Code	PEICT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the theory and operational principles of various instrumental methods for qualitative and quantitative evaluation by different types of spectroscopy
2. To review of important methods in the analysis of industrial gases and also to impart fundamental knowledge on chromatography

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	UV-Vis, IR and fluorescence spectroscopy UV and Visible Spectroscopy: Colorimeters/ photometers: Single beam and double beam filter photometer – Spectro photometers: Single beam and double beam spectrophotometers. Infrared Spectrophotometers (NIR-Near IR): wave number, basic principle using diatomic harmonic structure, radiation sources, detectors. Fourier Transform Infrared Spectroscopy (FTIR). Raman spectroscopy: Raman Effect, basic principle, Raman spectrometer. Fluorescence spectroscopy: Principle of fluorescence – Measurement of fluorescence –Single beam and double beam filter fluorimeter- Ratio fluorimeter. Spectro fluorimeters.	8
2	Photometry, Atomic Absorption spectrometer, and ESR Spectrometer Flame Photometry: Principle and constructional details of flame photometer- Emission system –Optical system – Detectors. Atomic Absorption spectrometer: basic block diagram, atomizers, nebulizers, plasma excitation sources, radiation sources	12

	Electron Spin Resonance Spectrometers (ESR): Electron spin resonance, basic ESR spectrometer, microwave bridge, Klystrone, modulation, sample cavities. Mass, NMR spectroscopy and Gas Analysers Molecular Mass spectrometry: Principle of operation, Magnetic deflection mass spectrometers, Components of a mass spectrometer. Proton Nuclear Magnetic Resonance spectroscopy (NMR): Nuclear spin, resonance condition, relaxation condition Industrial Gas analysers- pH meters- Conductivity meters - Dissolved oxygen meters-Sodium analyser	
3	Chromatography Chromatography: Chroma to graphic process—Classification- Terms in chromatography Gas chromatography: Block diagram, Principle, Constructional details, Column details, GC detector, katharometer, flame ionization detector, flame photometric detector, photo ionization detector, electro capture detector, argon ionization detector. Liquid Chromatography: Types of liquid chromatography (adsorption chromatography, partition chromatography, gel permeation chromatography, ion-exchange chromatography) High pressure Liquid Chromatography (HPLC): Principle, Constructional details.	8
4	UV-Vis, IR and fluorescence spectroscopy UV and Visible Spectroscopy: Colorimeters/ photometers: Single beam and double beam filter photometer – Spectro photometers: Single beam and double beam spectrophotometers. Infrared Spectrophotometers (NIR-Near IR): wave number, basic principle using diatomic harmonic structure, radiation sources, detectors. Fourier Transform Infrared Spectroscopy (FTIR). Raman spectroscopy: Raman Effect, basic principle, Raman spectrometer. Fluorescence spectroscopy: Principle of fluorescence – Measurement of fluorescence –Single beam and double beam filter fluorimeter- Ratio fluorimeter. Spectro fluorimeters.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)
Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discuss the basic knowledge about analytical instruments, its concepts, and its technique.	K1
CO2	Explain the principle and operation of UV-Visible, IR and fluorescence spectroscopic analysis	K2
CO3	Explain the basic principle and working of Photometry, Atomic Absorption, and ESR Spectrometer. Discuss on the Mass, NMR spectroscopy and different types of Gas Analysers.	K2
CO4	Explain the concept of different Chromatography techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Analytical Instruments	R.S. Khandpur	Tata McGraw-Hill	3rd edition, 2015
2	Principles of Instrumental Analysis	Skoog.D.A, Holler.F.J and Niemann.T.A.	Saunders	7th edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	X-ray Method	Wiston.C	John Wiley	1 ST 1991
2	Instrumental Methods of Analysis	Ewing.G.W	McGraw-Hill	5th edition, 1985.
3	Instrumental Analysis	Mann.CK., Vickers.T.J, and Gullick.W.H	Harper and Row Publications	
4	Introduction to Instrumental Analysis	Robert.D. Braun	McGraw-Hill	1 ST EDITION 1987
5	Handbook of Instrumental Techniques for Analytical Chemistry	Frank.A.Settle	Prentice Hall	1997
6	Instrument Methods of Analysis	Willard, Merritt Dean and Settle	East-west Press	7th edition, 1988
7	Analytical Instruments	S. Abirami& D. Vijayanandh	Anuradha Publications	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/104/106/104106122/
2	https://archive.nptel.ac.in/courses/104/104/104104084/
3	https://archive.nptel.ac.in/courses/103/108/103108124/
4	https://www.youtube.com/watch?v=gaBXQW9rCDA

SEMESTER S5
BIOMEDICAL INSTRUMENTATION

Course Code	PEICT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Gain fundamental knowledge of biopotential acquisition and analysis:
2. Develop an understanding of signal conditioning principles for biomedical applications
3. Become familiar with the measurement of various physiological parameters

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	BASIC ANATOMY AND PHYSIOLOGY: Origin of bio potential and its propagation, Nervous system, Muscular system, Cardiovascular system, Respiratory System. BIOPOTENTIAL ELECTRODES: Electrode-electrolyte interface, electrode-skin interface, half-cell potential, Contact impedance, polarization effects of electrode – non polarizable electrodes. Types of electrodes - surface, needle and micro electrodes and their equivalent circuits. Recording problems - motion artifacts- Monopolar and bipolar electrode configurations	9
2	Bio signals characteristics – frequency and amplitude ranges. ECG – Einthoven ‘s triangle, standard 12 lead system, Principles of vector cardiography. EEG – 10-20 electrode system, unipolar, bipolar and average mode. EMG– unipolar and bipolar mode. SIGNAL CONDITIONING CIRCUITS: Need for bio-amplifier - differential bio-amplifier, Design of bio-amplifiers	9

	for the measurement of physiological variables. Power line interference, Band pass filtering. Working of isolation amplifiers, (transformer, capacitive and optical isolation), isolated DC amplifier and AC carrier amplifier. (Circuit diagram and working only)	
3	MEASUREMENT OF NON-ELECTRICAL PARAMETERS: Temperature and pulse rate measurements. Blood Pressure: indirect methods - Auscultatory method, oscillometric method, direct methods: electronic manometer, Pressure amplifiers, Systolic, diastolic, mean detector circuit. Blood flow and cardiac output measurement: Indicator dilution, thermal dilution and dye dilution method, Electromagnetic and ultrasound blood flow measurement. Pulmonary function measurements-Spirometer, Photo plethysmography, Body plethysmography.	7
4	ELECTRICAL SAFETY IN MEDICAL ENVIRONMENT: Types of Electrical hazards, Physiological effects of electric current – Threshold Values for Electrical Shock, shock hazards from electrical equipment – leakage current-Instruments for checking safety parameters of biomedical equipment, Methods of accident prevention, Electrical Safety Standards ASSISTIVE AND THERAPEUTIC EQUIPMENTS: Pacemakers, defibrillators, ventilators, diathermy, Heart-Lung machine and dialyzers. (Diagrams and Basic working only) BIOMEDICAL IMAGING X-ray, Computed Tomography, Ultrasound, Magnetic resonance Imaging. (Diagrams and Basic working only)	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the origin of bioelectric potentials, and working of human nervous system, muscular system, cardiovascular system and respiratory system, and describe the theory, construction and working of bio-electrodes	K2
CO2	Explain the measurement of biomedical signals such as ECG, EEG and EMG, and explain working of signal conditioning circuits and isolations amplifiers.	K2
CO3	Describe the measurement of non-electric parameters of heart and lungs	K2
CO4	Perceive the concepts of assistive & therapeutic equipment such as Pacemakers, defibrillators, ventilators, diathermy, Heart-Lung machine & dialyzers, electrical safety in medical environment and basing working of imaging modalities such as X-Ray, CT, US and MRI	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2						3
CO2	3					2						3
CO3	3					2						3
CO4	3					2						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Instrumentation and Measurements	L. Cromwell, F. J. Weibell, and L. A. Pfeiffer	Pearson Education, Delhi	2 nd , 2015
2	Medical Instrumentation Application and Design	John G Webster	Wiley India Private Limited	5 th , 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Biomedical Equipment Technology	J. J. Carr and J. M. Brown	Pearson Education, Delhi	4 th , 2002
2	Hand Book of Bio-Medical Instrumentation	R.S. Khandhpur	McGraw Hill Education (India) Private Limited	3 rd , 2014
3	Principles of Applied Bio-Medical Instrumentation	L.A. Geddes and L.E.Baker	Wiley India Pvt Ltd	3 rd , 2008
4	Principles of Medical Electronics and Bio-medical Instrumentation	C. Rajarao and S.K. Guha	Universities press (India) Ltd	2000

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI
2	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI
3	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI
4	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI

SEMESTER S5
POWER PLANT INSTRUMENTATION

Course Code	PEICT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT402	Course Type	Theory Elective

Course Objectives:

1. Provide a comprehensive overview of various methods of power generation, including hydro, thermal, nuclear, solar, and wind power, with an emphasis on their processes, instrumentation, and control mechanisms.
2. Equip students with knowledge about the critical role of instrumentation and control systems in power plant operations, including detailed study of boiler and turbine systems, and their measurement and control parameters.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Brief survey of methods of power generation-hydro, Thermal, Nuclear, Solar and Wind power, Power generation and distribution, Introduction to thermal power plant processes, Ideal steam cycles. Importance of instrumentation in power generation, Nuclear power plant instrumentation, Piping and instrumentation diagram of different types in nuclear power plant. Nuclear reactor control loops, Control and safety instrumentation, reliability aspects	8
2	Boiler – types, Boiler - turbine units and its range systems, feed water systems, steam circuits, air preheating, air to fuel ratio. Soot blowers, combustion process, products of combustion, fuel systems, treatment of flue gases, steam turbine, alternator, feed water conditioning, turbine bypass valves, details of boiler processes combined cycle power plant, burner tilting, and bypass damper.	9
3	Measurement in boiler and turbine: Metal temperature measurement in boilers, Measurements of non-electrical parameters in power	9

	plants – flow of feed water, fuel, air and steam with correction factor for temperature – steam pressure and steam temperature measurement, smoke and dust monitor, flame monitoring. Smoke density measurement, Drum level measurement, radiation detector, Distributed control system in power plants, Interlocks in boiler operation, Cooling system.	
4	Controls in boiler: feed water control, soot blowing operation, steam temperature control, Coordinated control, Combustion control – Main pressure air/fuel ratio, furnace draft and excess air control. Drum level control- two element and three-element control. Main and reheat steam temperature control Boiler following mode operation, turbine following mode operation, selection between boiler and turbine following modes. Introduction to turbine supervising system, Automatic turbine runs up systems,	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the principles and processes of different power generation methods, including hydro, thermal, nuclear, solar, and wind power.	K2
CO2	Understand the importance and implementation of instrumentation in power generation, with a focus on nuclear power plant instrumentation, control loops, and safety systems.	K2
CO3	Explain the various components and processes involved in boiler and turbine operations, including feed water systems, steam circuits, and combustion processes.	K2
CO4	Understand the measurement and control of non-electrical parameters in power plants, including flow, pressure, temperature, and density, as well as the use of distributed control systems and interlocks in boiler operations.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					2						2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power Plant Performance	Gill A.B	Butterworth, London	1984
2	Modern Power Station Practice	P.C Martin, I.W Hannah	Pergamon,	3/e.
3	The Control of Boilers	Sam. G.Dukelow	ISA Press, New York	2nd Edition ,1991

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Boiler Control Systems	David Lindsley	McGraw Hill, New York	1991
2	Power Station Instrumentation	Jervis M.J,	Butterworth Heinemann, Oxford	1993
3	Instrumentation, Controls and Testing	Modern Power Station Practice	Pergamon Press, Oxford	Vol.6, 1971

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107291/
2	https://archive.nptel.ac.in/courses/112/107/112107291/
3	https://archive.nptel.ac.in/courses/112/107/112107291/
4	https://archive.nptel.ac.in/courses/112/107/112107291/

SEMESTER S5
NUMERICAL COMPUTATION USING PYTHON

Course Code	PEICT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0 :0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory (Elective)

Course Objectives:

1. To develop students' skills in applying numerical methods to analyse and interpret engineering data and solve real-world problems.
2. To provide students with a solid foundation in numerical methods for solving engineering problems using Python.
3. To enable students to implement numerical techniques for nonlinear equations, interpolation, integration, differentiation, and ordinary differential equations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Root Finding Methods (6 Hours) (Derivation or proof not required) Bisection method; Newton-Raphson method; Secant method; False Position Method; Fixed-point iteration; Convergence and error analysis of root-finding methods. Solving nonlinear equations using the above methods in Python (2 Hours)	8
2	Interpolation (9 Hours) (Derivation or proof not required) Finite difference interpolation: Newton's finite and backward difference interpolation methods. Polynomial Interpolation: Lagrange interpolation and Newton's divided difference interpolation. Spline Interpolation: Cubic splines and B-splines.	11

	Error analysis in interpolation. Implementing interpolation methods in Python (2 Hours)	
3	Numerical Integration and Differentiation (6 Hours) Numerical Differentiation: Finite difference methods; Error analysis in numerical differentiation. Numerical Integration: Trapezoidal rule, Simpson's rule, Gaussian quadrature, Adaptive quadrature methods. Writing Python programs to perform numerical differentiation and integration (2 Hours)	8
4	Solution of Ordinary Differential Equations (7 Hours) Initial Value Problems: Euler's method; Runge-Kutta methods; Multistep methods (Adams-Bashforth and Adams-Moulton methods). Boundary Value Problems: Shooting method; Finite difference method. Solving ODEs using Python (2 Hours)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement root-finding algorithms in Python to solve nonlinear equations and analyse the convergence behaviour of these methods.	K3
CO2	Apply interpolation techniques in Python for data fitting and solving engineering problems and perform error analysis.	K3
CO3	Develop Python programs to implement numerical differentiation and integration techniques.	K3
CO4	Implement numerical methods in Python to solve ordinary differential equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				3							3
CO2	3				3							3
CO3	3				3							3
CO4	3				3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical Methods in Engineering with Python 3	Jaan Kiusalaas	Cambridge University Press	3 & 2013
2	Numerical Methods for Engineers	Steven C. Chapra and Raymond P. Canale	McGraw-Hill Education	7 & 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical Analysis	Richard L. Burden and J. Douglas Faires	Cengage Learning	10 & 2015
2	Applied Numerical Methods with MATLAB for Engineers and Scientists	Steven C. Chapra	McGraw-Hill Education	4 & 2017
3	Introductory Methods of Numerical Analysis	S.S.Sastry	Prentice Hall India Learning Private Limited	5 & 2012
4	Numerical Method and Programming	Bikas Chandra Bhui & Dipak Chatterjee	Vikas	2 & 2016
5	Numerical Methods Using Matlab	Mathews and Fink	Pearson Education India	4 & 2015
6	Numerical Analysis	Richard L. Burden and J. Douglas Faires	Cengage Learning	10 & 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur
2	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur
3	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur
4	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur

SEMESTER S5 VLSI DESIGN

Course Code	PEICT526	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course aims to develop the skill to design various VLSI circuits

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of CMOS device fundamentals . The CMOS inverter: - Voltage Transfer Characteristics, SPICE Description, Static Behavior - Switching Threshold - Noise Margins, Dynamic behavior - Device Capacitances - Propagation Delay - Power Consumption.	9
2	CMOS fabrication Processes: -N-Tub, P-Tub and Twin Tub. MOS Circuit Layout - Stick diagrams, Layout design rules, Transistor layout - PMOS and NMOS, Gate Layout - Inverter, NAND, NOR and XOR, Layout generation using MICROWIND tool (For Assignments/Projects only).	9
3	Combinational logic Circuits: - Static MOS - Complementary MOS - Ratioed logic - Pass Transistor logic - Differential Pass Transistor Logic - Transmission gate logic, Dynamic MOS - Basic Principles - Speed and power Dissipation - Signal Integrity issues.	9
4	Sequential Logic Circuits: -Timing Metrics for sequential circuits, Static Latches and Registers, Dynamic Latches and Registers. Datapath Subsystems: - Adder - Ripple carry Adder - Full adder - Carry chain Adder - Carry-Bypass Adder - Carry Select Adder - Carry Look Ahead Adder, Multiplier - Array Multiplier - Carry Save Multiplier, Shifter - Barrel Shifter - Logarithmic Shifter.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	explain the fundamentals of CMOS devices and analyze the behavior of CMOS inverters	K2
CO2	Understand CMOS fabrication processes and develop skills in MOS circuit layout	K2
CO3	design and analyse sequential logic circuits, including static and dynamic latches and registers, and understand timing metrics for sequential circuits.	K3
CO4	design and analyze various combinational logic circuits using different MOS logic styles	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3				3							2
CO3	3											2
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Integrated Circuits- A Design Perspective	J.M. Rabaey, A. Chandrakasan and B. Nikolic	PHI.	2 nd edition 2002
2	Basic VLSI Design	Douglas A. Pucknell & Kamran Eshraghian	PHI	3 rd edition
3	CMOS digital integrated circuits: Analysis and design	Sung-Mo Kang, Yusuf Lalebici	TATA McGraw-Hill	4 th edition 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CMOS Logic Circuit Design	John P. Uyemura	Springer India Pvt. Ltd. New Delhi.	1 st edition 2001
2	Analysis & Design of Digital Integrated Circuits	David A. Hodges, Horace G. Jackson, R. Saleh	McGraw Hill.	3 rd edition
3	CMOS VLSI Design, a Circuits and Systems Perspective,	Neil H. E. Weste, David Money Harris, Addison-Wesley	Pearson	4 th edition 2010
4	VLSI Circuit Layout: theory and design	T.C Hu, Ernest S. Kuh	IEEE Press	1 st edition 1985

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ee09/preview
2	https://archive.nptel.ac.in/courses/108/101/108101089/
3	https://archive.nptel.ac.in/courses/117/106/117106086/
4	https://www.youtube.com/watch?v=ibQBb5yEDIQ

SEMESTER: S5
MICROCONTROLLER LAB

Course Code	PCICL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBICT504	Course Type	Lab

Course Objectives:

1. Proficiency in Microcontroller Programming
2. Hands-on Experience with Microcontroller Interfacing
3. Introduction to Arduino-Based Systems Development

Expt. No.	Experiments
A) Assembly Language programming of 8051 Family in Keil/Ride environment: (All experiments compulsory)	
1	Examples of arithmetic operations
2	Examples of logical operation
3	Examples using simulated ports, timers
4	Examples of serial communication
B) C language programming of 8051 Family in Keil/Ride environment (All experiments compulsory)	
5	Examples of arithmetic and logical operations.
6	Examples using simulated ports, timers.
7	Examples of serial communication.
C) Using microcontroller kits (Minimum 3 experiment)	
8	Interfacing LED
9	Interfacing LCD displays
10	Keypad scanning
11	Interfacing buzzers
12	Interfacing ADC and DAC
13	Interfacing stepper motor

D) Using Arduino Board (Minimum 1 experiment)	
14	Interfacing LED
15	Interfacing LEDs, switches and various sensors
16	Design systems to control physical actuation devices like hydraulic piston, Motors etc

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Do 8051 microcontrollers programming in assembly language.	K3
CO2	Do 8051 microcontrollers programming in C language.	K3
CO3	Interface Microcontroller with peripheral devices and I/O operations	K3
CO4	Design programming in the Arduino board and implement microcontroller-based real-life projects.	K3
CO5	Demonstrate skills in technical communication and team work.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3	3	3		3							3
CO4	3	3	3	2	3						3	3
CO5										3		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller and Embedded Systems using Assembly and C	Muhammed Ali Mazidi and Janice Gillispie Mazidi	Pearson Education	3rd edition, 2007
2	AVR Microcontroller and Embedded Systems: Using Assembly and C	Muhammad Ali Mazidi, Zarmad Naimi	Pearson Education	1 st edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller	Kenneth Ayala	Cengage Learning India	3rd edition, 2007
2	Microcontroller and Applications	Mrs. Manjushri S. Joshi, Mrs. Vijaya V. Pangave	Nirali Prakashan	1 st edition, 2019
3	Arduino – A Technical Reference: A Handbook for Technicians, Engineers, and Makers	J.m Hughes	O'Reilly;	1 st edition, 2016
4	The 8051 Microcontroller	Kenneth Ayala	Cengage Learning India	3rd edition, 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117104072
2	https://nptel.ac.in/courses/108105102

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER: S5
INDUSTRIAL INSTRUMENTATION & IOT LAB

Course Code	PCICL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT503,PCICT 402	Course Type	Lab

Course Objectives:

1. Enable students to gain hands-on experience in measuring temperature, viscosity, pressure, level, and speed using various instruments like Optical Pyrometer, viscometers, bourdon pressure gauge, Capacitive transducer, and Cup anemometer.
2. Equip students with the ability to design and implement temperature measurement instruments and interface them with computers.
3. Provide students with insights into the characteristics of control valves, transmitters (level, flow, temperature, pressure), and the dynamic response of second-order systems

Expt. No.	Experiments
A) Measurement experiments (4 out of 6 experiment is mandatory)	
1	Measurement of temperature using Optical Pyrometer.
2	Measurement of Viscosity using red wood viscometer.
3	Measurement of Viscosity using rotational viscometer.
4	Calibration of bourdon pressure gauge using Dead weight pressure gauge tester.
5	Measurement of (1) level using Capacitive transducer (2) speed using Cup anemometer.
6	Study the principle of operation of a gyroscope and sonic range finder
B) Design of Industrial instruments (3 out of 5 experiment is mandatory)	
7	Design a thermometer using platinum RTD as sensor with the following specification. Temp range: - 2000c to +2000c Output Voltage: - 2Volt to 2Volt Also interface the thermometer with computer using any data acquisition system (preferably NI DAQ and LabView software) (mandatory experiment)
8	Design a thermometer using NTC thermistor as the sensor with the following specification

	Temperature Range: 0oc to 100oc, Output voltage range: 0V to 1V
9	Design and implement a thermometer (temperature sensing instrument) using solid state temperature sensing device as the sensor with the following specification. Temperature Range: 0oc to 100oc Output voltage range: 0V to 4V
10	Design and implement a thermometer (temperature sensing instrument) using thermocouples temperature sensing device as the sensor with the following specification. Temperature Range: 0oc to 400oc, Output voltage range: 0V to 4V Reference junction temp- 25oc (room temp)
11	Design and implementation of signal conditioning circuit for conductivity sensor.
C) Transmitter and final control element characteristics & second order system characteristics (2 out of 4 experiment is mandatory)	
12	Control valve inherent flow characteristics (with and without positioner)
13	Study the characteristics of elements of final control operation.
14	Study the transmitter characteristics of level, flow, temperature, and pressure
15	Analysis of dynamic response of second order system.
D) Industrial IoT based Experiments (1 out of 4 experiment is mandatory)	
16	Design an IoT based temperature controlled automatic fan
17	Upload Temperature and humidity data to thingspeak cloud
18	Retrieve Temperature and humidity data to thingspeak cloud

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Learn to accurately measure temperature using an Optical Pyrometer, viscosity, level, speed, range, acceleration and calibration of pressure gauge	K3
CO2	Design thermometers using different temperature sensors with specified temperature ranges and output voltages, and interface them with computers and develop skills in designing signal conditioning circuits for conductivity sensors.	K3
CO3	Study the dynamic response of second order system, control valve characteristics and characteristics of level, flow, temperature, and pressure transmitters	K3
CO4	Demonstrating proficiency in integrating IoT with industrial processes for enhanced automation and monitoring.	K3
CO5	Demonstrate skills in technical communication and team work.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3		3							3
CO3	3											3
CO4	3		3									3
CO5									3	3		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Transducer theory and applications	Allen Stuart , John A. Allocca	Reston Pub Co	1 st edition 1984
2	Fundamentals of Industrial Instrumentation and Process Control	William Dunn	McGraw-Hill Education	1 st edition, 2005
3	LabVIEW Based Advanced Instrumentation Systems	Sai Sumathi, P. Surekha	Springer-Verlag Berlin and Heidelberg GmbH & Co. K	2007
4	Instrumentation, Automation, IoT and Emerging Technologies for Engineers: Handbook	Madhukar Varshney, Sanjay Galhan	NA	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrumentation Reference Book	Walt Boyes	Elsevier Science	4th Edition, 2009
2	Introduction to Industrial Internet of Things and Industry 4.0	Sudip Misra, Chandana Roy, Anandarup Mukherjee	Routledge	1 st Edition, 2020
3	IoT System Design: Project Based Approach	Alice James, Avishkar Seth, Subhas Chandra Mukhopadhyay	Springer	1 st Edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105064

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

**INSTRUMENTATION AND CONTROL
ENGINEERING**

SEMESTER S6

DISCRETE TIME SIGNAL PROCESSING

Course Code	PCICT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT403 Signals and Systems	Course Type	Theory

Course Objectives:

1. Equip students with the foundational knowledge of discrete time signal processing techniques and digital filtering.
2. Enable students to apply digital signal processing techniques in their engineering careers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	DISCRETE FOURIER TRANSFORM Review of Discrete Time Fourier Transform. Frequency domain sampling and reconstruction of Discrete time signals - DFT- properties of DFT. Circular convolution - Definition, Linear convolution using DFT. Computation of DFT - Direct computation of DFT – DFT as linear Transformation – Fast Fourier Transform (Radix – 2 FFT only) – Decimation in time and decimation in frequency algorithms. Problem solving using software tools.	11
2	DIGITAL FILTER STRUCTURES Z-Transform representation of FIR and IIR filters- FIR filter structures: - Direct form, Cascade form and Lattice structures IIR filter structure: Direct form Transpose form, Cascade form and Parallel form structures. FINITE WORD LENGTH EFFECTS Quantization noise, Round off effects, Limit cycle oscillations.	11

3	IIR DIGITAL FILTERS Digital Filter Specifications – Design of IIR digital filters from analog filters – Butterworth and Chebyshev filters – Design of Low pass filters – Spectral transformation of IIR filters –Analogue to digital transfer function: impulse invariant method, bilinear transformation method. Problem solving using software tools.	11
4	FIR DIGITAL FILTERS Design of FIR filters: Estimation of filter order - FIR filter design - Impulse response of ideal filters - Gibbs Phenomenon – Fixed window functions: Rectangular, Hamming, Blackman –Adjustable window function: Kaiser window. Problem solving using software tools.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Compute DFT using direct computation and FFT methods, and perform circular convolution of discrete time signals	K3
CO2	Construct FIR and IIR Filter structures	K3
CO3	Design IIR Digital Filters	K3
CO4	Design FIR Digital filters	K3
CO5	Solve filter design problems using software tools	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO5 is to be evaluated as a part of CIE. Assignments shall be given to solve discrete time signal processing using software tools

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3	2										3
CO4	3	2										3
CO5	3	2			3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Discrete Time Signal Processing	Oppenheim and Schaffer	Pearson Education India	Discrete Time Signal Processing
2	Digital Signal Processing Principles, Algorithms and Applications	Proakis and Manolakis	Pearson Education India	Digital Signal Processing Principles, Algorithms and Applications
3	Digital Signal Processing	Monson Hayes	McGraw Hill Education (Schaum's)	Digital Signal Processing

			Outline Series)	
4	Discrete Time Signal Processing	Oppenheim and Schaffer	Pearson Education India	Discrete Time Signal Processing

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal Processing	Sanjit K Mitra	McGraw Hill Education	4 th /2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PL9567DFCA3A66F299&si=IzyfKkqrgphr2x1b
2	https://youtube.com/playlist?list=PLyqSpQzTE6M9py2rcGOLss74ZxBUvaC49&si=k7sSfgErFbzYDkcH

SEMESTER S6

INSTRUMENTATION SYSTEM DESIGN

Course Code	PCICT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	(PCICT502 & PCICT503)	Course Type	Theory

Course Objectives:

1. Improve the ability of students to comprehend and design various types of instrumentation systems.
2. To get a thorough knowledge about electric and intrinsic aspect of safety while designing an instrumentation system

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Instrumentation system design- System configuration- Problem analysis Experimental engineering analysis- Instrument design- The designer's viewpoint. Electronic processing modules for handling transducer output- P/I Conversion- I/P conversion- V/I and I/V - conversion -Carrier amplifier, Charge amplifier- Impedance converters - Phase sensitive detector -RMS converters- Ratio metric conversion- Logarithmic compression Design of thermo couple circuitry with cold junction compensation, linearization, amplification and conversion of its output to 4 to 20 ma current. Design of RTD with lead wire compensation and conversion of its output to 4 to 20 ma current. Design of Capacitance based Level Transmitter for both conducting and non-conducting liquids - Signal conditioning circuit for pH measurement.	12
2	Piping and instrumentation diagram. Introduction to process piping- Material selection- Pipe fitting. Cabling- General requirements- Cable types- Cable segregation Earthing – pre installation testing- piping and cable testing- loop testing	8

	Instrument project control- Documents to be produced- Process flow- Mechanical flow. Instrument index and instrument specification sheets- Loop wiring diagrams-Panel drawing and specification- Instrument specification and standard- Piping specifications- Electrical specification- Bid documents- Project procedures- Project scheduler work- Co-ordination. Project manager- Process engineers- Equipment engineers – Piping design supervisor- Job execution- Planning hints- scheduling- Specifying instruments-	
3	Specification of various measurement and control loops (flow, pressure, level, temperature etc.) Processes connections- Location of taps- sealing Instruments from process- Manifolds and gauge valves- Selection of units- Charts and ranges- Instrument identification- Winderising Packaged equipment system. Construction - organizing- Documents required- Planning schedule- Cost control- Ordering and receiving equipment and method- Purchase order- Material status- Instrument installation and commissioning -Instrument piping and tubing system- Air supplies- Pneumatic signal - Checklist of installation practices - calibration- testing- typical flow transmitter checkout procedures – typical control valve-check out procedure. Plant commission - Testing and pre commissioning	8
4	Control panels – Introduction- Control room lay out – Instrument power requirements and distribution Control room lighting- Communication system- Control panel types. Auxiliary racks and cabinet- Panel piping and tubing – Air headers- Graphic displays- Panel bid Specification. Safety- Introduction- Electric circuits risk- Flammable atmosphere- Other safety aspects such as electrocution, let go current. Electrical Safety- Purging and pressurizing endorser- Safety precautions and parameters for industrial jobs- Intrinsic safety- Process safety and Safety Instrumented systems , Intrinsic safety barrier installation with suitable examples.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design the commonly using electronic circuits in instrumentation system.	K3
CO2	Explain instrumentation diagrams and documentation for instrumentation project control.	K2
CO3	Discuss how to perform better supervision, Construction, installation, and commissioning in instrumentation industry.	K2
CO4	Explain about the electric and intrinsic aspect of safety while designing an instrumentation system	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											2
CO3	3											2
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied instrumentation in the process industries	Andrew.W.G and Williams.H.B	vol-1,2,3", Gulf Publishing Co.	
2	Hand book on Industrial Safety and Fire Management	Ravi Kant Pandey		
3	Applied instrumentation in the process industries	Andrew.W.G and Williams.H.B	vol-1,2,3", Gulf Publishing Co.	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrumentation for Process measurement and control	Anderson.N.A	CRC Press.	
2	Instrument Engineer's hand book-	Liptak.B.G	volume on process control", ISA CRC Press.	
3	Instrumentation reference book"	Noltingk.B.E.	Butter worth-Heinemann, 1995.	
4	Introduction to Intrinsic Safety	P R electronics		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/7iOPteIA2m0
2	https://youtu.be/sOb2EGMgaZ8
3	https://youtu.be/FE0njwfM36g
4	https://youtu.be/jFDWIKayrTc

SEMESTER S6
REFINERY INSTRUMENTATION

Course Code	PEICT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To give an exposure on the important parameters to be monitored and analysed in a refinery and impart knowledge about various techniques used for the measurement and control in the refineries

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Origin of petroleum Oil and gas traps. Physical and chemical characteristics of crude oil, source rock and maturation. Migration of oil mechanism pattern and barriers, Reservoir rocks and cap rocks, Entrapment of oil-types and mechanism. Application of remote sensing in petroleum resource Development , Basin and exploration strategies. The model approach to exploration strategy, Basin mapping methods, Depositional systems such as marine, non-marine, coastal, shelf, carbonate evaporates, Basin evolution, Sedimentation and Plate tectonics.	6
2	Important unit operations Drying-Separation-Heat Transfer-Distillation-Thermal cracking catalytic cracking catalytic reforming- hydro cracking – hydro treating. Chemical oxidation-Chemical reduction- Polymerisation. Alkylation-Isomerisation.	10
3	Production of important petrochemicals Chemicals from petroleum -Methane derivatives- Acetylene derivatives-	9

	Ethylene derivatives. Propylene derivatives - Cyclic petrochemicals. Production of Ethylene, Acetylene- and propylene from petroleum.	
4	Controlling methods Process control in refinery and petrochemical industry-Control of distillation column, catalytic cracking unit, catalytic reformer, pyrolysis unit. Optimal control of cracking units.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basics of methodologies applied for recovery and processing of petroleum.	K2
CO2	Discuss different unit operations involved in Petroleum industry.	K2
CO3	Describe different production routes for important petrochemicals.	K2
CO4	Explain the Controlling methods used in refinery	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Control Structures and applications	Balchan .J.G. and Mumme K.L	Van Nostrand Reinhold Company, New York	1998
2	Instrumentation in process industries	Liptak B.G.	Chilton Book Co,	1994
3	Chemical from petroleum	Waddams A.L	Butter and Janner Ltd	1968.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Industries	Austin G.T. Shreeves	McGraw Hill Education	
2	Process / Industrial Instruments and Control Handbook	Considine D.M	McGraw Hill	
3	Chemical Process Industries	Austin G.T. Shreeves	McGraw Hill Education	

SEMESTER S6

MICRO ELECTRO MECHANICAL SYSTEMS

Course Code	PEICT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To develop knowledge on Micro electro mechanical systems and materials used in the manufacture
2. To understand various micro fabrication procedures, techniques and packaging considerations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Micro Electronic Mechanical Systems: Materials for MEMS- silicon- structure and orientations, properties. Other materials- shape memory alloy (SMA), polymers, silicon compounds etc, material properties. Commercial MEMS structures and systems: Overview of commonly used mechanical structures in MEMS - Beams, Cantilevers, Plates, Diaphragms – Typical applications. Passive MEMS structures- fluid nozzles- ink jet nozzles. Sensors- piezoresistive pressure sensors- mass flow sensors- acceleration sensors, actuators- micro mechanical valves-micro pumps.	9
2	Process of micromachining: Process of micromachining- pattern transfer using additive technique. Silicon growth- oxidation of silicon-molecular beam epitaxy. Physical vapour deposition (PVD)- sputtering- laser deposition- ion plating- cluster beam technology- Chemical vapour deposition (CVD)- step coverage- plasma enhanced CVD, Electron cyclotron resonance CVD (ECRCVD)- sol-gel deposition technique.	9
3	Lithography	9

	Photolithography- Introduction- masks- spinning- resist and soft backing- exposure and post exposure treatment- development- descumming and post baking- resists- wafer priming, wafer cleaning and contamination- clean room- resist striping. Next generation lithography-extreme ultra violet lithography (EUVL)- x-ray lithography- LIGA- charged particle beam lithography- electron beam lithography- ion beam lithography. Etching: Introduction- dry etching- physical etching- ion etching or sputtering- ion beam etching, plasma etching- deep reactive ion etching (DRIE)- isotropic wet etching- anisotropic wet etching.	
4	MEMS packaging: Key design and packaging consideration- wafer stack thickness- wafer dicing concerns, thermal management- stress isolation- protective coating and media isolation. Wafer level packaging and die level packaging. Hermetic packaging- calibration and compensation- die attach process- solder alloy- bonding process using solder alloy- organic adhesive wiring and inter connections- Wire bonding- flip chip bonding- micro fluidics interconnections. Packaging material properties- mechanical- moisture penetration- interfacial adhesion, thermal- chemical- electrical.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define MEMS and elaborate on various materials used for MEMS fabrication and to describe different kinds of commercial applications like micropumps, valves and sensors.	K2
CO2	Describe the various techniques involved in micromachining and discuss techniques for deposition of materials to create structures.	K2
CO3	Discuss lithography and etching techniques.	K2
CO4	Develop a deep understanding of design and packaging considerations during manufacture of MEMS devices	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Micro electromechanical Systems Engineering	Nadim Maluf	Artech house Boston London	6/e, 2006
2	Fundamentals of Micro-fabrication	Mark Madou	CRC Press, New York	3/e, 2011
3	Foundations of MEMS	Chang Liu	Pearson	2/e, 2011
4	MEMS and Microsystems Design and Manufacture	Tai Ran Hsu	McGraw Hill	4/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electromechanics and MEMS	Thomas B. Jones, Nenad G. Nenadic	Cambridge University Press	Illustrated edition, 2013
2	Microsystem design	Stephen D. Senturia,	Springer (India)	2nd edition, 2006
3	Smart Sensors and MEMS	Stoyan Nihtianov, Antonio Luque	Woodhead Publishing Series	2nd edition, 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117105082
2	https://nptel.ac.in/courses/117105082
3	https://nptel.ac.in/courses/117105082
4	https://nptel.ac.in/courses/117105082

SEMESTER S6

BIOMEDICAL SIGNAL PROCESSING

Course Code	PEICT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PEICT 522 Biomedical Instrumentation	Course Type	Theory

Course Objectives:

1. Understand the fundamental concepts and characteristics of biomedical signals.
2. Understand signal acquisition systems and advanced signal processing techniques.
3. Apply signal processing knowledge to practical biomedical engineering challenges, including cardiological and neurological signal processing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Biomedical Signals Introduction to Biomedical Signals, Characteristics of Biomedical Signals, Time-domain vs. frequency-domain representation Acquisition of Biomedical Signals Signal Acquisition Systems, acquisition of biomedical signals like ECG, EEG and EMG , Analog-to-digital conversion (ADC), Nyquist theorem and its importance in signal acquisition, Aliasing, Signal conversion circuits	9
2	The Fourier transform, Correlation, Convolution, Power spectrum estimation Adaptive filters: Principle of noise canceller model, 60 Hz adaptive cancelling using a sine wave model Adaptive and optimal filtering - Modelling of Biomedical signals - Detection of biomedical signals in noise - removal of artifacts of one signal embedded in another - Maternal-Fetal ECG Signal Averaging: Basics of signal averaging, signal averaging as a digital filter, a typical averager.	9

	Turning point algorithm, AZTEC algorithm, CORTES algorithm, Fan algorithm, Huffman coding	
3	Cardiological Signal Processing Basics of ECG, Anatomy of the heart and electrical activity Basics of ECG waveform and lead systems, Methods and devices for ECG signal acquisition, Noise & artifacts, Preprocessing of ECG Signals Noise removal and baseline correction, Feature Extraction-detection of P, QRS, and T waves, Heart rate variability analysis Advanced ECG Processing Arrhythmia Detection- Algorithms for arrhythmia detection and classification	9
4	Neurological signal processing Basics of EEG and its characteristics, Brain anatomy and electrical activity, basics of EEG waveform and electrode placement, methods and devices for EEG signal acquisition, detection of EEG rhythms, template matching for EEG spike and wave detection Preprocessing of EEG Signals- Noise removal and artifact reduction Feature Extraction- Time-frequency analysis, power spectral density Advanced EEG Processing Brain-Computer Interface (BCI), Introduction to BCI and its applications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts and characteristics of biomedical signals and acquisition techniques	K2
CO2	Apply the concepts of Fourier transform, Correlation, Convolution and Power spectrum estimation in biomedical signals and explain the concepts of adaptive filtering and signal averaging	K3
CO3	Perceive the concepts and methods used for cardiological signal processing	K2
CO4	Perceive the concepts and methods used for neurological signals	K2
CO5	Implement the methods learnt in the subject using advanced signal processing tools	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2						3
CO2	3					2						3
CO3	3	2				2						3
CO4	3	2				2						3
CO5	3	2			3	2						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Signal Analysis	Rangayyan	Wiley	3 rd , 2024
2	Biomedical Signal Processing & Signal Modeling	Bruce	Wiley	1 st , 2001
3	Discrete time Signal Processing	Alan V Oppenheim, Schafer	Pearson	3 rd , 2014
4	Bioelectrical Signal Processing in Cardiac & Neurological Applications	Sörnmo	Elsevier	1 st , 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Biomedical Engineering	Enderle	Elsevier	2 nd , 2005
2	Biomedical Signal Processing: Principles and techniques	D.C.Reddy	Tata McGraw Hill	1 st , 2005
3	Biosignal and Biomedical Image Processing	Semmlow, Marcel Dekker	Taylor & Francis	3 rd , 2014
4	Biomedical Signal Processing A Modern Approach	Sergio Cerutti, Carrlo Marchesi	CRC press	1 st , 2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
	https://youtu.be/ezfPl8kUdbg?si=QYbbxDPh0vjkPunl

SEMESTER S6

COMMUNICATION SYSTEM AND TELEMETRY

Course Code	PEICT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand and analyse the principles of communication and telemetry.
2. Telemetry automatically collects, transmits and measures data from remote sources, using sensors and other devices to collect data. It uses communication systems to transmit the data back to a central location.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to communication systems- Modulation-Principles of Amplitude modulation and Single side band techniques. Detection of AM waves.Principles of Frequency modulation and FM Reception Comparison of AM and FM. Pulse modulation techniques: Generation and demodulation of Pulse amplitude modulation- Pulse time modulation- Pulse width modulation Principles of Pulse Code Modulation-Delta Modulation-PSK, ASK, FSK	9
2	Introduction to waveguides-Types of waveguides-Principles of Radio wave Propagation Principles of antennas-Basic Definitions-Types of Antennas-Antennas for Telemetry. Principles of Satellite Communications-Geostationary Orbits-Attitude control- Satellite station keeping-Transponders-Power Budget Calculations.	9
3	Telemetry: functional blocks of telemetry and telecontrol systems,Telemetry methods- Classical ones- Pneumatic and electrical telemetry- Electrical telemetry systems- Voltage, current, position	9

	telemetry-Optical telemetry. Transducers for Telemetry Methods of Improving signal to noise ratio- Signal filters- Signal averaging- Box car integration-Signal correlation	
4	Grounding and shielding techniques-Introduction to Magnetic Tape Recording Modern telemetry techniques- Radio telemetry- Multiplexing – TDM- FDM – comparison of TDM and FDM.Methods of data transmission in telemetry- FM/FM, PCM/FM and PAM/FM techniques Introduction to telemetry standards- Introduction to satellite telemetry- Typical telemetry and Telecontrol schemes related to industry and space exploration	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To provide students with the knowledge of communication systems	K2
CO2	To understand different modulation techniques in communication systems	K2
CO3	To understand principles of communication systems	K2
CO4	To explain telemetry and different methods of telemetry	K2
CO5	To understand modern telemetry techniques	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1					3					1
CO2						2	3					1
CO3		1					3					1
CO4						2	3					1
CO5		1					3					1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic communication systems	Kennedy & Davis	TMH	6th edition 2018
2	Telemetry principles	Patranabis	TMH	2 nd edition 1999
3	Introduction to telemetry	O.J Strock	ISA	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Telemetry and Communications	K. Radhakrishnan	IK International publishing house	2014
2	electronics communications	D. Rodyy, J. Coolen	PHI	4th edition 1998
3	Principles of Communication Systems	Taub and Schilling	TMH	3 rd edition 2008
4	Instrument Engineers Hand book	B.G. Liptak,	CRC Press	4 th edition 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=h2P2XaVbeVw
2	https://onlinecourses.nptel.ac.in/noc24_ce25/preview
3	https://www.youtube.com/watch?v=62HI dnAtPrE
4	https://www.youtube.com/watch?v=AVKfVS3e6RU

SEMESTER S6

TOTAL QUALITY MANAGEMENT

Course Code	PEICT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Study the concepts of total quality management, employee involvement and quality planning.
2. Study the importance of quality of service ,role of reliability and statistical tools for measuring quality.
3. Identify the importance of continuous process improvement in quality, suppliers, benchmarking and quality function deployment.
4. Identify the importance of total productive maintenance, failure mode and effect analysis, need for ISO systems and quality auditing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Principles of TQM Definition of Quality, Dimensions of Quality, Quality Planning, Basic concepts of Total Quality Management. Leadership Concepts, Role of Senior Management, Quality Council, Strategic Planning, Barriers to TQM Implementation. Employee Involvement Motivation, Empowerment, Teams, Recognition and Reward, Performance Appraisal, Benefits. Quality planning SWOT analysis	9

2	Service Quality Customer Perception of Quality, Service Quality, Customer satisfaction, Customer complaints, Customer Retention. Reliability Definition of reliability– key elements; failure analysis–failure density– failure rate– probability of failure– Bath tub curve. Mean time to failure (MTTF). Quality Tools The seven tools of quality, Statistical Fundamentals–Measures of central tendency and dispersion, Population and Sample, Normal Curve.	9
3	Continuous Process Improvement Juran Trilogy, PDCA Cycle, 5S, Kaizen. Supplier Partnership Partnering, sourcing, Supplier Selection, Supplier Rating, Relationship Development. Benchmarking Reasons to Benchmarking, Benchmarking Process. QFD Quality Function Deployment (QFD), QFD Process, Benefits.	9
4	TPM and FMEA Total Productive Maintenance (TPM) – Concept. Stages of FMEA – Concept . ISO Need for ISO 9000 and Other Quality Systems, ISO 9000:2000, Quality System– Elements, Implementation of Quality System. Quality Auditing Documentation, ISO 14000 –Concept, Requirements and Benefits.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Study the concepts of total quality management, employee involvement and quality planning.	K2
CO2	Study the importance of quality of service ,role of reliability and statistical tools for measuring quality.	K2
CO3	Identify the importance of continuous process improvement in quality, suppliers, benchmarking and quality function deployment.	K2
CO4	Identify the importance of total productive maintenance, failure mode and effect analysis, need for ISO systems and quality auditing.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1									2		2	3
CO2									2			3
CO3				2							2	3
CO4									2		2	3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	Dale H. Besterfield, et al.,	Pearson Education India, 4/e.	4TH EDITION

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering,	L.S.Srinath,	East West, 2005.	1ST EDITION
2	The Management and Control of Quality	James R.Evans & William M.Lindsay,	South-Western College Pub, 6/e.	9TH EDITION 2014
3	Total Quality Management	Feigenbaum. A.V	McGraw-Hill Professional, 4/e.	4TH EDITION
4	Total Quality Management	Oakland.J.S	Routledge, 4/e.	5TH EDITION 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/110104080
2	https://archive.nptel.ac.in/courses/110/104/110104085/
3	https://nptel.ac.in/courses/110104080
4	https://archive.nptel.ac.in/courses/110/104/110104085/

SEMESTER 6

PLC AND DCS

Course Code	PEICT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hr.30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Learn the roles, architectures, and interfacing techniques of computer-based measurement and control systems, including HMI and hardware integration.
2. Gain hands-on experience with PLC programming and simulation, and understand the functionalities and interfacing of Distributed Control Systems for process control.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to computer based control system -Role of computers in measurement and (process) control Basic components of computer based measurement and control systems Architecture – computer based process control system –Centralised, Distributed and Hierarchical. Human Machine Interface (HMI) Hardware for computer based process control system, Interfacing computer system with process. Architecture of DDC, SCADA and DCS. Programmable logic Controller (PLC): Introduction, Evolution, Relay VS PLC VS Computer	9
2	PLC- Hardware and Internal Architecture-Input –output devices .Basics of Ladder Programming, on/off instructions, internal relay, jump instructions, data handling instruction, data manipulation instructions, Arithmetic and Comparison ,PID and other important instructions	9
3	Timers and Counters in PLC. Problems. Design Development and Simulation of PLC Programme Program on Temperature control Valve sequencing, Conveyor belt control and Control of a process.	9

	PLC Installation, trouble shooting and maintenance, Design of Alarms and Interlocks, Networks of PLC Distributed Control System- DCS - Evolution– Various Architectures – Comparison – Local control unit	
4	DCS -LCU Languages-Process interfacing issues-communication facilities- Operator interface-Low level and High level Operator interface- Displays - Engineering interfaces – Low level and high level engineering interfaces – Factors to be considered in selecting DCS – Other key issues in DCS – Packaging and Power system issues.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Micro projects on automation using PLC and DCS for student group comprising of 3 students.

Report – 5 marks

Working Model – 15 Marks

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic architecture and components of computer-based measurement and control systems.	K2
CO2	Understand the human-machine interfaces (HMI) and learn the hardware and interfacing techniques needed to integrate computer systems with process controls.	K2
CO3	Create and troubleshoot PLC programs using ladder logic for various applications.	K5
CO4	Understand and apply the architecture and interfaces of Distributed Control Systems in various process control settings.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3				2							
CO4	3											

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineer's Handbook – Process Control,	B G Liptak	CRC Press	4 th edition
2	Understanding Distributed Processor Systems for Control,	Samel M. Herb	ISA Publication	1 st edition 1999
3	Programmable Logic Controllers – Principles and Applications.	John W.Webb & Ronald A. Reiss,	PHI	5 th edition
4	Computer Control of Processes,	M. Chidambaram	Alpha Science International Ltd	1 st edition 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Software and Digital Networks, CRC Press.	B G Liptak	CRC	3 rd edition
2	Programmable Logic Controllers – Programming Methods and Applications, Pearson Education.	John R. Hackworth & Frederick D. Hackworth Jr	Pearson	1 st edition 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me67/preview
2	https://onlinecourses.nptel.ac.in/noc21_me67/preview
3	https://onlinecourses.nptel.ac.in/noc21_me67/preview
4	https://onlinecourses.nptel.ac.in/noc21_me67/preview

SEMESTER S6

MODERN CONTROL ENGINEERING

Course Code	PBICT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT501	Course Type	Theory

Course Objectives:

1. To equip students with advanced techniques in designing cascade compensators using root-locus and Bode plot methods.
2. To enable students to model and analyse continuous linear time-invariant (LTI) systems using state-space representations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Design of Cascade Compensators (10 Hours) Dominant pole; realisation of basic compensators: lead, lag, lag, and lag-lead compensator. Cascade compensation using root-locus: lead, lag, and lead-lag compensation. Cascade compensation using bode plot: lag, lead, and lag-lead compensation.	10
2	State Space Modelling of Continuous Linear Time-Invariant Systems (9 Hours) Concept of state, state variable, state vector and state space; State models for linear continuous-time systems; State variable representation of continuous-time systems using physical variables. Transfer function from state variable model; transformation of state variables and invariance property. State space representation using phase variable (state space representation from transfer function): Controllable canonical form, observable canonical form, Jordan Canonical Form.	9

3	State Variable Analysis of Continuous LTI Systems (8 Hours) Solution of state equation; state transition matrix; properties of state transition matrix. Computation of the state transition matrix using Lyapunov's Method and Cayley Hamilton theorem; Response of homogeneous and non-homogeneous state-space model. Concept of controllability and observability; Kalaman test.	8
4	State Variable Design (Continuous) (6 Hours) Design of state variable feedback if the given model is in the controllable canonical form; design of arbitrary state variable feedback if the given model is not in controllable canonical form. Design of full order state observer; application of duality Principle; concept of reduced order state-observer. Stability analysis of Continuous LTI Systems using Phase Plane Method (3 Hours) Singular points, classification of singular points.	9

Suggestion on Project Topics

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of dominant pole placement to design and implement lead, lag, and lead-lag compensators using root-locus and Bode plot techniques.	K3
CO2	Construct state-space models for continuous-time LTI systems and derive transfer functions from state variable representations.	K3
CO3	Compute and utilise the state transition matrix to analyse the response of continuous-time LTI systems.	K4
CO4	Design and implement state variable feedback controllers and state observers for continuous-time LTI systems to achieve desired system performance.	K3
CO5	Evaluate the stability and performance of a continuous LTI system using phase plane methods and design compensators or state feedback mechanisms to enhance system stability and performance based on project requirements.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									3
CO2	3											3
CO3	3	2										3
CO4	3	1										3
CO5	3	3	3	2	3			3	3	3	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Control Systems Engineering	I. J. Nagrath, Madan Gopal	New Age International Private Limited	7 & 2021
2	Modern Control Engineering	Katsuhiko Ogata	Pearson Education India	5 & 2015
3	Automatic Control Systems	Farid Golnaraghi and Benjamin C. Kuo	McGraw Hill Education (India) Pvt. Ltd	10 & 2017
4	Control Systems: Principles and Design	M Gopal	McGraw Hill Education	4 & 2012
5	Linear Control Systems with Matlab Applications	B. S. Manke	Khanna Publishers	12 & 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Control Systems	Richard Dorf, Robert Bishop	Pearson	14 & 2021
2	Problems and Solutions of Control Systems: With Essential Theory	A.K. Jairath	CBS Publishers & Distributors Pvt Ltd	6 & 2015
3	Modern Control Engineering	Choudhury, Dr. Roy	Prentice Hall India Learning Private Limited	1 & 2005
4	Nise's Control Systems Engineering	Norman S. Nise	Wiley	1 & 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Swayam - Control Systems by Prof. C. S. Shankar Ram, IIT Madras
2	NPTEL - Advanced Linear Continuous Control Systems: Applications with MATLAB Programming and Simulink by Prof. Yogesh Vijay Hote, IIT Roorkee
3	NPTEL - Advanced Linear Continuous Control Systems: Applications with MATLAB Programming and Simulink by Prof. Yogesh Vijay Hote, IIT Roorkee
4	NPTEL - Advanced Linear Continuous Control Systems: Applications with MATLAB Programming and Simulink by Prof. Yogesh Vijay Hote, IIT Roorkee

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project \
- Creativity in solutions and approaches

SEMESTER S6

DESIGN THINKING AND PRODUCT DEVELOPMENT

(Common to Group A & Group B)

Course Code	GXEST605	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	60
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To guide students through the iterative stages of design thinking, including empathizing with users, defining problems, ideating solutions and developing Proof of Concepts (PoC) and technical feasibility studies.
2. To promote the development of critical thinking skills by engaging students in integrative inquiry, where they ask meaningful questions that connect classroom knowledge with real-world applications.
3. To equip students with the ability to involve in product design considering the sustainability, inclusivity, diversity and equity aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of design thinking and product development: Overview of stages of product development lifecycle; Design thinking -Definition-Design thinking for product innovation; Bringing social impact in ideation-Identifying societal needs-understanding multi-faceted issues-community engagement and empathetic design- technological innovation meeting societal needs; Understanding and Bridging the divide using Human Centered Design (HCD); Designing for inclusivity in product development-embracing user diversity - Long term impact - sustainability encompassing environmental,economic and social dimensions; Technology Readiness Level in the Innovation Life-cycle; Performing a self-check on innovative	6

	ideas - Originality of idea- understanding innovation landscape - patentability - understanding the economic landscape - Unique Selling Proposition (USP) - Repeatability and Manufacturability - Sustainability - Leveraging business models for comprehensive analysis	
2	Empathize: Design thinking phases; Role of empathy in design thinking; Methods of empathize phase - Ask 5 Why/ 5 W+H questions; Empathy maps - Things to be done prior to empathy mapping - Activities during and after the session; Understanding empathy tools - Customer Journey Map - Personas. Define: Methods of Define Phase: Storytelling, Critical items diagrams, Define success.	6
3	Ideation : Stages of ideation; Techniques and tools - Divergent thinking tools - Convergent thinking tools - Idea capturing tools; Cross-industry inspiration; Role of research in ideation - Market research - consumer research - leveraging research for informed ideation; Technological trends - navigating the technological landscape - Integrating emerging technologies; Feasibility studies - technical, economic, market, operational, legal, and ethical feasibility; Ideation session- techniques and tips. Proof of Concept (PoC) : Setting objectives; Risk assessment; Technology scouting; Document and process management; Change management; Knowledge Capture; Validating PoC; Story telling in PoC presentation	6
4	Design: Navigating from PoC to detailed design; Developing Specification Requirement Document (SRD)/Software Requirement Specification (SRS); Design for manufacturability; Industrial standards and readability of code; Design to cost; Pre-compliance; Optimized code; Design Failure Mode and Effects Analysis (DFMEA); Forecasting future design changes. Prototyping: Alpha prototypes; Beta prototypes; Transition from design to prototype; Goals and expectations for Alpha and Beta prototypes; Effective strategies for maintaining timeline in prototyping; Testing and refining Alpha prototypes; Transitioning to Beta prototypes. Pilot build: Definition and purpose of a pilot build; setting objectives; Identification and selection of manufacturing partner for pilot build; Testing procedures in pilot build; Scaling from pilot build to full-scale production / implementation.	6

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignments	Internal Examination	Reflective Journal and Portfolio	Total
5	20	10	5	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Empathize to capture the user needs and define the objectives with due consideration of various aspects including inclusivity, diversity and equity	K5
CO2	Ideate using divergent and convergent thinking to arrive at innovative ideas keeping in mind the sustainability, inclusivity, diversity and equity aspects.	K6
CO3	Engage in Human Centric Design of innovative products meeting the specifications	K5
CO4	Develop Proof of Concepts (PoC), prototypes & pilot build of products and test their performance with respect to the Specification Requirement Document.	K4
CO5	Reflect on professional and personal growth through the learnings in the course, identifying areas for further development	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2	3	3	3	2	2		3
CO2	3	2	3		2	3	3	3	2	2		3
CO3	3	2	3		2	3	3	2	2	2		3
CO4	3	2	2		3	3	3	2	2	2		3
CO5	3					3	3	2	2	2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Sense: Engineering your ideas into reality	Dr. K R Suresh Nair	NotionPress.com	2024
2	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers Ltd.	2009
3	Design Thinking for Strategic Innovation	Idris Mootee	John Wiley & Sons Inc.	2013

Sample Assignments:

1. Evaluate and prepare a report on how the aspects including inclusivity, diversity and equity are taken into consideration during the empathize and define phases of the Miniproject course.
2. Evaluate and prepare a report on how the aspects including sustainability, inclusivity, diversity and equity are taken into consideration during the ideate phase of the Miniproject course.
3. Evaluate and prepare a report on how User-Centric Design (UCD) is used in the design and development of PoC of the product being developed in the Miniproject course.
4. Prepare a plan for the prototype building of the product being developed in the Miniproject course.
5. Report on the activities during the empathize phase including the maps & other materials created during the sessions.
6. Report on the activities during the define phase including the maps & other materials created during the sessions.
7. Report of all the ideas created during the ideation phase of the Miniproject course through the tools including SCAMPER technique, SWOT analysis, Decision matrix analysis, six thinking hats exercise
8. Prepare a full scale production plan for the product being developed in the Miniproject course.
9. Create a Stanford Business Model Canvas related to the Miniproject.

An industrial visit of at least a day for experiential learning and submit a report on the learnings, for example industry standards and procedures.

SEMESTER S6

BASICS OF INSTRUMENTATION ENGINEERING

Course Code	OEICT 611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the Fundamentals of Instrumentation and Control Engineering and functional elements of measurement systems.
2. Describe about functions of a control system and master Instrument Characteristics.
3. Study input-output configuration of instruments and develop Proficiency in Calibration
4. Analyse and Mitigate Measurement Errors and measurement noise.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION TO INSTRUMENTATION & CONTROL ENGINEERING- Definition-scope and importance- application of instrumentation in different fields like petrochemical, Biomedical, pharmaceutical and food processing. FUNCTIONAL ELEMENTS OF MEASUREMENT SYSTEM Introduction to instrument systems - Typical applications of instrument systems. Basic description of the functional elements of the measurement system with examples. TRANSDUCERS AND INSTRUMENT TYPES Definition of transducers. Role of transducers in instrumentation. Classification of instruments: Active and passive instruments - Null-type and deflection-type instruments - Analogue and digital instruments - Indicating instruments and instruments with a signal output -Manually operated and automatic type, Self-generating and power operated types, Contacting and	9

	non-contacting types - Smart and non-smart instruments.	
2	FUNCTIONAL ELEMENTS OF CONTROL SYSTEM Need of Control systems – Role of error detectors – Block diagram for a control system with examples. MEASUREMENT SYSTEM PERFORMANCE Static characteristics of instruments. Accuracy and precision, static sensitivity, linearity, hysteresis, threshold, dead time, dead zone, resolution or discrimination. static error, static correction. Scale range and span, reproducibility and drift, repeatability. Dynamic characteristics of instruments. Understanding of (1) Zero order instrument (2) First order instrument (3) Second order instrument.	9
3	CALIBRATION OF MEASURING SENSORS AND INSTRUMENTS Standards and calibration - Necessity of calibration- Careful instrument design, Principles of calibration - Control of calibration environment - Calibration chain and traceability - Calibration records - Intelligent instruments. TYPES OF INPUTS Input output configuration of measuring instruments and measurement systems. Desired inputs, interfering inputs, modifying inputs. Methods of correction for interfering and modifying inputs. Loading effects. Input and output impedances. Input impedances, input admittance, output impedances, output admittance.	9
4	MEASUREMENT NOISE Noise, signal to noise ratio, sources of noise, Johnson noise, power spectrum density. Sources of measurement noise - Inductive coupling - Capacitive (electrostatic) coupling - Noise due to multiple earths - Noise in the form of voltage. ERRORS DURING THE MEASUREMENT PROCESS Errors in measurements, true value, Limiting errors (Guarantee errors).	9

	Relative (fractional) limiting error. Combination of quantities with limiting errors. Known errors, types of errors, gross errors, systematic errors, instrumental errors, environmental errors, observational errors. Random (residual) errors. Reduction of systematic errors - Careful instrument design - Method of opposing inputs - Calibration - Manual correction of output reading.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate a comprehensive understanding of instrumentation and control engineering principles and the performance characteristics of different types of instruments.	K2
CO2	Develop a thorough understanding of the static and dynamic characteristics of instruments enabling effective evaluation and utilization of instrumentation in various applications.	K2
CO3	Attain a deep understanding of the principles of calibration and analysing input-output relationships in measurement systems	K2
CO4	Demonstrate a comprehensive understanding of noise and errors in measurement.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Measurements and Instrumentation Principles.	Alan S Morris	Elsevier Butterworth Heinemann	3 rd edition, 2001
2	A course in Mechanical Measurement and Instrumentation.	A.K Sawhney	Dhanpat Rai & Co.	12 th edition, 2017
3	Instrumentation Devices & Systems.	C.S. Rangan, G.R. Sarma, V.S.V. Mani,	Tata McGraw- Hill publishing company Ltd.	2017
4	Mechanical Measurements	S P Venkateshan	Ane Books	2nd edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Measurement systems: Application and design,	Ernest. O Doebelin	McGraw Hill Higher Education	5 th edition, 2003
2	Transducers and Instrumentation	D V S Murty	Prentice Hall India Learning Private Limited	2nd edition, 2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
2	https://nptel.ac.in/courses/108102191

SEMESTER S6

MICRO ELECTRO MECHANICAL SYSTEMS

Course Code	OEICT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To develop knowledge on Micro electro mechanical systems and materials used in the manufacture
2. To understand various micro fabrication procedures, techniques and packaging considerations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Micro Electronic Mechanical Systems: Materials for MEMS- silicon- structure and orientations, properties. Other materials- shape memory alloy (SMA), polymers, silicon compounds etc, material properties. Commercial MEMS structures and systems: Overview of commonly used mechanical structures in MEMS - Beams, Cantilevers, Plates, Diaphragms – Typical applications. Passive MEMS structures- fluid nozzles- ink jet nozzles. Sensors- piezoresistive pressure sensors- mass flow sensors- acceleration sensors, actuators- micro mechanical valves-micro pumps.	9
2	Process of micromachining: Process of micromachining- pattern transfer using additive technique. Silicon growth- oxidation of silicon-molecular beam epitaxy. Physical vapour deposition (PVD)- sputtering- laser deposition- ion plating- cluster beam technology- Chemical vapour deposition (CVD)- step	9

	coverage- plasma enhanced CVD, Electron cyclotron resonance CVD (ECRCVD)- sol-gel deposition technique.	
3	Lithography Photolithography- Introduction- masks- spinning- resist and soft backing- exposure and post exposure treatment- development- descumming and post baking- resists- wafer priming, wafer cleaning and contamination- clean room- resist striping. Next generation lithography-extreme ultra violet lithography (EUVL)- x-ray lithography- LIGA- charged particle beam lithography- electron beam lithography- ion beam lithography. Etching: Introduction- dry etching- physical etching- ion etching or sputtering- ion beam etching, plasma etching- deep reactive ion etching (DRIE)- isotropic wet etching- anisotropic wet etching.	9
4	MEMS packaging: Key design and packaging consideration- wafer stack thickness- wafer dicing concerns, thermal management- stress isolation- protective coating and media isolation. Wafer level packaging and die level packaging. Hermetic packaging- calibration and compensation- die attach process- solder alloy- bonding process using solder alloy- organic adhesive wiring and inter connections- Wire bonding- flip chip bonding- micro fluidics interconnections. Packaging material properties- mechanical- moisture penetration- interfacial adhesion, thermal- chemical- electrical.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define MEMS and elaborate on various materials used for MEMS fabrication and to describe different kinds of commercial applications like micropumps, valves and sensors.	K2
CO2	Describe the various techniques involved in micromachining and discuss techniques for deposition of materials to create structures.	K2
CO3	Discuss lithography and etching techniques.	K2
CO4	Develop a deep understanding of design and packaging considerations during manufacture of MEMS devices	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Micro electromechanical Systems Engineering	Nadim Maluf	Artech house Boston London	6/e, 2006
2	Fundamentals of Micro-fabrication	Mark Madou	CRC Press, New York	3/e, 2011
3	Foundations of MEMS	Chang Liu	Pearson	2/e, 2011
4	MEMS and Microsystems Design and Manufacture	Tai Ran Hsu	McGraw Hill	4/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electromechanics and MEMS	Thomas B. Jones, Nenad G. Nenadic	Cambridge University Press	Illustrated edition, 2013
2	Microsystem design	Stephen D. Senturia,	Springer (India)	2nd edition, 2006
3	Smart Sensors and MEMS	Stoyan Nihtianov, Antonio Luque	Woodhead Publishing Series	2nd edition, 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117105082
2	https://nptel.ac.in/courses/117105082
3	https://nptel.ac.in/courses/117105082
4	https://nptel.ac.in/courses/117105082

SEMESTER S6

INSTRUMENTATION FOR AGRICULTURE

Course Code	OEICT 613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To expose the students about various instruments used in agriculture
2. To teach the design of greenhouses using various instruments
3. To impart ideas about modern technologies used in agriculture

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Soil Science & Sensors Soil science : fundamental definitions & relationships properties of soil : index properties of soil, Engineering properties of soil, physical properties of soil permeability & seepage analysis, Mohr's circle of stress Necessity of instrumentation & control for agriculture Sensors and transducers: definitions, introduction to sonic anemometers, hygrometers, thermocouples, close path gas analysers, introduction to various biosensors	9
2	Instrumentation in industrial Continuous & Batch processes Idea about continuous and batch processes Flow diagram of sugar plant, sensors & instrumentation set up Flow diagram of fermenter, sensors & instrumentation set up Flow diagram of dairy industry & instrumentation set up for it	9

	Juice extraction control process & instrumentation set up. Oil extraction plant and instrumentation set up for it. Pesticides manufacturing process and control	
3	Instrumentation in Irrigation Necessity of irrigation in agriculture Evapotranspiration and its measurement methods irrigation methods: overhead, centre pivot, lateral move, micro irrigation systems & the comparison of different irrigation systems soil moisture measurement methods: resistance-based method, voltage based method, thermal based method Water distribution & management control, Auto drip & sprinkler irrigation systems, Irrigation canal management systems upstream & downstream control concept Introduction to SCADA , Applications of SCADA in agriculture	11
4	Greenhouse Parameters & Instrumentation Greenhouse effect, Concept & construction of green houses, ventilation, cooling & heating, wind speed, temperature & humidity, soil moisture, rain gauge, carbon dioxide enrichment measurement & control implementation of hydraulic, pneumatic & electronics control circuits in harvesters cotton pickers, tractor Classification of pumps: pump characteristics, pump selection & installation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret the soil properties and sensors used in agriculture field	K2
CO2	Explain the continuous and batch processes.	K2
CO3	Realize the various methods involved in Irrigation system	K2
CO4	Explain the parameters and instrumentation in greenhouses	K2
CO5	Explain the pumps used in agriculture	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3					3
CO2	3					3	3					3
CO3	3					3	3					3
CO4	3					3	3					3
CO5	3					3	3					3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Instrumentation	D. Patranabis	McGraw Hill Education	3/e, 2017
2	Instrumentation handbook-process control	B.G.Liptak	Chilton	3/e, 2013
3	Advanced agricultural instrumentation	Tom Weather Jr and Cland C. Hunter	Springer	2011
4	Process control and instrumentation technology	C.D. Johnson	Pearson education	7/e, 2005
5	Industrial Instrumentation	D. Patranabis	McGraw Hill Education	3/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Agricultural Engineering	RadheyLal	Saroj Publication	2008
2	Mineral Processing Technology	Wills B.A	Butterworth-Heinemann	7/e, 2006
3	Application of Vision Technology and Artificial Intelligence in Smart Farming	Xiuguo Zou, Zheng Liu	journal <i>Agriculture</i> by MDPI	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108108147
2	https://nptel.ac.in/courses/108108147
3	https://nptel.ac.in/courses/126105010 https://nptel.ac.in/courses/126105019 https://nptel.ac.in/courses/108106022
4	https://nptel.ac.in/courses/108108147

SEMESTER: S6

PROCESS CONTROL INSTRUMENTATION LAB

Course Code	PCICL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PROCESS CONTROL INSTRUMENTATION	Course Type	Lab

Course Objectives:

1. Develop a comprehensive understanding of various control modes and their applications through hands-on experiments
2. Acquire practical skills in advanced process control and automation technologies

Expt. No.	Experiments
Part A	Process Control Modes (All experiments are mandatory. Students must compare the performance of different process stations and the effect of adding different process control modes.)
1	Familiarisation of ON-OFF, proportional, proportional plus integral, and proportional plus integral plus derivative control mode using pressure process station or flow process station and Construction of ISA diagram.
2	Familiarisation of ON-OFF, proportional, proportional plus integral, and proportional plus integral plus derivative control mode using level process station or temperature process station and Construction of ISA diagram.
3	Tuning of P, PI, and PID control mode for any one process station using Ziegler Nichols open-loop method.
4	Tuning of P, PI, and PID control mode for any one process station using Ziegler Nichols closed-loop method.
5	Familiarisation of cascade controller and feedforward controller.
Part B	Process Automation (Minimum two experiments are mandatory)

6	Familiarisation of I/O, Timer, counter, addition, and binary instruction set of PLC.
7	Familiarisation of PLC and SCADA interface.
8	Control of process stations using DCS and Construction of ISA diagram.
Part C	Design of control mode using LabVIEW (A minimum four experiments are mandatory)
9	Control of RC circuit using LabVIEW.
10	Control of a process station using LabVIEW.
11	Design of Proportional, PI, and PID control modes for DC motor in LabVIEW.
12	Design of electronic proportional controller using OPAMP and monitoring both controlled variable and manipulating variable using LabVIEW.
13	Design of electronic proportional plus integral controller using OPAMP and monitoring both controlled variable and manipulating variable using LabVIEW

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Choose a process control mode with respect to the process characteristics.	K3
CO2	Use process automation software like PLC, DCS, and SCADA.	K3
CO3	Use Virtual Instrumentation software for analysis and control of systems	K3
CO4	Demonstrate skills in technical communication and teamwork.	K3
CO5	Choose a process control mode with respect to the process characteristics.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									3
CO2	3				3							3
CO3	3	2	2		3							3
CO4									2	3		
CO5	3	2	2									3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Control Instrumentation Technology	Curtis Johnson	Pearson	8/e.
2	Programmable Logic Controllers	Frank D.Petruzella	Mcgraw Hill Higher Education	Sixth edition
3	Modern Control Systems with LabVIEW	Robert H. Bishop	NTS Press	Edition 13,2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook on Process Control	B.G.Liptak	Chiller Book and Co	Fourth, Volume 1, 2003
2	Chemical Process Control an Introduction to Theory and Practice	George Stephanopoulos	Pearson Education India	1/e.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105064/
2	https://archive.nptel.ac.in/courses/108/105/108105062/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

**INSTRUMENTATION AND CONTROL
ENGINEERING**

SEMESTER S7
DIGITAL CONTROL

Course Code	PEICT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBICT604	Course Type	Theory

Course Objectives:

1. Introduce fundamental concepts of digital control systems, including discrete-time system representation, sampling processes, and the effects of sampling periods on system behavior.
2. Provide methods for modeling discrete-time systems, focusing on Z-transforms, s-plane to z-plane mapping, and stability analysis techniques such as Jury stability test and bi-linear transformation.
3. Develop skills for designing digital control systems using frequency response plots, root locus plots, and z-plane synthesis, addressing both transient and steady-state responses.
4. Explore discrete state-space models, including state variable representations, stability analysis via Lyapunov's theorem, and advanced design techniques such as pole placement, observer design, and state feedback design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to digital control-Discrete time system representation- Sampling process- Mathematical modelling of sampling process- Data reconstruction- Effect of sampling period- Modelling discrete-time systems –Review of Z-transform-Mapping of s-plane to z-plane- Pulse transfer function-Warping effect	9
2	Stability analysis of discrete time system:- Jury stability test, stability analysis using bi-linear transformation. Time response of discrete systems: Transient and steady state responses, time response parameters of a prototype second order system. Design of digital control systems--Digital compensator design using frequency response plots,	12

	Digital compensator design using root locus plots, z plane synthesis.	
3	Discrete state space model- Introduction to state variable model- canonical forms-Characteristic equation, state transition matrix-Solution to discrete state equation. Stability- Lyapunov stability theorem	7
4	Pole placement and observer design-Controllability and observability-State feedback design-Pole placement by state feedback-Set point tracking controller-Full order observer-Reduced order observer- Output feedback design	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discuss sampling and model discrete-time systems using difference equations and Z-transform techniques.	K2
CO2	Analyze the characteristics and stability of discrete-time systems and develop state variable models.	K3
CO3	Design digital compensators, state variable feedback controllers, output feedback controllers, and state observers.	K3
CO4	Evaluate discrete systems' controllability, observability, and stability using state transition matrices and Lyapunov criteria.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Discrete Time Control Systems,	K Ogata	Prentice Hall	2 nd edition 2005
2	, Digital Control and State Variable Methods	M Gopal	Tata Mcgraw Hill, 2/e, 2003	2 nd edition 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Control Systems	K C Kuo	Oxford Universal press	2 nd edition 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/103/108103008/
2	https://archive.nptel.ac.in/courses/108/103/108103008/
3	https://archive.nptel.ac.in/courses/108/103/108103008/
4	https://archive.nptel.ac.in/courses/108/103/108103008/

SEMESTER S7
CYBER SECURITY FOR INDUSTRIAL AUTOMATION

Course Code	PEICT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Study the importance of cybersecurity in industries.
2. Understanding the architecture and communication protocols used in ICS.
3. Conducting risk assessments specific to industrial automation systems.
4. Developing risk mitigation strategies and contingency plans.
5. Bridging the gap between Information Technology (IT) and Operational Technology (OT) security practices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of industrial automation systems (e.g., SCADA, DCS, PLC). Importance of cybersecurity in protecting critical infrastructure. Key differences between IT (Information Technology) and OT (Operational Technology) networks. Fundamentals of Cybersecurity Basics of cybersecurity: threats, vulnerabilities, and risks.	8
2	Industrial Control Systems (ICS) Architecture Components of ICS: sensors, actuators, controllers, and communication protocols Understanding network topologies in industrial environments. Typical architectures (e.g., Purdue model). Threat Landscape for Industrial cyber threats to ICS (e.g., malware, ransomware, insider threats). Case studies of historical cyber incidents in industrial settings.	9
3	Cyber attacks Impact of cyber-attacks on industrial operations and safety. Security Challenges and Vulnerabilities in ICS, Specific vulnerabilities in ICS components and protocols (e.g., Insecure	10

	protocols, lack of authentication). Human factors: challenges in securing ICS due to operational needs and legacy systems. Risk Assessment and Management for ICS Conducting risk assessments specific to industrial environments. Identifying critical assets and vulnerabilities. Developing risk mitigation strategies and incident response	
4	Secure Design and Configuration of ICS Best practices for secure system design and architecture. Secure configuration of ICS components (e.g., PLCs, HMIs). Principles of secure coding and application development for ICS. Network security Network Security for ICS Segmentation and isolation strategies for ICS networks. Secure communication protocols (e.g., TLS, VPNs) and encryption methods. Intrusion detection and prevention systems (IDPS) for ICS. encryption methods. Emerging Technologies and Future Trends IoT (Internet of Things) and IIoT (Industrial Internet of Things) in industrial cybersecurity.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance of cyber security in industries.	K2
CO2	gain a thorough understanding of the unique cybersecurity challenges and requirements specific to industrial automation environments, including Industrial Control Systems (ICS) and Operational Technology (OT) networks.	K2
CO3	conduct risk assessments, develop risk management strategies, and implement proactive measures to mitigate cybersecurity risks in industrial environments.	K2
CO4	Understand Knowledge and skills in emerging technologies and future trends in cyber security.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3			3		2					2
CO2	3	3			3		2					2
CO3	3	3			3		2					2
CO4	3	3			3		2					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cybersecurity for Industrial Automation Systems	Tyson Macaulay, Bryan L. Singer, Brian J. O'Hara	CRC Press	1st edition (2020)
2	Industrial Network Security: Securing Critical Infrastructure Networks for Smart Grid, SCADA, and Other Industrial Control Systems	Eric D. Knapp, Joel Langill	Syngress	1st edition (2011)
3	Protecting Industrial Control Systems from Electronic Threats	Joseph Weiss	Momentum Press	1st edition (2010)
4	Cybersecurity for Industrial Automation Systems	Tyson Macaulay, Bryan L. Singer, Brian J. O'Hara	CRC Press	1st edition (2020)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hacking Exposed Industrial Control Systems: ICS and SCADA Security Secrets & Solutions	Clint Bodungen, Bryan L. Singer, Aaron Shbeeb	McGraw-Hill Education	1st edition (2016)
2	Practical Cyber Security for Industrial Control Systems: Design, Implementation and Assessment	Joseph Muniz, Aamir Lakhani, Ron Sangster	Packt Publishing	1st edition (2016)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me67/preview https://www.jcip1.org/uploads/1/3/6/5/136597491/cybersecurity_preparedness_1.pdf
2	https://www.wevolver.com/article/ics-scada
3	https://www.sciencedirect.com/science/article/abs/pii/S0140366419319991
4	https://onlinecourses.nptel.ac.in/noc20_cs69/preview

SEMESTER S7
INDUSTRIAL PROCESS CONTROL

Course Code	PEICT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of industrial process control and its applications in various industrial sectors.
2. To familiarise students with instrumentation symbols and standards as per ISA and the control of key industrial processes.
3. To equip students with the skills to analyse, design, and implement control strategies for reactors, distillation towers, heat exchangers, pumps, and compressors.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Industrial Process Control and Instrumentation Standards (8 Hours) Introduction to Industrial Process Control. Instrumentation Symbols as per ISA Standards: Purpose, scope, identification of the systems, function identification, loop identification, instrument line symbols, general instrument or function symbols, control valve body symbols, actuator symbols, symbols of self-actuated regulators, symbols for actuator action, primary element symbols, symbols of heat transfer elements, symbols of pump and compressor, symbols of process pressure vessels.	8
2	Instrumentation and Control of Chemical Reactors and Distillation Towers (10 Hours) Instrumentation and Control of Chemical Reactors: Flow sheet symbol, temperature control, simple temperature control scheme, the valve used in the temperature control scheme, the temperature response of an uncontrolled chemical reactor to step changes in load, recirculated cooling	10

	water configuration, temperature gradient, cascade control, pressure control, maximised production by constraint optimisation. Instrumentation and Control of Distillation Towers: Distillation equipment, column, bubble point and dew point, process of distillation tower, flow sheet symbol, column variables, guidelines for identifying the manipulating variable, column pressure control, column pressure control with liquid distillate and inert, column pressure control with liquid distillate and negligible inert, column pressure control with vapour distillate and inert, vacuum system, feed control, feed flow rate, variable column feed, feed temperature, boil-up control, reflux control.	
3	Control of Heat Exchangers (9 Hours) Different types of heat exchangers, process characteristics, terminology, degrees of freedom (DOF), liquid-to-liquid heat exchanger, control of liquid-to-liquid heat exchanger using two-way valve, control of liquid-to-liquid heat exchanger using bypass control, balancing the three-way valve, replacing three-way valve using two two-way valve, steam heater, steam throttling, condensate throttling, level control, bypass control in steam heater, cascade loop on steam heater, condenser, control of condenser, reboiler, control of reboiler.	9
4	Control of Pumps and Compressors (9 Hours) Control of Pumps: System curves, constant static head, friction pressure component, control of centrifugal pumps, head capacity curve, operating point of pump, on-off level control, on-off flow control, on-off pressure control, throttling flow control, control of rotary pump, on-off control, throttling and safety control. Control of Centrifugal Compressors: Capacity control method used, curves of centrifugal compressor, control of centrifugal compressor, suction throttling constant pressure system, suction throttling mostly friction system, discharge throttling mostly friction system, surge control (fixed setpoint), antisurge control.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the significance and application of ISA standards for instrumentation symbols in industrial process control.	K2
CO2	Explain the control strategies and instrumentation used in chemical reactors and distillation towers.	K2
CO3	Discuss the control techniques for various types of heat exchangers, including steam heaters, condensers, and reboilers.	K2
CO4	Describe the methods for controlling pumps and centrifugal compressors, including on-off and throttling controls.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineers' Handbook, Process Control and Optimization, Volume II	B.G.Liptak	CRC Press	4 & 2003
2	Instrumentation Symbols and Identification	ISA	ISA	1 & 1992

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Process Control Systems	Dale R. Patrick and Stephan W.Fardo	Vikas Publishing House	1 & 1986

SEMESTER S7
AUTOMOBILE INSTRUMENTATION

Course Code	PEICT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To expose the students to the concepts of various sensors used in automobile systems
2. To teach the basic and advanced controls in automotive systems
3. To impart ideas on aircraft and navigation instruments

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Vehicles Engine basic concepts (4 strokes), Starting & charging system, Electronic dashboard instruments, on board diagnostic system, Open loop and closed loop systems components for electronic engine management, Vehicle motion control, Digital cruise control, current trends in modern automobiles.	9
2	Engine management systems Basic sensor arrangement, types of sensors such as oxygen sensors, crank angle position sensors, Fuel metering/ vehicle speed sensors, flow sensor, temperature, air mass flow sensors, throttle position sensor, solenoids etc., algorithms for engine control including open loop and closed loop control system, electronic ignition, EGR for exhaust emission control.	9
3	Automobile Chassis Electronic Control System Principle of electronic braking, automatic transmission electronic control circuit cruise control circuit, the electronic steering control theory ABS, ASR, ESP, and other electronic control method Auto Body Electronic Control Technology: Automotive central locking and	9

	anti-theft system control technology. Air bag technology	
4	Latest Automotive Instrumentation Concepts of hybrid technology Introduction to ADAS (Automatic Driver Assist Systems) Introduction to electric vehicles, classifications Modern Automotive Instrumentation Input and Output Signal Conversion Multiplexing Multi rate Sampling Advantages of Computer- Based Instrumentation Display Devices (any one). Fuel Quantity Measurement, Coolant Temperature Measurement Oil Pressure Measurement Vehicle Speed Measurement High-Speed Digital Communications (CAN) CAN Network Trip Information Computer Telematics GPS Navigation The GPS System Structure.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of automobiles.	K2
CO2	Explain the evaluate of sensor and measuring system of automobile	K2
CO3	Explain the design of the basic modelling and control scheme for automotive systems	K2
CO4	Explain about the Latest trends in automotive instrumentation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Understanding Automotive Electronics	William B. Riddens	Butterworth Heinemann Woburn	5/e, 1998
2	Automotive Control Systems	U.Kiencke, and L. Nielson	Springer	2/e, 2005
3	Automotive Computers and Control System	Tom Weather Jr and Cland C. Hunter	Prentice Hall Inc. New Jersey	2009
4	Aircraft instruments- Principles and applications	Pallet.E.H.J	Pitman Pub	2/e, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automotive Electric Systems	Walter E, Billiet and Leslie. F, Foings	American Technical Society, Chicago	2009
2	Modern Electric Equipments for Automobiles	Judge.A.W	Chapman and Hall, London	1975
3	Elements of electronic navigation	Nagaraja.M.S	Tata McGraw Hill	2/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/107106088
2	https://nptel.ac.in/courses/107106088
3	https://nptel.ac.in/courses/107103084
4	https://youtu.be/Pwz76htytQA?si=Ux0DVPlv_YMnEP1G https://nptel.ac.in/courses/101108056

SEMESTER S7
MULTIVARIABLE CONTROL

Course Code	PEICT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBICT604, PCICT501	Course Type	Theory

Course Objectives:

1. Understand the fundamentals of multivariable control systems, including process behavior, control aims, feedback, and model considerations.
2. Develop skills in linear system representation, analysis, and solutions to control problems with practical applications such as a simple distillation column.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Multivariable Control: Introduction: Process and Instrumentation, Process Variables, The Process Behavior, Control Aims ,Modes of Operation, The Need for Feedback, Model-free vs. Model-based Control, The Importance of Considering Modelling Errors, Multivariable Systems, Implementation and Structural Issues.....	9
2	Linear System Representation: Models and Equivalence Introduction: Objectives of Modelling, Types of Models, First-principle Models: Components, Internal Representation: State Variables, Linear Models and Linearization, Input/Output Representations, Polynomial Representation, Transfer Matrix, Systems and Subsystems: Interconnection, Series, Parallel and Feedback Connection, Generalised Interconnection, Generalised Interconnection	9
3	Linear Systems Analysis ; Introduction, Linear System Time-response, Stability Conditions, Relative Degree of Stability, Discretisation, Gain, Static Gain, Instantaneous Gain, Directional Gain, Frequency response, System Internal Structure, Reachability (State Controllability), Observability, Output Reachability, Remarks on	9

	Reachability and Observability, Canonical Forms, Block System Structure (Kalman Form), Minimal Realisation, Balanced Realisation, Poles and Zeros, Input/Output Properties, Input/Output Controllability, Model Reduction, Time Scale Decomposition, Balanced Reduction, Key Issues in MIMO Systems Analysis, Case Study: Simple Distillation Column	
4	Solutions to the Control Problem: The Control Variables Structures, Feedback, Closed, Interactions, Generalised Plant, Performance Analysis, Feedforward Control, Manual Control, Open-loop Inversion and Trajectory Tracking, Disturbances, Two, Hierarchical, Key, Case.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand multivariable control	K2
CO2	To develop linearised model of systems.	K3
CO3	To analyse linear systems.	K3
CO4	To solve control problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3			3							3
CO2	3	3			3							3
CO3	3	3			3							3
CO4	3	3			3							3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Multivariable Control Systems	P. Albertos and A. Sala	Springer-Verlag London Limited	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Matrix Analysis and Applied Linear Algebra.	C.D. Meyer	Society for Industrial & Applied Mathematics (SIAM)	2001.
2	Robust Systems, Theory and Applications	R.S. Sánchez-Peña and M. Sznajder.	Wiley, NY	1998.
3	Modelling and Control of Robot Manipulators.	L. Sciavicco and B. Siciliano.	Springer-Verlag, second edition	2000.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=mMtFuYeJp5A
2	http://acl.digimat.in/nptel/courses/video/108106150/L01.html
3	https://www.youtube.com/watch?v=aL-RAgy2mJA
4	https://www.youtube.com/watch?v=VC1vFkFm9Wg

SEMESTER S7

INDUSTRIAL NETWORKS

Course Code	PEICT747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT 303	Course Type	Theory

Course Objectives:

1. Understand the key components and characteristics of Industry 4.0, including sensors, actuators, IoT connectivity, and smart factory technologies.
2. Develop knowledge of field communication networks, automation protocols, and the implementation and architecture of industrial bus systems such as PROFIBUS and MODBUS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Industry 4.0 Sensors-Characteristics-Static Characteristics-Dynamic characteristics, Active sensor, passive sensor, zero order, First order and Second order sensors. Actuators-Micro Actuators-Macro Actuators-Classification of actuators-Electric linear actuators-Electric rotary actuators-Fluid Power linear actuators-Fluid power rotary actuators, Linear chain actuators, Manual linear actuators, manual rotary actuators	9
2	Introduction of IoT IoT Connectivity-Communication Protocol-IEEE 802.15.4-variants, Zigbee-Layers and communication,6LowPAN-Layers, Wireless HART-features-Topology, ISA 100, Blue tooth, Near field connection (NFC)-RFID. Smart Factories: Connectivity field devices, Mobile devices, Operating devices, Agility-Supporting technology-Bigdata-Cloud Computing-Smart grid. Automation Pyramid-Virtual reading-augmented reality-Uses and applications	9

3	Field Communication networks Field design Installation: Bus, Basics, design, Installation, commissioning, documentation-global system architecture, foundation fieldbus architecture, Physical layer, Datalink layer, Application layer, User layer PROFIBUS basics, architecture, bus access method, block model, transducer block model, transducer blocks, applications in hazardous areas. MODBUS Networks: Communication transactions, transmission modes, MODBUS message framing-data highway.	9
4	Buses and networks Introduction to networks in process automation: Information flow requirements, hierarchical communication model, Network requirements, OSI reference model-Open networks: types of cable optical fibre cable Industry open standards (RS-232C, RS-422, RS-485). Fieldbus-functions -cyclic data exchange, Redundance and failure considerations-Advantages and disadvantages.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand about the different types of sensors and actuators.	K2
CO2	Understand about various types of connectivity.	K2
CO3	Understand about Industrial communication devices.	K2
CO4	Understand Industrial communication protocols.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process software and Digital Networks	B.G LIPTAK	CRC Press	4 TH 2011
2	LABVIEW graphical programming	Gary Jhonson	McGrawhill	5 TH 2017

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied instrumentation in Process Industries	Andrews W G	Gulf Professional Publishing.	2 ND 2006
2	Process Control instrumentation technology	C D Johnson	Prentice Hall of India	8 TH 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.digimat.in/nptel/courses/video/106105195/L01.html https://www.digimat.in/nptel/courses/video/106105195/L02.html
2	https://www.digimat.in/nptel/courses/video/106105195/L03.html https://www.digimat.in/nptel/courses/video/106105195/L04.html
3	https://www.digimat.in/nptel/courses/video/106105195/L05.html https://www.digimat.in/nptel/courses/video/106105195/L07.html
4	https://www.digimat.in/nptel/courses/video/106105195/L09.html https://www.digimat.in/nptel/courses/video/106105195/L10.html

SEMESTER S7
VIRTUAL INSTRUMENTATION

Course Code	PEICT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop proficiency in using LabVIEW for graphical system design, virtual instrumentation, and modular programming to create, manage, and execute complex engineering processes.
2. Gain hands-on experience with data acquisition, instrument control, and advanced data visualization techniques, integrating hardware and software components effectively for comprehensive engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Graphical system design Graphical system design (GSD) model-design flow with GSD-virtual instrumentation-virtual instrument and traditional instrument-hardware and software in virtual instrumentation-virtual instrumentation for test, control and design-virtual instrumentation in the engineering process-virtual instruments beyond personal computer- graphical system design using LabVIEW-graphical programming and textual programming. Introduction to LabVIEW Advantages of LabVIEW- software environment-creating and saving a vi - front panel toolbar- block diagram toolbar-palettes -shortcut menus - property dialog boxes - front panel controls and indicators- block diagram-data types -data flow program-LabVIEW documentation resources - keyboard shortcuts. Modular programming Modular programming in LabVIEW - build a vi front panel and block diagram-icon and connector pane - creating an icon - building a connector pane- displaying subVIs and express vis as icons or expandable nodes - creating subVIs from sections of a vi - opening and editing subVIs -	9

	placing subVIs on block diagrams- saving subVIs - creating a stand-alone application.	
2	Repetition and loops For loops - while loops - structure tunnels- terminals inside or outside loops - shift registers-feedback nodes- control timing- communicating among multiple loops - local variables - global variables. Arrays Arrays in LabVIEW - creating one-dimensional array controls, indicators and constants - creating two-dimensional arrays-creating multidimensional arrays- initializing arrays - deleting elements, rows, columns and pages within arrays- inserting elements, rows, columns and pages into - replacing elements, rows, columns, and pages within arrays- array functions - auto indexing - creating two-dimensional arrays using loops - identification of data structure (scalar and array) using wires - using auto-indexing to set the for loop count - matrix operations with arrays-polymorphism. Clusters Creating cluster controls and indicators- creating cluster constant - order of cluster elements - cluster operations - assembling clusters- disassembling clusters - conversion between arrays and clusters - error handling- error cluster.	10
3	Plotting data types of waveforms - waveform graphs-waveform charts-waveform data type- XY graphs-intensity graphs and charts-digital waveform graphs-3d graphs - customizing graphs and charts-customizing graphs-customizing 3d graphs - customizing charts-dynamically formatting waveform graphs - configuring a graph or chart - displaying special planes on the XY graph. Structures Case structures-sequence structures-customizing structures- timed structures- formula nodes - event structure - LabVIEW mathscript. Strings and file i/o Creating string controls and indicators - string functions-editing, formatting and parsing strings - formatting strings - configuring string controls and indicators- basics of file input/output - choosing a file I/O format - LabVIEW data directory - file i/o VIs - creating a relative path.	9
4	Instrument Control Introduction - GPIB communication - hardware specifications - software	8

	architecture - instrument i/o assistant – visa- instrument - serial port communications - data transfer rate - serial port standards - using other interfaces. Data acquisition Transducers - signals- signal conditioning- DAQ hardware configuration - DAQ hardware - analog inputs - analog outputs - counters - digital I/O (DIO) – DAQ software architecture- DAQ assistant- channels and task configuration - selecting and configuring a data acquisition device- components of computer based measurement system	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define virtual instrumentation concepts and Compare traditional and virtual instrumentation.	K2
CO2	Study the basic building blocks of virtual instrumentation, usage of loops, arrays, clusters using LabVIEW.	K2
CO3	Study the usage of structures, string operations and file operations.	K2
CO4	Study the usage of structures, string operations and file operations.	K2
CO5	Design a Virtual Instrument.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3									3
CO2	3		3									3
CO3	3		3									3
CO4	3		3		3							3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Virtual Instrumentation using LabVIEW	Jovitha Jerome	PHI Learning Pvt. Ltd, New Delhi.	2010
2	Learning with Lab-view	Robert H. Bishop	Pearson	1 edition, 2014
3	LabVIEW based Advanced Instrumentation	S.Sumathi&P.Surekha	Springer	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Virtual Instrumentation using LabVIEW	Sanjay Gupta, Joseph John	McGraw Hill Education	2 nd edition
2	LabVIEW Graphical Programming	Gary W. Johnson, Richard Jennings	McGrawHill Education	4th edition 2006

SEMESTER S7
AEROSPACE ENGINEERING AND NAVIGATION
INSTRUMENTATION

Course Code	PEICT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT 205PBICT604, PCICT504	Course Type	Theory

Course Objectives:

1. Explore the history of aviation and space flight, including airplane and space vehicle anatomy, aerodynamics, flight performance, and rocket propulsion principles.
2. Understand the operation and application of various flight and navigation instruments, including GPS, GNSS, automatic pilots, and advanced inertial sensors used in guidance and avionics systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	History of aviation and space flight -anatomy of airplane and space vehicle with emphasis on control surfaces-air foil nomenclature-basics of aerodynamics to illustrate lift and drag-types of drag-finite wings-swept wings -flaps airplane performance-thrust-power-rate of climb-absolute and service ceiling-range and endurance, Kepler's laws-rocket engines, propellants and staging	9
2	Basic engine instruments - capacitive fuel content-gauges. Standard atmosphere-altimeters-aneroid and radio altimeters, Aircraft Compass-Remote Indicating magnetic compass-Rate of climb indicator-Pilot static system-Airspeed Indicator-Mach Meters-Integrated flight Instruments	9
3	GPS and GNSS -Automatic Pilots-Aircraft flight simulation instrumentation, Introduction to guidance, Navigation and avionics-Radio navigational aids-automatic direction finder-VHF-Phase-Comparison direction finder	9
4	Introduction to navigation and guidance instrumentation -Principle, Construction and application of inertial sensors-gyroscope and	9

	accelerometers-Ring laser gyroscope-Fibre optic gyroscope, MEMS gyroscopes and accelerometers-Pulse doppler RADAR	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the aerodynamic systems	K2
CO2	Understand various measuring instruments in aviation systems	K2
CO3	Understand the Operation of Navigation systems	K2
CO4	Understand about vary frequency used measurements in Aviation	K2
CO5	Understand the operation of stability control sensors	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2
CO5	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Aircraft Instruments-Principles and applications	Pallet.E.H.J	Pitman publications	3 RD 1992
2	Elements of electronic navigation	Nagaraja M S	Tata McGraw Hill	2 ND 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to flight	John D Anderson	McGraw Hill	9 TH 2002
2	Radio aids to navigation	San Darite		
3	Automatic control of aircraft and missiles.	John H Blakelock	John wiley and sons	2 ND 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.digimat.in/nptel/courses/video/101105059/L01.html https://www.digimat.in/nptel/courses/video/101105059/L02.html
2	https://www.digimat.in/nptel/courses/video/101105059/L03.html https://www.digimat.in/nptel/courses/video/101105059/L04.html
3	https://www.digimat.in/nptel/courses/video/101105059/L05.html https://www.digimat.in/nptel/courses/video/101105059/L06.html
4	https://www.digimat.in/nptel/courses/video/101105059/L07.html https://www.digimat.in/nptel/courses/video/101105059/L08.html

SEMESTER S7

MODERN METHODS OF INSTRUMENT ANALYSIS

Course Code	PEICT 752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NONE	Course Type	Theory

Course Objectives:

1. Gain a solid understanding of the fundamental principles behind modern analytical instruments.
2. Learn the theoretical foundations and practical applications of various modern analytical techniques used in chemical and biochemical analysis.
3. Learn how to select appropriate instruments based on analytical needs.
4. Develop skills in data acquisition, interpretation, and analysis using modern software and computational tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Instrumental analysis - Review What is analytical science, Quantitative and qualitative measurements- Interpretation of analytical results, Basic tools available-Concentration and various methods of expressing concentration and standard followed during chemical analysis- Various online and offline sources available- Overview of Food and Drug Laws and Regulations -Regulatory affairs, Agencies for implementing regulation.Various regulatory protocols, USFDA, ICH Guidelines etc. WHO guidelines-The Central Drugs Standard Control Organisation (CDSCO- International Organization for Standardization (ISO).	10
2	Thermal Analytical Techniques - Thermogravimetry (TG)-Differential Thermal analysis (DTA)-Differential Scanning calorimetry (DSC)- Theories of different thermal methods, TG, DTA and DSC methods, Instrumentation, Different furnaces, Crucibles, Gas handling system, mass flow controllers, crucible and ceramic materials used in thermal analysis. Interpretation of data,	9

	combined techniques like TG-MS.	
3	Electro analytical methods -(CV, DPV, LSV)-Common electrochemical methods, Cyclic voltametry, systems of reversible, irreversible and quasi reversible systems, analysis of redox molecules using CV. voltametry analysis basics and application, Application of LSV in hydrogen and oxygen evolution reaction.	9
4	Water quality Water quality testing parameters, temperature, dissolved oxygen, pH, conductivity, ORP, and turbidity. Standard protocols for testing water, WHO standards on water quality, Water contamination and purification methods	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Gain Awareness on food and drug laws and regulations	K2
CO2	Gain Knowledge about different Thermal analytical methods	K3
CO3	Gain Knowledge about different Electro analytical methods	K3
CO4	Gain Awareness of water quality management.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				1	2						2
CO2	3				3	2						2
CO3	3				3	2						2
CO4	3					3	3					2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Instrumental Analysis	Douglas A. Skoog, F. James Holler, and Stanley R. Crouch	Cengage Learning	Edition: 7 th edition. Year: 2016
2	Analytical Chemistry: An Introduction	Douglas A. Skoog, Donald M. West, and F. James Holler	Cengage Learning	Edition: 7 th edition Year: 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrumental Analysis	Douglas A. Skoog, Donald M. West, F. James Holler, and Stanley R. Crouch	Cengage Learning	Edition: 7 th Year: 2017
2	Quantitative Chemical Analysis	Daniel C. Harris	W . H . Freeman and company	Edition: 9th Year: 2015
3	Modern Analytical Chemistry	David Harvey	McGraw-Hill Education	Edition:2 ⁿ d Year:2000

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.nptelvideos.com/course.php?id=173 https://www.sciencedirect.com/science/article/abs/pii/B978032391817600005X
2	https://onlinecourses.nptel.ac.in/noc23_me31/preview
3	https://archive.nptel.ac.in/courses/103/108/103108162/
4	https://elearn.nptel.ac.in/shop/iit-workshops/completed/water-quality-an-approach-to-peoples-water-data/?v=c86ee0d9d7ed

SEMESTER S7
INSTRUMENTATION FOR AGRICULTURE

Course Code	PEICT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To expose the students about various instruments used in agriculture
2. To teach the design of greenhouses using various instruments
3. To impart ideas about modern technologies used in agriculture

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Soil Science & Sensors Soil science : fundamental definitions & relationships properties of soil : index properties of soil, Engineering properties of soil, physical properties of soil permeability & seepage analysis, Mohr's circle of stress Necessity of instrumentation & control for agriculture Sensors and transducers: definitions, introduction to sonic anemometers, hygrometers, thermocouples, close path gas analysers, introduction to various biosensors	9
2	Instrumentation in industrial Continuous & Batch processes Idea about continuous and batch processes Flow diagram of sugar plant, sensors & instrumentation set up Flow diagram of fermenter, sensors & instrumentation set up Flow diagram of dairy industry & instrumentation set up for it Juice extraction control process & instrumentation set up. Oil extraction plant and instrumentation set up for it. Pesticides manufacturing process and control	9

3	Instrumentation in Irrigation Necessity of irrigation in agriculture Evapotranspiration and its measurement methods irrigation methods: overhead, centre pivot, lateral move, micro irrigation systems & the comparison of different irrigation systems soil moisture measurement methods: resistance-based method, voltage based method, thermal based method Water distribution & management control, Auto drip & sprinkler irrigation systems, Irrigation canal management systems upstream & downstream control concept Introduction to SCADA , Applications of SCADA in agriculture	11
4	Greenhouse Parameters & Instrumentation Greenhouse effect, Concept & construction of green houses, ventilation, cooling & heating, wind speed, temperature & humidity, soil moisture, rain gauge, carbon dioxide enrichment measurement & control implementation of hydraulic, pneumatic & electronics control circuits in harvesters cotton pickers, tractor Classification of pumps: pump characteristics, pump selection & installation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret the soil properties and sensors used in agriculture field	K2
CO2	Explain the continuous and batch processes.	K2
CO3	Realize the various methods involved in Irrigation system	K2
CO4	Explain the parameters and instrumentation in greenhouses	K2
CO5	Explain the pumps used in agriculture	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3					3
CO2	3					3	3					3
CO3	3					3	3					3
CO4	3					3	3					3
CO5	3					3	3					3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Instrumentation	D. Patranabis	McGraw Hill Education	3/e, 2017
2	Instrumentation handbook-process control	B.G.Liptak	Chilton	3/e, 2013
3	Advanced agricultural instrumentation	Tom Weather Jr and Cland C. Hunter	Springer	2011
4	Process control and instrumentation technology	C.D. Johnson	Pearson education	7/e, 2005
5	Industrial Instrumentation	D. Patranabis	McGraw Hill Education	3/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Agricultural Engineering	RadheyLal	Saroj Publication	2008
2	Mineral Processing Technology	Wills B.A	Butterworth-Heinemann	7/e, 2006
3	Application of Vision Technology and Artificial Intelligence in Smart Farming	Xiuguo Zou, Zheng Liu	journal <i>Agriculture</i> by MDPI	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108108147
2	https://nptel.ac.in/courses/108108147
3	https://nptel.ac.in/courses/126105010 https://nptel.ac.in/courses/126105019 https://nptel.ac.in/courses/108106022
4	https://nptel.ac.in/courses/108108147

SEMESTER S7
INDUSTRIAL DRIVES AND CONTROL

Course Code	PEICT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the fundamental concepts of electric drives, including types, factors influencing selection, thermal characteristics, and load variations for optimal power rating selection.
2. Gain knowledge of mechanical and speed-torque characteristics of different motors, braking methods, starting techniques, and advanced speed control methods for DC and AC drives, with practical applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Basic elements-types of electric drives-factors influencing electric drives-heating and cooling curves loading conditions and classes of duty-Selection of power rating for drive motors with regard to thermal overloading and load variation factors	9
2	Drive motor characteristics: Mechanical characteristics- speed- torque characteristics of various types of load and drive motors - braking of electrical motors-dc motors: shunt, series, compound motors-single phase and three phase induction motors	9
3	Starting methods: Types of d.c motor starters-typical control circuits for shunt and series motors-three phase squirrel and slip ring induction motors	9
4	Conventional and solid-state speed control of D.C and A.C Drives Speed: control of DC series and shunt motors-Armature and field control, ward-Leonard control system using controlled rectifiers and DC choppers – applications Speed control of three phase induction motor-Voltage control, voltage/frequency control, slip power recovery scheme-using inverters and AC voltage regulators-applications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the basic elements of electric drives.	K2
CO2	To understand mechanical characteristics of the load and drive motors.	K2
CO3	To understand about the types of d.c motor starters	K2
CO4	To analyse the Conventional and solid-state speed control of D.C and A.C Drive	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2				1							2
CO2	2				1							2
CO3	2				1							2
CO4		3			2							2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric drives (concepts and applications)	Vedam subramaniam	Tata McGraw-Hill.	2001
2	Electrical machines	Nagarath.i. j & kothari. D.p	Tata McGraw-Hill.	1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A first course on Electric drives	Pillai.S. K	Wiley Eastern Limited	1998
2	Power electronics	M.D. Singh, K.B. Khanchandani	Tata McGraw-Hill.	1998
3	Art and science and utilization of electrical energy	H.Partab	Dhanpat Rai and sons	1994

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://acl.digimat.in/nptel/courses/video/108104140/L24.html
2	http://nitttrc.edu.in/nptel/courses/video/108104140/L20.html
3	https://www.youtube.com/watch?v=1OfLgpFq6Rc
4	https://www.youtube.com/watch?v=VoN0e3n6EGA

SEMESTER S7
NON LINEAR DYNAMICS AND CHAOS

Course Code	PEICT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the fundamental concepts of nonlinear dynamics and chaos.
2. Analyze dynamical systems using phase plane techniques.
3. Explore the behavior of systems near bifurcation points.
4. Investigate the routes to chaos and the characteristics of chaotic systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Dynamical Systems -Chaos, Fractals and Dynamics – History of Dynamics – Nonlinear systems. One-Dimensional Flows: Fixed Points and Stability-Linear Stability Analysis-Existence and Uniqueness – Potentials.	9
2	Bifurcations Saddle-Node Bifurcation-Transcritical and Pitchfork Bifurcations Imperfect Bifurcations and Applications Flows on the Circle Introduction to Circle Maps -Phase-Locked Solutions-Bifurcations and Chaos in Circle Maps	9
3	Two-Dimensional Flows Phase Plane Analysis-Linear Systems and Phase Portraits-Nonlinear Systems and Limit Cycles-Limit Cycles and Oscillators-Van der Pol	9

	Oscillator-Relaxation Oscillations-Perturbation Methods Bifurcations Hopf Bifurcation-Global Bifurcations and Homoclinic Orbits Bifurcation Diagrams and Routes to Chaos	
4	Chaos Lorenz Equations and Strange Attractors-Lyapunov Exponents-Fractals and Dimension One Dimensional Maps Introduction –Fixed points and Cobwebs - Logistic Map: Numerics- Logistic Map: Analysis – Periodic windows – Liapunov Exponents.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	explain the concepts of fixed points and stability in one-dimensional flows, including performing linear stability analysis and understanding the implications of existence and uniqueness theorems in nonlinear systems.	K2
CO2	demonstrate an understanding of various types of bifurcations, such as saddle-node, transcritical, and pitchfork bifurcations, including recognizing imperfect bifurcations and their applications in different dynamical systems.	K2
CO3	describe and analyse phase plane dynamics in two-dimensional systems, including constructing and interpreting phase portraits for both linear and nonlinear systems, and understanding the behavior of limit cycles and oscillators.	K3
CO4	understand chaotic systems, including the Lorenz equations, strange attractors, and Lyapunov exponents and the concept of fractals and dimensions, and analyze the behavior of one-dimensional maps, such as the logistic map, using techniques like cobwebbing and identifying periodic windows and Lyapunov exponents.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering"	Steven H. Strogatz	CRC PRESS	2 ND EDITION 2018
2	Chaos: An Introduction to Dynamical Systems	Kathleen T. Alligood, Tim D. Sauer, and James A. Yorke	SPRINGER	1 ST EDITION 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Applied Nonlinear Dynamical Systems and Chaos	Stephen Wiggins	SPRINGER	2 ND EDITION 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/106/108106085/
2	https://archive.nptel.ac.in/courses/108/106/108106085/
3	https://nptelvideos.com/course.php?id=461
4	https://nptelvideos.com/course.php?id=461

SEMESTER S7
SYSTEM IDENTIFICATION AND ADAPTIVE
CONTROL

Course Code	PEICT757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCICT403, PCICT501, PBICT604	Course Type	Theory

Course Objectives:

1. To introduce various model structures for system identification
2. To impart knowledge on parametric and non-parametric identification
3. To introduce the concept of adaptation techniques and control

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	System Identification Models for identification: Models of LTI systems: Linear Models-State space Models-OE model- Model sets, Structures and identifiability-Models for Time-varying and Non-linear systems: Models with Nonlinearities – Non-linear state-space models-Black box models, Fuzzy models	9
2	Non-parametric and parametric identification: Transient response and Correlation Analysis – Frequency response analysis – Spectral Analysis – Least Square – Recursive Least Square –Forgetting factor- Maximum Likelihood – Instrumental Variable methods	9
3	Introduction to adaptive control Linear feedback; effects of process variations; adaptive schemes; Introduction to deterministic self-tuning regulators; pole placement design; indirect self-tuning regulator; design of minimum variance and moving average controllers; adaptive predictive control	9
4	Model Reference Adaptive Systems Introduction to model-reference adaptive systems; Lyapunov theory; design	9

	of MRAS using Lyapunov theory; Bounded-input bounded-output stability; applications to adaptive control.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain about various models of identification	K2
CO2	Use non-parametric and parametric methods to identify the system models.	K2
CO3	Design self-tuning regulators.	K3
CO4	Design Model-reference Adaptive control	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				2							3
CO2	3		2		2							3
CO3	3		2		2							3
CO4	3		2		2							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Identification: Theory for the user	Lennart Ljung	Prentice Hall	2nd,1999
2	System Identification	T. S. Soderstrom, Petre G. Stoica	Prentice Hall	2001
3	Adaptive control	Karl J. Astrom, Bjorn Wittenmark	Dover Publications	1994
4	Adaptive Control– Stability, Convergence and Robustness”	Sastry,S. and Bodson, M	Prentice Hall inc., New Jersey	1989

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Identification: An Introduction	Karel J. Keesman	Springer	
2	Principles Of System Identification	Arun K. Tangirala	CRC Press	2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/106/103106149/
2	https://archive.nptel.ac.in/courses/103/106/103106149/
3	https://archive.nptel.ac.in/courses/108/102/108102113/
4	https://archive.nptel.ac.in/courses/108/102/108102113/

SEMESTER 7

IOT AND APPLICATIONS

Course Code	PEICT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None		

Course Objectives:

1. Gain a comprehensive understanding of IoT technology, including its definitions, architecture, functional blocks, communication models, enabling technologies, and deployment templates.
2. Learn about IoT and M2M differences, sensor networks, communication protocols, IoT privacy and security solutions, and practical applications such as smart city IoT architecture and Raspberry Pi interfacing and programming.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT technology: Definitions and Characteristics of IoT, IoT Architectural View, Physical Design of IOT, Logical Design of IoT- IoT Functional blocks, IoT communication models, IoT Enabling Technologies, IoT Levels & Deployment Templates.	9
2	IoT and M2M- M2M, Difference between IoT and M2M, SDN and NFV for IoT, Smart Objects: The “Things” in IoT: Sensors, Actuators, and Smart Objects, Sensor Networks- Wireless Sensor Networks (WSNs), Communication Protocols for Wireless Sensor Networks- Connecting Smart Objects- Communication Criteria.	9
3	Unified Data Standards –Protocols –IEEE 802.15.4 -The Physical Layer, The Media-Access Control Layer, Uses of 802.15.4, The Future of 802.15.4: 802.15.4e and 802.15.4g–Modbus– ZigBee-Zigbee Architecture- LoRaWAN -Standardization and Alliances, Physical Layer, MAC Layer, Topology, LTE-M, NB-IoT-Network layer –The next generation: IP-based protocols	9
4	Raspberry-Pi-Raspberry-Pi Interfaces (serial, SPI, I2C). Raspberry Pi interfacing and programming examples using python (LED, switch, sensor,	9

	serial, SPI, I2C devices). IoT privacy, security and vulnerabilities solutions, vulnerabilities, security requirements, threat analysis, security tomography, layered attacker model, Identity management, access control, secure message communication. Smart City IoT Architecture-Street Layer, City Layer, Data Center Layer, Services Layer.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Microproject-10 (Eg:-Communication among devices, Programming of Raspberry pi processor etc.)

Assignment -10 (Eg: -Circuit design, Network topology selection etc.)

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the IoT fundamentals and architecture modelling	K1
CO2	Understand the smart things in IoT and functional blocks	K2
CO3	To understand the communication networks and protocols used in IoT.	K2
CO4	To understand the cloud resources, data analysis and applications.	K3
CO5	To apply the IoT processes in embedded applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		1			2				2
CO2	3	3	3		3			2				2
CO3	3	3	3		3			2	3			2
CO4	3								3			2
CO5	3								3			2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things (A Hands - on-Approach)	Vijay Madiseti and ArshdeepBahga	VPT	I st Edition,2014
2	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry,	Cisco Press	I st Edition,2017
3	Internet of Things: Architecture and Design Principles	Rajkamal	McGraw Hill (India) Private Limited.	
4	Raspberry Pi Cookbook, Software and Hardware Problems and solutions	Simon Monk, O'Reilly (SPD)	ISBN	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“The Internet of Things –Key applications and Protocols”,	Olivier Hersent, David Boswarthick, Omar Elloumi	Wiley,	2012
2	Internet of things: A survey on enabling technologies, protocols, and applications.	Al-Fuqaha <i>et.al</i>	IEEE Communications Surveys & Tutorials	(2015), pp. 2347-2376.
3	The Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems	Ovidu Vermesan and Peter Friess	River Publishers	
4	Internet of Things - From Research and Innovation to Market Deployment-	Ovidu Vermesan	River Publishers	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_cs53/preview
2	https://nptel.ac.in/courses/106105081
3	https://nptel.ac.in/courses/108108098
4	https://onlinecourses.nptel.ac.in/noc21_cs14/preview

SEMESTER S7
AUTOMOBILE INSTRUMENTATION

Course Code	OEICT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To make them understand fundamentals of 4 stroke engine
2. To expose the students to the concepts of various sensors used in automobile systems.
3. To teach the basic and advanced controls in automotive systems
4. To impart ideas on hybrid and electric vehicles, ADAS, GPS

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Vehicles Engine basic concepts (4 strokes), Starting & charging system, onboard diagnostic system, Vehicle motion control, Digital cruise control, and current trends in modern automobiles.	9
2	Engine management systems Basic sensor arrangement and types of sensors: oxygen sensors, crank angle position sensors, vehicle speed sensors, flow sensors, temperature, air mass flow sensors, throttle position sensors, electronic ignition, and EGR for exhaust emission control.	9
3	Automobile Chassis Electronic Control System Principles of electronic braking, automatic transmission electronic control circuit, cruise control circuit, electronic steering control theory, ABS, ASR, ESP, and other electronic control methods. auto Body Electronic Control Technology, Automotive central locking, and anti-theft system control technology. Air bag technology	9

4	Latest Automotive Instrumentation Concepts of hybrid technology and its classification. Introduction to electric vehicles and classifications. Introduction to ADAS (Automatic Driver Assist Systems). GPS Navigation and GPS System Structure.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of automobiles.	K2
CO2	Explain the evaluate of sensor and measuring system of automobile	K2
CO3	Explain the design of the basic modelling and control scheme for automotive systems	K2
CO4	Explain about the Latest trends in automotive instrumentation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	2											2
CO3	2											2
CO4	2											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Understanding Automotive Electronics	William B. Riddens	Butterworth Heinemann Woburn	5/e, 1998
2	Automotive Control Systems	U.Kiencke, and L. Nielson	Springer	2/e, 2005
3	Automotive Computers and Control System	Tom Weather Jr and Cland C. Hunter	Prentice Hall Inc. New Jersey	2009
4	Aircraft instruments- Principles and applications	Pallet.E.H.J	Pitman Pub	2/e, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year

1	Automotive Electric Systems	Walter E, Billiet and Leslie. F, Foings	American Technical Society, Chicago	2009
2	Modern Electric Equipments for Automobiles	Judge.A.W	Chapman and Hall, London	1975
3	Elements of electronic navigation	Nagaraja.M.S	Tata McGraw Hill	2/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/107106088
2	https://nptel.ac.in/courses/107106088
3	https://nptel.ac.in/courses/107103084
4	https://youtu.be/Pwz76htytQA?si=Ux0DVPIv_YMnEP1G https://nptel.ac.in/courses/101108056

SEMESTER S7
MEASUREMENTS AND TRANSDUCERS

Course Code	OEICT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	OEICT 611	Course Type	Theory

Course Objectives:

1. Learn various transducer types and their principles. Differentiate between primary and secondary transducers.
2. Understand principles and functions of electrical measuring instruments, current and potential transformers and resistance measurement techniques with bridges and ohmmeters.
3. Master techniques for measuring force, acceleration, torque, power, and fluid properties.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Transducers Definition of transducers - primary and secondary transducers. Principles of variable resistance transducers – Potentiometers - Strain gauges Piezo electric transducers, materials and properties, modes of deformation, Hall effect transducers. Principle, type and construction of variable inductive transducers, Different types of self and mutual inductance transducers, LVDT and RVDT. Uses, advantages and disadvantages of inductive transducers. Principle, types and construction of different types of variable capacitance transducers. Uses, advantages and disadvantages of capacitive transducers, LDR.	9
2	Electrical Measuring Instruments Indicating Instruments: Principle, types of control and damping; Moving coil Instruments: Types (permanent magnet, dynamometer type meters);	9

	Moving Iron Instruments: Attraction and repulsion type, Principles and torque equation; Wattmeters: Dynamometer type wattmeter, Principles and torque equation; Measurement of single phase and three phase power; true RMS meter; Errors and Compensation.	
3	Transformers Current transformers and Potential transformers; use of instrument transformers with wattmeter. Bridges and Potentiometers Measurement of resistance: Ohmmeter, Megger, Wheatstone bridge; Kelvin's double bridge; Phasor diagram; AC bridges: Measurements of inductance using Maxwell and Anderson bridges; Measurements of capacitance using Schering Bridge; Potentiometer: General principle; Modern form of dc potentiometers; Vernier dial principle; Standardization.	9
4	Force Measurement - Platform balance - Force to displacement conversion - Proving ring - Conversion of force to hydraulic pressure - Piezoelectric force transducer. Measurement of acceleration Preliminary ideas - Characteristics of a spring - mass - damper system - Piezoelectric accelerometer - Laser Doppler Vibrometer - Fiber Optic Accelerometer Measurement of torque and power - Mechanical brake arrangement - Prony brake - Electric generator as a dynamometer - Measure shear stress on the shaft - Tachometer - Mechanical Device - Non-contact optical RPM meter. Fluid properties: Density, surface tension, capillarity and viscosity; Newton's law of viscosity; Fluid Statics; Pascal's law, Centre of pressure, Buoyancy, Metacentre; Basic equations of fluid flow; continuity, momentum and energy equations; Bernoulli's equations	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles, types, construction, and applications of various transducers, enabling them to effectively select, design, and implement appropriate sensing technologies across engineering disciplines.	K2
CO2	possess comprehensive understanding of the principles, types, construction, and applications of electrical measuring instruments.	K2
CO3	Understand the theory, operation, and practical application of transformers, instrument transformers with wattmeters and various types of bridges.	K2
CO4	Understand the principles and operation of measuring systems used for force, and torque.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical Measurements (for module 4)	S.P.Venkateshan	Ane Books	2 nd edition, 2015
2	A course in Electrical and Electronics Measurements and Instrumentation	A K SAWHNEY	Dhanpat Rai & Co. (P) Limited	19th edition, 2015
3	Mechanical Measurements	Thomas G. Beckwith, Roy D. Marangoini, John H. Lienhard V	Pearson	5th edition, 1993
4	Mechanical Measurements (for module 4)	S.P.Venkateshan	Ane Books	2 nd edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Metrology	R.K. Jain	Khanna Publishers	2022
2	Electrical and Electronics Measurements and Instrumentation	Prithwiraj Purkait, Budhaditya Biswas, Santanu Das	McGraw Hill Education	2017

SEMESTER S7
BIOMEDICAL INSTRUMENTATION

Course Code	OEICT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Gain fundamental knowledge of biopotential acquisition and analysis:
2. Develop an understanding of signal conditioning principles for biomedical applications
3. Become familiar with the measurement of various physiological parameters

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	BASIC ANATOMY AND PHYSIOLOGY: Origin of bio potential and its propagation, Nervous system, Muscular system, Cardiovascular system, Respiratory System. BIOPOTENTIAL ELECTRODES: Electrode-electrolyte interface, electrode-skin interface, half-cell potential, Contact impedance, polarization effects of electrode – non polarizable electrodes. Types of electrodes - surface, needle and micro electrodes and their equivalent circuits. Recording problems - motion artifacts- Monopolar and bipolar electrode configurations	9
2	Bio signals characteristics – frequency and amplitude ranges. ECG – Einthoven ‘s triangle, standard 12 lead system, Principles of vector cardiography. EEG – 10-20 electrode system, unipolar, bipolar and average mode. EMG– unipolar and bipolar mode. SIGNAL CONDITIONING CIRCUITS: Need for bio-amplifier - differential bio-amplifier, Design of bio-amplifiers for the measurement of physiological variables. Power line interference, Band pass filtering.	9

	Working of isolation amplifiers, (transformer, capacitive and optical isolation), isolated DC amplifier and AC carrier amplifier. (Circuit diagram and working only)	
3	MEASUREMENT OF NON-ELECTRICAL PARAMETERS: Temperature and pulse rate measurements. Blood Pressure: indirect methods - Auscultatory method, oscillometric method, direct methods: electronic manometer, Pressure amplifiers, Systolic, diastolic, mean detector circuit. Blood flow and cardiac output measurement: Indicator dilution, thermal dilution and dye dilution method, Electromagnetic and ultrasound blood flow measurement. Pulmonary function measurements-Spirometer, Photo plethysmography, Body plethysmography.	7
4	ELECTRICAL SAFETY IN MEDICAL ENVIRONMENT: Types of Electrical hazards, Physiological effects of electric current – Threshold Values for Electrical Shock, shock hazards from electrical equipment – leakage current-Instruments for checking safety parameters of biomedical equipment, Methods of accident prevention, Electrical Safety Standards ASSISTIVE AND THERAPEUTIC EQUIPMENTS: Pacemakers, defibrillators, ventilators, diathermy, Heart-Lung machine and dialyzers. (Diagrams and Basic working only) BIOMEDICAL IMAGING X-ray, Computed Tomography, Ultrasound, Magnetic resonance Imaging. (Diagrams and Basic working only)	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the origin of bioelectric potentials, and working of human nervous system, muscular system, cardiovascular system and respiratory system, and describe the theory, construction and working of bio-electrodes	K2
CO2	Explain the measurement of biomedical signals such as ECG, EEG and EMG, and explain working of signal conditioning circuits and isolations amplifiers.	K2
CO3	Describe the measurement of non-electric parameters of heart and lungs	K2
CO4	Perceive the concepts of assistive & therapeutic equipment such as Pacemakers, defibrillators, ventilators, diathermy, Heart-Lung machine & dialyzers, electrical safety in medical environment and basing working of imaging modalities such as X-Ray, CT, US and MRI	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2						3
CO2	3					2						3
CO3	3					2						3
CO4	3					2						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Instrumentation and Measurements	L. Cromwell, F. J. Weibell, and L. A. Pfeiffer	Pearson Education, Delhi	2 nd , 2015
2	Medical Instrumentation Application and Design	John G Webster	Wiley India Private Limited	5 th , 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Biomedical Equipment Technology	J. J. Carr and J. M. Brown	Pearson Education, Delhi	4 th , 2002
2	Hand Book of Bio-Medical Instrumentation	R.S. Khandhpur	McGraw Hill Education (India) Private Limited	3 rd , 2014
3	Principles of Applied Bio-Medical Instrumentation	L.A. Geddes and L.E.Baker	Wiley India Pvt Ltd	3 rd , 2008
4	Principles of Medical Electronics and Bio-medical Instrumentation	C. Rajarao and S.K. Guha	Universities press (India) Ltd	2000

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI
2	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI
3	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI
4	https://youtube.com/playlist?list=PLm_MSClsnwm-kjSqlfwaEhKwfnnfLUUBM&si=UkPImh4yMqtPBNVI

SEMESTER 8

**INSTRUMENTATION AND CONTROL
ENGINEERING**

SEMESTER S8
OPTOELECTRONICS AND INSTRUMENTATION

Course Code	PEICT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the principles of interferometers, Lasers and its applications
2. To gain knowledge about fiber optics in instrumentation and its applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Interferometers: Fabry-Perot and Michelson interferometers – Interference filters – Interferometric method of measurement of optical components – Optical spectrum analyser. Raman Spectroscopy, Raman effect.	7
2	Modulation of light: Birefringence – Optical activity – Electro-optic effect – Magneto-optic devices – Acousto-optic effect. Display devices: Electroluminescence – Injection Luminescence – Light emitting diode – Plasma displays – Liquid crystal displays. Opto-couplers ,Optrodes	9
3	Lasers: Principles of operation –Einstein relations – population inversion– Classes of lasers – Solid state: Nd-YAG laser, Ruby laser, gas: He-Ne laser, Argon ion laser, CO2 laser and liquid dye lasers – Q-switching and mode locking ,cavity dumping Application of LASER - Distance, length, velocity, acceleration, current and voltage measurements. Material processing: Laser heating, melting, scribing, splicing, welding and trimming of materials, removal and vaporization, calculation of power requirements. Laser gyroscope. Holography – Principles and applications – Pollution monitoring, industry and biomedical applications	9
4	Optical fibers: Introduction to optical fiber – Fiber characteristics, Light guidance through fibers – Different types of fibers– Losses in the optical	11

	fiber – Dispersion – optical time domain reflectometer-Advantages and disadvantages of optical fibers, Connectors and splices Applications of optical fibres: optical fibre communication- Selection of optical fibers – fibre optic sensors – measurement of temperature, pressure, displacement, strain, acceleration and fluid level	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate the principles of interferometers and optical components	K2
CO2	Outline the theory and operation of display devices	K2
CO3	Explain LASER operating principle, structures and applications	K2
CO4	Outline the basics of fibre optics	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											2
CO3	3											2
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Optoelectronics: An Introduction	J. Wilson and J.F.B. Hawkes	Prentice Hall of India.	2e/d 1998
2	Optical Fiber Communications- Principles and Practice	John M. Senior	Pearson Education Limited.	3e/d, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Laser Technology	BreckHitz, J.J Ewing, Jeff Hecht	Wiley-IEEE Press,	4 th edition
2	Engineering applications of Lasers and Holography	W.E. Cock	Plenum Press	1975

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117/104/117104127/
2	https://nptel.ac.in/courses/117/104/117104127/
3	https://nptel.ac.in/courses/117/101/117101054/
4	https://nptel.ac.in/courses/117/101/117101054/

SEMESTER S8
SAFETY INSTRUMENTATION SYSTEM

Course Code	PEICT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Gain a comprehensive understanding of safety hazards, risks, and the principles of Safety Instrumented Systems (SIS), including their design, implementation, and standards compliance (IEC 61508, IEC 61511, ISA 84.01-2003).
2. Develop the ability to identify, assess, and mitigate occupational safety and health hazards, utilizing accident theories and models to prevent incidents and ensure a safe working environment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Safety Hazards and Risks Safety Instrumented System-Definition-Introduction-Hazards-risks-Measurement of risks-Reduction of risks-Design Principles-Risk reduction factor (RRF), Safety Integrity Level (SIL)-Occupational Safety hazard (OSH)-Occupational Health Hazard (OHH)-Factors of affecting OSH. - Accident-Theories-Domino theory-Bird's law causation model, Tye Pearson model	9
2	Safety standards Safety Terminology-Critical control systems-Safety shutdown systems-Protective instrumented systems-Equipment protection systems-Emergency shutdown systems-safety critical systems-Interlocks-Safety instrumented systems-ISA, IEC-Standards:IEC61508-Fundamental concepts, IEC61511, ISA84.01-2003,Safety life cycle.	9
3	Safety Integrity parameters Safety Integrity Level Selection- Defining SIL-SIL Selection Process-Representing Tolerable Risk for SIL Selection- Layer of Protection	9

	Analysis- Conceptual Design / SIL Verification- Component Selection- Fault Tolerance- Functional Test Interval- Common Cause Failures- Diagnostic Coverage- PFD Calculation	
4	Safety Testing Safety Requirements Specifications- Detailed Design and Specifications- Procedure Development- Construction, Installation, and Commissioning- Pre-Startup Acceptance Testing- Operations and Maintenance- Management of Change.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts of Risk, Hazard and safety.	K2
CO2	Understand about Emergency shutdown systems	K2
CO3	Understand the Industrial safety standards	K2
CO4	Understand the safety and its process level selections	K2
CO5	Understand the design specifications of Safety systems	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3											2
CO3	3											2
CO4	3											2
CO5	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	SIL Manual, Safety Instrumented systems	Steve Mackay	GM International	3 RD EDITION
2	Safety instrumented system Engineering Handbook	Kevin J. Mitchell, Peter Herena, Todd M. Longendelpher, Matthew C. Kuhn	Kenexis Consulting corporation	2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Functional Society: A straight forward guide to IEC61508	David J Smith		2 ND EDITION 2004
2	Practical guide to safety instrumented systems (SIS) standard IEC 61511	Mark A Brown		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.slideshare.net/slideshow/module-1-introduction-to-industrial-safety/5443462,
2	https://www.kenexis.com/wp-content/uploads/2014/05/www.kenexis.com_resources_TechnicalPapers_SIS_Handbook.pdf
3	https://www.kenexis.com/wp-content/uploads/2014/05/www.kenexis.com_resources_TechnicalPapers_SIS_Handbook.pdf
4	https://www.kenexis.com/wp-content/uploads/2014/05/www.kenexis.com_resources_TechnicalPapers_SIS_Handbook.pdf

SEMESTER S8
REMOTE SENSING AND CONTROL

Course Code	PEICT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To offer students to gain a glimpse into the complexity alongside the potential of remote sensing
2. To provide a basic understanding of remote sensing and GIS and their application in various fields including research

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to remote sensing Introduction to remote sensing: Definition, History, Principle, Advantage, Challenges, Scope; Satellite Remote Sensing: Principle, Sensor measurement, resolutions, Basic Remote Sensing unit using Landsat data: History, Different types, Landsat Accessing level, Selection, Landsat calibration	9
2	Introduction to Image Processing Image: relevance, properties, Image Processing techniques: filtering pixels, applying algorithms, colour palettes, Practical software: SNAP, downloading, importing, creating the image, manipulating, filter, NDVI Algorithm.	9
3	Introduction to QGIS and Water cycle GIS, installation, and packages, QGIS: Importing remote sensing data into QGIS, Data handling techniques, CRS Adjustments within QGIS, Saving projects. Microwave data for water cycle monitoring, applications, coastal water, Bathymetric data, microwave signatures, water quality and protection	9
4	Urban Environments and Landscape evolution Introduction to optical signatures of urban surfaces, Thermal signatures of	9

	urban surfaces, Applications, Creep, dynamics, Air quality, Time series analysis for landscape evolution monitoring, Techniques, microwave data, applications. Future of Remote Sensing: Instruments, Data, Satellite, Cloud processing, analysis, software, applications.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the challenges associated with remote sensing.	K1
CO2	Explain the use of knowledge in Image processing techniques, its software and algorithms in Remote sensing.	K2
CO3	Explain various Data handling techniques involved.	K2
CO4	Explain quality-based time series analysis and future trends of remote sensing	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2						3					3
CO2	2						3					3
CO3	2						3					3
CO4	2						3					3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Practical Handbook of remote sensing”	Lavender, Andrew Lavender, Samantha,	CRC Press	1/e
2	“Introduction to the Physics and Techniques of Remote Sensing”	Charles Elachi, Jakob van Zyl, Jin Au Kong,	Wiley-Interscience	2/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Remote sensing”	Siamak Khorram, Stacy A.C. Nelson, Frank H. Koch, Cynthia F. Van der Wiele	Springer	2012
2	“Remote Sensing. Models and Methods for Image Processing”	Robert A. Schowengerdt	Academic Press	3/e

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=uQ-a5Ffy8SM
2	https://www.youtube.com/watch?v=ZItfCdwBySQ
3	https://www.youtube.com/watch?v=WYy-owOcfSY
4	https://www.youtube.com/watch?v=S22u7_Eq26g

SEMESTER S8
INSTRUMENTATION AND CONTROL IN LARGE SCALE
INDUSTRIES

Course Code	PEICT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PEICT 743, PCICT205	Course Type	Theory

Course Objectives:

1. Gain knowledge of various chemical and biological treatment processes, including oxidation, reduction, and neutralization, and their control mechanisms.
2. Learn the control systems and optimization techniques for evaporators, clean rooms, steam boilers, and oil industry instrumentation, focusing on maintaining efficiency, safety, and environmental compliance.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Instrumentation and Control of Effluent Water and Waste Water Treatment Basics, chemical oxidation, cyanide destruction process, batch oxidation, continuous oxidation, chlorinator, chemical reduction, reduction of hexavalent chromium, dichlorination, neutralisation, valve rangeability required, precise neutralisation, single reagent control, feedforward feedback neutralisation control, precipitation, lime softening, biological control, chlorination-based disinfection control	9
2	Control of Evaporator Evaporator terminology, single effect evaporator, multiple effect evaporator, economy, capacity, modelling, steady-state model, control systems for evaporators, feedback control, cascade control, feedforward control: process dynamics, process dynamics with feedback trim, steam enthalpy, absolute pressure.	9
3	Control and Optimisation of Clean Room Need of cleanroom, clean-room, subzone optimisation, pressure control,	9

	elimination of draft, temperature control, humidity control, flow control, optimisation of zones, envelope optimisation, plantwide optimisation, material balance control, heat balance control, mechanical design limitations, humidity control, exhaust air control.	
4	Control of Steam Boiler Flow sheet symbol, process of boiler, safety interlocks, basic boiler controls, load sharing control, boiler control with alternative pressure or flow master, feedforward control with feedback trim, fuel controls, air flow control, damper and fan controls, furnace draft control, fuel-air ratio, proportional compensation in air-fuel ratio control. Instrumentation and Control used in Oil Industry separators, test separators and well test, first stage separator, second stage separator, third stage or final stage separator, coalescer, electrostatic, gas treatment, heat exchanger, HVAC.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the control of chemical treatments used in effluent and water treatment.	K2
CO2	Understand the use of multi loop control with respect to the process dynamics.	K2
CO3	Illustrate the process, operation, control, and optimization of a clean room.	K2
CO4	Explain the trouble shooting, and master control signal flow of steam boiler.	K2
CO5	Illustrate the process, operation, instrumentation, and control of oil & natural gas industries	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineers' Handbook, Process Control and Optimization	B.G.Liptak	CRC Press	Volume II ,4/e
2	Oil and Gas Production Handbook	Harvard Devold	LULU Press	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Process Control Systems	Dale R. Patrick and Stephan W.Fardo	Vikas Publishing House	
2	Instrumentation Symbols and Identification	Mark A Brown	ISA	1992
3	Boiler Control System	David Lindesnev	McGraw Hill	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/126/105/126105020/
2	https://youtu.be/id1DQ0eRG0o?si=KAKjI6w7tjKhosBz
3	https://youtu.be/YAouXIS_FSU?si=fQoJeY5mr5YxHvD3
4	https://youtu.be/-cr5vfV4YAI?si=Q_JcvysEfG8q2sJf

SEMESTER S8
CORROSION CONTROL

Course Code	PEICT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. It aims to explain the concept of corrosion and different types of corrosion, and also helps the students to select a suitable corrosion control or prevention technique.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Corrosion Historical background, definitions, corrosive environment, consequences of corrosion, cost of corrosion, breakdown of spending on corrosion, corrosion science and corrosion engineering, functional aspects of corrosion. Basic Concepts in Corrosion Anodic and cathodic reaction, Anodic reactions characteristics, Cathodic reactions characteristics, types of corrosion cells, mechanism of corrosion of Iron, concept of free energy, reversible electrode potential, concentration cell, liquid junction potential, application of free energy to corrosion cells, Nernst equation, sign convention, reference electrodes	9
2	Corrosion Kinetics and Measurement Faraday's Laws, Corrosion Kinetics, Helmholtz double layer, reverse reaction (cathodic reaction), departure from equilibrium, measurement of corrosion, determination of corrosion rates by electrochemical measurements, polarisation linearization, kinetics of passivity, definition of important electrochemical parameters for active passive metals, control of passivity, effect of environment.	9

3	Cathodic Protection Basis of cathodic protection, working of cathodic protection system, factors leading to corrosion of underground metallic structures, electrical basis of cathodic protection, electrochemical theory of cathodic protection, anodic polarisation, cathodic protection system, components of Galvanic systems, impressed current system, protection of submarine pipelines, design parameters in cathode protection, measurements in cathodic protection.	9
4	Corrosion Control by Inhibition Scope of inhibitor, examples of application of inhibitors, important consideration in selection of inhibitors, important terms related to inhibitors, classification of inhibitors, description of inhibitors, inorganic anodic inhibitors, organic inhibitors, synergistic inhibitors, scavengers, neutralisers, scale inhibitors, rebar inhibitors, inhibitor application techniques, inhibitor efficiency and inhibitor concentration	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of corrosion.	K2
CO2	Calculate the corrosion kinetics and use measurement techniques.	K3
CO3	Discuss about the cathodic protection and measurements in cathodic protection.	K2
CO4	Explain how to control corrosion by inhibition.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											3
CO2	3	2										3
CO3	2											3
CO4	2											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Corrosion Engineering and Corrosion Control	Zaki Ahmad	Elsevier Science and Technology Books	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Corrosion Control	Samuel A. Bradford	Van Nostrand Reinhold	2/e, 2001
2	Corrosion Control in Petroleum Production	Harry G. Byars	TPC Publication,	5/e
3	Corrosion and Corrosion Control	R. Winston Revie and Herbert H. Uhlig	Wiley	4/e

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.digimat.in/nptel/courses/video/113104082/L01.html
2	https://www.youtube.com/watch?v=qmDoabQ7-ZI
3	https://www.digimat.in/nptel/courses/video/113101099/L06.html
4	https://www.youtube.com/watch?v=J4aiB6jAZR8

SEMESTER S8

IMAGE PROCESSING

Course Code	PEICT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PEICT 415	Course Type	THEORY

Course Objectives:

1. This course will strengthen fundamental knowledge about image processing techniques.
2. Digital image processing is used in almost all engineering fields and wide range of applications in industrial automation, medical, agriculture, security, entertainment, education and many more.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Image Fundamentals: Image representation, basic relationship between pixels, elements of DIP system, elements of visual perception-simple image formation model. Brightness, contrast, hue, saturation, mach band effect. Colour image fundamentals RGB, CMY, HIS models. 2D Image transforms DFT, its properties, Walsh transform, Hadamard transform, Haar transform.DCT and KL transform.	9
2	Image Enhancement Spatial domain methods: point processing- intensity transformations, histogram processing,image subtraction, image averaging.Spatial filtering-smoothing filters, sharpening filters.Frequency domain methods: low pass filtering, high pass filtering, homomorphic filter.	9
3	Image Restoration Unconstraint restoration- Lagrange multiplier and constraint restoration. Inverse filtering- removal of blur caused by uniform linear motion, Weiner filtering, Image segmentation Classification of Image segmentation techniques, region approach,	9

	clustering techniques. Classification of edges, edge detection, Hough transform.	
4	Image Compression Need for compression, redundancy, classification of image compression schemes, Huffman coding, arithmetic coding, dictionary-based compression, transform based compression, Image compression standards- JPEG& MPEG.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discuss image representation, and various concepts and mathematical transforms necessary for image processing.	K2
CO2	Differentiate and interpret the various image enhancement techniques	K2
CO3	Distinguish the various image restoration techniques.	K2
CO4	Illustrate image segmentation algorithm and Explain basic image compression techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

3	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3				3							3
CO4	3				3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Image Processing	Gonzalez Rafel C	Pearson Education	2009
2	Digital image processing	S Jayaraman, S Esakkirajan, T Veerakumar	Tata Mc Graw Hill.	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of digital image processing	Jain Anil K	PHI	1988
2	Digital image processing	Kenneth R Castleman	Pearson Education, 2/e	1996
3	Digital Image Processing	Pratt William K	John Wiley, 4/e	4 th ed/2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_ee116/preview
2	https://www.youtube.com/watch?v=LyDrGJRT0PI
3	https://www.youtube.com/watch?v=jl3JeshN2zA https://www.youtube.com/watch?v=3qJej6wgezA
4	https://www.youtube.com/watch?v=sckLJpjH5p8

SEMESTER S8
INDUSTRIAL AUTOMATION

Course Code	OEICT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Provide an overview of automation concepts, including historical evolution, principles, and the importance of automation in modern industries.
2. Introduce students to Programmable Logic Controllers (PLCs)
3. Explain Supervisory Control and Data Acquisition (SCADA) systems.
4. Discuss networking fundamentals relevant to industrial automation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Automation introduction Control Systems and Automation Strategy, Evolution of instrumentation and control, Role of automation in industries, Benefits of automation, Introduction to Descriptive automation tools PLC, DCS, SCADA, Hybrid DCS/PLC, Automation strategy evolution, Control system audit, and performance criteria.	8
2	Instrumentation Standard Protocols Definition of protocol, Introduction to Open System Interconnection (OSI) model, Communication standard (RS232, RS485), Modbus (ASCII/RTU), Introduction to third party interface, concept of OPC (Object linking and embedding for Process Control), HART Protocol: Introduction, frame structure, programming, implementation examples, benefits, advantages and limitation. Foundation Fieldbus H1: Introduction, frame structure, programming, implementation examples, benefits, advantages and limitation. Comparison of HART, Foundation Fieldbus, Device net, Profibus, Controlnet, Industrial Ethernet	11
3	Programmable logic controllers (PLC)	9

	Introduction, architecture, definition of discrete state process control, PLC Vs PC, PLC Vs DCS, relay diagram, ladder diagram, ladder diagram examples, relay sequencers, timers/counters, high speed counter, PTO, PWM and PID blocks in PLC, PLC design, study of at least one industrial PLC.	
4	Networking fundamentals Advance Applications of PLC and SCADA, PLC programming methods as per IEC 61131, PLC applications for batch process using SFC, Analog Control using PLC, PLC interface to SCADA/DCS using communication links (RS232, RS485) and protocols (Modbus ASCII/RTU) Distributed Control Systems DCS introduction, functions, advantages and limitations, DCS as an automation tool to support Enterprise Resources Planning, DCS Architecture of different makes, Latest trends and developments.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Provide an overview of automation concepts, including historical evolution, principles, and the importance of automation in modern industries.	K2
CO2	Information about Instrumentation standard protocols	K2
CO3	Develop Skills in Programmable Logic Controllers (PLCs), Supervisory Control and Data Acquisition (SCADA) systems.	K2
CO4	Knowledge of networking fundamentals relevant to industrial automation.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				2							2
CO2	3				2							2
CO3	3				2							2
CO4	3				2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Automation and Control System Security Principles	Tyler, Joel Thomas	CRC Press	Edition: 1st Edition Year: 2020
2	Industrial Communication Technology Handbook	Richard Zurawski (Editor)	CRC Press	Edition: 3rd Edition Year: 2016
3	Programmable Logic Controllers: A Practical Approach to IEC 61131-3 using CoDeSys	Dag H. Hanssen	Wiley-IEEE Press	Edition: 1st Edition Year: 2015
4	Industrial Automation and Control System Security Principles	Joel Thomas Tyler	CRC Press	Edition: 1st Edition Year: 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Industrial Automation and Process Control"	Jon Stenerson, Mark A. Howard, and Damian A. Flynn	Prentice Hall	Edition: 2nd Edition Year: 2010
2	"Programmable Logic Controllers: Principles and Applications"	John W. Webb, Ronald A. Reis	Pearson	Edition: 1st Edition Year: 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ee56/preview#:~:text=Nptel https://control.com/technical-articles/understanding-the-automation-lingo-plc-pac-rtu-dcs-and-scada/
2	https://www.scribd.com/presentation/571046284/Unit-2
3	https://archive.nptel.ac.in/content/storage2/courses/112102011/downloads/faq%20of%20module%204.pdf
4	https://controllogicbd.com/advance-plc-with-hmi-scada/ https://onlinecourses.nptel.ac.in/noc21_cs87/preview

SEMESTER S8
ENVIRONMENTAL INSTRUMENTATION

Course Code	OEICT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The aim of environmental instrumentation program is to offer students a basic knowledge of Environmental Instrumentation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Environmental Engineering An overview of environmental laws and regulations. Necessity of Instrumentation & Control for environment, sensor requirement for environment, Classification of pollution: Air pollution, water pollution, soil pollution, marine pollution, noise pollution, thermal pollution, nuclear hazards and hazardous waste, occupational safety and health. Causes, Effects and control of all types of Pollution.	9
2	Quality of water Standards of raw & treated water, sources of water & their natural quality, effects of water quality. Water quality parameters: Thermal conductivity, dissolved oxygen, anions, turbidity. Water Quality detectors, Opacity monitors, conductivity analysers & their application. Spectrometric methods- emission spectrograph- atomic absorption spectra.	9
3	Air monitoring Measurement of ambient air quality. Air pollution control of stationary sources: gaseous pollutants and particulate pollutants. Air pollution control of mobile sources: automobile emissions. Noise pollution: effect of noise pollution on people and community, noise area classification and hearing conservation. Noise sources and criteria, noise level measurement, noise absorption techniques and noise control.	9

4	Pollution control in industries pollution control in petroleum refineries, fertilizer industries, pulp and paper industries, cement industries. Pollution case studies - disaster management: floods, earthquake, cyclone and landslides, Environmental disaster monitoring. Handling and management of hospital wastes.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discuss the role of Instrumentation in Environmental aspects.	K2
CO2	Explain the design of systems for water and air pollution control.	K2
CO3	Discuss the safety aspects of Environmental Engineering and proper waste disposal method	K2
CO4	Discuss the applications like pollution control in petroleum, paper and cement industries.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2						3					2
CO2	2						3					2
CO3	2						3					2
CO4	2						3					2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Instrumentation & Analysis Handbook	Randy D. Down	WileyInterscience	1/e
2	Air Pollution Engineering	M. N. Rao & H. V. N. Rao	McGraw Hill Education	1/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Instrumental Analysis	Skoog, Holler, Nieman	Cengage Learning	7/e
2	Introduction to Instrumental Analysis	Braun, Robert D., Pharma	Pharma MEd Press/BSP Books	2019
3	Analytical Instrumentation	Sherman R.E. and Rhodes L.J		ISA.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://acl.digimat.in/nptel/courses/video/103107215/L09.html
2	https://www.digimat.in/nptel/courses/video/105107207/L32.html
3	https://www.youtube.com/watch?v=ITZPRJtLdeg
4	http://digimat.in/nptel/courses/video/103107215/L49.html

SEMESTER: S8
INDUSTRIAL INSTRUMENTATION

Course Code	OEICT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the students about various types of temperature, pressure, level, pH, density, viscosity, turbidity and conductivity measurement systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Temperature Measurement Definitions and units- Standards of temperature – Thermometry and pyrometry- Thermocouples- Peltier effect- Thomson effect- Seebeck effect- Types of thermocouples (Chromel- copper. Chromel-alumel.copper-constantan, iron-constantan, platinum-rhodium- platinum) Cold junction compensation-Signal conditioning for thermocouple- Thermowells— Thermopiles- Resistance thermometers- Principle of operation- Platinum RTD- Construction of RTDs Thermistors, principle and sensor types, manufacturing techniques, measuring circuits. Thermometers- Expansion thermometers- Bimetallic thermometers – Liquid filled -thermometers- Gas filled thermometers. Pyrometry- Stefan Boltzmann’s law- Black body radiation- Total radiation pyrometersOptical radiation pyrometers.	10
2	Pressure measurement Units of pressure – Manometers: Different types, Elastic type pressure gauges: Bourdon tube, Bellows, Diaphragms and Capsules – Electrical methods: Elastic elements with LVDT and strain gauges – Capacitive type pressure gauge – Piezo resistive pressure sensor-Resonator pressure sensor – High-Pressure sensors: Dead-Weight piston gauge, Bulk modulus cellsMeasurement of vacuum: McLeod gauge, Thermal conductivity gauge, lionization gauges, Cold cathode type and hot cathode type – Pressure gauge selection	8

3	Measurement of viscosity Definitions, units, Newtonian and Non-Newtonian behaviour, measurement of viscosity using laboratory viscometers, industrial viscometers, Viscometer selection and application. Measurement of density Definitions, units, liquid density measurement, gas densitometers, its application and selection. Level measurement Level measurement: introduction, float level devices, displacer level devices, rotating paddle switches, diaphragm and differential pressure detectors, resistance, capacitance and RF probes, radiation, conductivity, field effect, thermal, ultrasonic, microwave level switches, radar and vibrating type level sensors. Level sensor selection and application	10
4	Flow measurement Introduction, definitions and units, classification of flow meters, pitot tubes, positive displacement liquid meters, positive displacement gas flow meters, variable area flow meters. Anemometers: Hot wire/hot film anemometer, laser Doppler anemometer (LDA) Electromagnetic flow meter, turbine and other rotary element flow meters, ultrasonic flow meters, doppler flow meters, cross correlation flow meters, vortex flow meters	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain working principle and measurement techniques of various types of temperature measuring instruments	K2
CO2	Explain working principle and measurement techniques of various types of pressure measuring instruments.	K2
CO3	Describe working principle and measurement techniques of various types of level, viscosity and density measuring instruments.	K2
CO4	Explain working principle and measurement techniques of various types of level measuring instruments.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineers Handbook	Liptak.B.G	Chilton book.co.	Fourth, Volume 1, 2003
2	Instrument technology	Jones.E.B	Scientific Pub	Volume1, Edition3, 1974
3	Instrumentation Reference Book	Walt Boyes	Elsevier	4/e.2010
4	Applied Instrumentation in Process Industries	Andrew and Williams	Gulf publications	Volume 1,2, Edition2, 1979

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical and Industrial Measurements	Jain.R.K	Khanna Publishers	Edition11, 1995
2	Principles of Industrial Instrumentation	Patranabis.D	Tata McGraw Hill	Edition2, 2008
3	Measurement System - Application and Design	Doebelin.E. O	McGraw hill	Edition4, 1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec28.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec29.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec27.mp4
2	https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec21.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec22.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec23.mp4
3	https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec46.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec47.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec48.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec49.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod11lec54.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod11lec55.mp4
4	https://archive.nptel.ac.in/courses/103/103/103103147/